STA2001H Probability

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Course Information

Lecture times: This course is held from January 2025-May 2025. Lecture times at 0900-1030 Wednesday and Friday (Teaching D, 307). The notes are available on blackboard.

Office Hour: My office hour is at 1400 Thursday at my office 511B Dao Yuan. I am not available at other times, unless there is a time-table clash. If you would like to see me or have a question, please email me or the TAs. I will try to respond as quickly as I can.

Assessment: The course is assessed by: (1) Mid-Term (35%), and (2) Final exam (65%). The mid-term is scheduled for the seventh week of teaching and consists of two (2) questions. The mid-term is 1 hour long, you are not allowed any type of calculation device (eg smart phone) but you may bring a single piece of A4 help sheet. On that help sheet you may write any information (on both sides) that you think is necessary. The final exam is 2 hours long, and consists of four (4) questions. You are not allowed any type of calculation device but you may bring a single piece of A4 help sheet. In the case of the mid-term and final, there will be a practice (example) paper for you to prepare from. There are 10 problem sheets that are not assessed.

Course Details: *These notes are not sufficient to replace lectures.* In particular, many examples and clarifications are given during the class. This course structure is explained in the subsequent lesson plan. The final chapter of the notes provides some mathematical background for the course, which you are strongly advised to read in the first week. You are expected to know everything that is in this chapter. In particular the notion of double summation and integration is used routinely in this course and you should spend some time recalling these ideas.

References: The recommended reference for this course is [1]. This book can be read here. The book [2] provides an entertaining and non-technical introduction to probability and can help to illuminate some of the topics that we cover.

Lesson Plan

The following lesson plan is subject to change. The main notes document will be constantly updated and this particularly concerns the lesson plan.

- Week 1: Review of basic mathematics for probability (Chapter 6). Course overview. Why study probability (Section 1.1 and [1, pp. 1-3])? An informal introduction to probability (Section 1.2).
- Week 2: An informal introduction to probability continued (Section 1.2). Probability Triples (Section 1.3). Conditional Probability (Section 1.4).
- Week 3: Conditional Probability continued (Section 1.4). Simpson's Paradox ([1, pp. 76-79]). Independence (Section 1.5).
- Week 4: (Discrete) random variables and distribution functions (Section 2.2). Probability mass functions (Section 2.3). Independence (Section 2.4). Some probability mass functions (Section 2.5).
- Week 5: Some probability mass functions continued (Section 2.5). Expectation (Section 2.6). Probability density functions (Section sec:pdf).
- Week 6: Probability density functions continued (Section 3.2). Independence (Section 3.3). Expetation (Section 3.4).
- Week 7: Mid-term review and in-class mid-term exam.
- Week 8: Joint discrete distributions (Section 4.2).
- Week 9: Joint discrete distributions continued (Section 4.2). Multinomial distribution ([1, pp. 332-334].).
- Week 10: Joint continuous distributions (Section 4.2).
- Week 11: Symmetry of i.i.d. continuous variables (Section 4.4). Order Statistics (Section 4.5). Connection between the exponential distribution and Poisson Process (Section 4.6).
- Week 12: Multivariate normal distribution (Section 4.7). Probability inequalities (Section 5.2).
- Week 13: Convergence in Distribution (Section 5.3). Convergence in Probability (Section 5.4).
- Week 14: Central Limit Theorem (Section 5.5). Final review.

Chapter 1

Introduction to Probability

1.1 Introduction

This Chapter provides a basic introduction to the notions that underlie the basics of probability. The level of notes is below that required for complete mathematical rigor, but does provide a technical development of these ideas. To understand *why* we study probability this is very well explained in [1, pp. 1-3] and we shall discuss these concepts now[1].

We begin with trying to give an initial and informal approach to probability. By itself it does not suffice in having a mathematical theory, but provides useful intuition. The basic ideas of probability theory start with a *probability triple* of an 'event-space' (e.g. in flipping a coin a head or tail), sets of events (e.g.'tail' or 'tail and head') and a way to compare the 'likelihood' of events by a probability distribution (the chance of obtaining a tail). Moving on from these notions, we consider the probability of events, given other events are *known* to have occurred (*conditional probability*) (the probability a coin lands tails, given we know it has two 'heads'). Some events have probabilities which decouple in a special way and are called *independent* (for example, flipping a fair coin twice, the outcome of the first flip, may not influence that of the second).

The structure of this Chapter is as follows: In Section 1.2 we start with an informal way to approaoch probability; In Section 1.3, we discuss the idea of a probability triple; in Section 1.4 conditional probability is introduced and in Section 1.5 we discuss independence.

1.2 An Informal Introduction to Probability

This section contains some intutive ideas, which by themselves are not enough to construct the calculus of probability. However, they can provide an interesting introduction and several concepts will be useful as we move through the notions of probability.

1.2.1 Initial Approach to Probability

Constructing the calculus of probabilities requires a high degree of mathematical sophistication. Indeed, even at this level, we will not be able to really study how probability is defined in a formal mathematical sense; our development will be correct, but some important details which are based on *measure theory* a branch of pure mathematics. None-the-less we will ultimately be able to derive some rules which will provide the foundation of probability. Before we do this, we start with some informality, to give us some intuition from which to begin our study.

Consider a state-space Ω (a set) of possible outcomes associated to an experiment for which there is some uncertainty.

Example 1.2.1. *The classical example: flipping a 'fair' two sided coin, once one side that is 'head' the other that is 'tail. We do not know which side the coin will land on. If H represents Head and T represents tails then $\Omega = \{H, T\}$.*

Associated to Ω are 'events' that is things that could happen and we wish to understand how likely each of these events will occur.

[1] Students, please open the book at the relevant pages

Example 1.2.2. *The collection of events which we denote as $\mathscr{F}$ in the case of the coin flipping experiment of Example 1.2.1 could be $\{H\}$, $\{T\}$ but it also could be either $\{H\}$ or $\{T\}$ and it could be $\{H\}$ and $\{T\}$. In words the events are heads, tails, heads or tails, heads and tails. This can be written mathematically as*

$$\mathscr{F} = \{\{H\}, \{T\}, \Omega, \emptyset\}$$

where $\emptyset$ is the empty set. Now intuitively the probability of each event $\mathbb{P}(A)$ where $A \in \mathscr{F}$ can be defined as:

$$\mathbb{P}(A) = \frac{\text{Card}(A)}{\text{Card}(\Omega)} = \frac{\text{number of elements in } A}{\text{number of elements in } \Omega}.$$

This is because with a fair coin one may expect that head or tail is equally likely. Therefore, for instance,

$$\mathbb{P}(\{H\}) = \frac{1}{2} = \mathbb{P}(\{T\}).$$

Examples 1.2.1 and 1.2.2 have given our first forray into probability: we need a state-space of outcomes,. events of those outcomes and a way to measure the chance of those outcomes. At this stage, we can think of the probability of an event, $A \in \mathscr{F}$, being

$$\frac{\text{number of elements in } A}{\text{number of elements in } \Omega}$$

when all elements of Ω are equally likely. This concept will sometimes serve us well, but in general is not enough to provide us with a definition. However, for now we shall consider using it and develop some notions of probability. This naturally takes us to the idea of being able to count outcomes.

1.2.2 Counting Outcomes

We start with the basic idea of the multiplication rule:

> *Given two possible events A and B each with a and b elements respectively and the fact that A will occur and then B will occur. The total number of possible outcomes is ab.*

Example 1.2.3. *Consider a race with 10 people and that ties are not possible. How many possible first, second and third place runners are there? There are 10 that could finish first, then 9 that could finish second and then 8 that could finish third, so the total is $10 \times 9 \times 8 = 720$. Note that one need not start off with first place (e.g. you could start with second, first then third) as the result is the same irrespective of the order you choose.*

Example 1.2.4. *Consider n balls in a box and taking a ball out of the box, returning it and repeating k times. In this case there are n^k possible outcomes.*

Example 1.2.5. *Consider n balls in a box and taking a ball out of the box, and repeating $k \leq n$ times. In this case there are $n(n-1) \times \cdots \times (n-k+1)$ possible outcomes.*

Consider the idea in Example 1.2.5 and the concept of n numbers. A permutation of the numbers $1, \ldots, n$ is the idea of taking one number out and repeating n times so we have a particular order of the numbers. For instance if $n = 3$ then a permutation could be $1, 2, 3$ or $3, 2, 1$. By using the ideas in Example 1.2.5, the number of possible permutations is then $n \times (n-1) \times \cdots \times 1 =: n!$. We define $0! := 1$.

Example 1.2.6. *The Birthday Problem: In a set of $n \leq 365$ randomly selected people, what is the probability that at least two people share the same birthday? Here random means that each persons birthday is equally likely to be any of the 365 days of the year: we exclude the possibility of being born on February 29.*

We can think of the idea of n people having a birthday on a given day as the process of sampling one of 365 days with replacement n times; hence there are 365^n possible outcomes. Let us start by counting the number of ways that amongst n people, no two have the same birthday. Then, denoting by A the event that in n randomly selected people, what is the probability that at least two people share the same birthday, B the event that amongst n people, no two have the same birthday we would have that:

$$\mathbb{P}(A) = 1 - \mathbb{P}(B).$$

This is because the sum of the number of elements in A and B must be that of Ω. Counting the number of elements in B is essentially that of choosing the birthday for each person, but sampling the days independently: so there are $365 \times 364 \times \cdots \times (365 - n + 1)$ elements in B. Hence the probability of interest is:

$$1 - \frac{365 \times 364 \times \cdots \times (365 - n + 1)}{365^n}.$$

One can easily compute this probability for a range of n and indeed if $n = 23$ the chance is already higher than 0.5.

Example 1.2.7. *We have four people: in how many ways can we take 2 people out of the four? Labelling the people as $1, 2, 3, 4$ this can be solved by enumeration: however in general this can be tediuos or (almost) impossible even using a super-computer. Perhaps a more general way is using the multiplication rule with a correction. Clearly we can choose the first person in four ways and the second person in three ways. But this will over-count as this permits (for example) the case $1, 2$ and $2, 1$ whereas these are the same subset of two people. So we have over-counted by a factor of 2 and hence there are $(4 \times 3)/2$ possible ways.*

Proposition 1.2.8. *Let $n \in \mathbb{N}$ be given and $k \in \{0, 1 \ldots, n\}$. The number of subsets of size k from n is:*

$$\frac{n(n-1) \times \cdots (n-k+1)}{k!} = \frac{n!}{(n-k)!k!} := \binom{n}{k}.$$

$\binom{n}{k}$ is called the Binomial coefficient.

Proof. Consider sampling k items out of n without replacement this has $n(n-1) \times \cdots (n-k+1)$ elements and this is an ordered list. However, this overcounts the total number of possibilities as there are $k!$ possible orders so the total number of subsets is hence

$$\frac{n(n-1) \times \cdots (n-k+1)}{k!}.$$

The equality is easily established by using the definition of the factorial. □

Another way to think of the Binomial coefficient is this: the number of ways of drawing k 'identical' items out of n ($0 \leq k \leq n$) when the order does not matter.

Example 1.2.9. *Suppose we have a society of $n > 3$ people. The number of ways of choosing a president, vice president and treasurer is $n(n-1)(n-2)$. The number of ways of taking three people with undetermined titles is $\binom{n}{3}$. As can be seen, in the second case, we do not care about the order and hence this latter number is smaller.*

Example 1.2.10. *A full house in poker consists of three cards of a given rank and two cards of another rank. For instance three kings and two queens. You sample (at random) 5 cards (wihout replacement) from a full deck: what is the probability that it is a full house?*

We note that we choose 5 cards out 52 and the order does not matter; so $\binom{52}{5}$ are the number of poker hands that you could have. Now for the rank three, there are 13 ranks and one must choose 3 out of 4 of the ranks in any order: so there are $13 \times \binom{4}{3}$ possible ways of choosing the rank 3. Then for the rank 2, there are 12 suits left and $\binom{4}{2}$ possible ways of choosing the associated required cards. Hence using our definition of probability the chance of drawing a full house is:

$$\frac{13 \times \binom{4}{3} \times 12 \times \binom{4}{2}}{\binom{52}{5}}.$$

1.3 Probability Triples

As mentioned in the introduction, probability begins with a triple:

- A sample space (possible outcomes)
- A collection of sets on outcomes ($\sigma-$field)
- A way to compare the likelihood of events (probability measure)

We will slowly introduce and analyze these concepts.

Notation	Set Terminology	Probability Terminology
Ω	Collection of objects	Sample Space
ω	Member of Ω	Elementary event
A	Subset of Ω	Event that an outcome in A occurs
A^c	Complement of A	Event that no outcome in A occurs
$A \cap B$	Intersection	Both A and B
$A \cup B$	Union	A or B
$A \setminus B$	Difference	A but not B
$A \Delta B$	Symmetric difference	A or B but not both
$A \subseteq B$	Inclusion	If A then B
$\emptyset$	Empty set	Impossible event

Table 1.1: Terminology of Set and Probability Theory

1.3.1 Sample Space

The basic notion we begin with is that of an experiment, with a collection of possible outcomes to which we cannot (usually) determine exactly what will happen. For example, if we flip a coin, or if we watch a football game and so on; in general we do not know for certain what will happen. The idea of a sample space is as follows.

Definition 1.3.1. *The set of all possible outcomes of an experiment is called the* ***sample space*** *and is denoted by* Ω.

Example 1.3.2. *Consider rolling a six sided fair die, once. There are 6 possible outcomes, and thus:* $\Omega = \{1, 2, \ldots, 6\}$. *We may be interested (for example, for betting purposes) in the following events:*

1. *we roll a 1*
2. *we roll an even number*
3. *we roll an even number, which is less than 3*
4. *we roll an odd number*

One thing that we immediately realize is that each of the events in the above example are subsets of Ω, that is each event (1)-(4) correspond to:

1. {1}
2. {2,4,6}
3. {2}
4. {1,3,5}

As a result, we think of events as subsets of the sample space Ω. Such events can be constructed by unions, intersections and complements of other events (we will formally explain why, below). For example, letting $A = \{2, 4, 6\}$ in the case (2) above, we immediately yield that the event (4) is A^c. Similarly letting A be the event in (2) and B be the event of rolling a number less than 4 we obtain that event (3) is $A \cap B$. For those of you that have forgotten basic set notations and terminologies, see Table[2] 1.1. In particular, we will think of Ω as the certain event (we will roll a $1, 2, \ldots, 6$) and its complement $\Omega^c = \emptyset$ as the impossible event (we must roll *something*).

1.3.2 Collection of Events

Now we have a notion of an event, in particular, that they are subsets of of Ω. A particular question of interest is then: are all subsets of Ω events? The answer actually turns out to be *no*, but the technical reasons for this lie far out of the scope of this course. We will content ourselves to use a particular collection of sets $\mathscr{F}$ (a 'set of sets') of Ω which contains all the events that we can make probability statements about.

[2]For those of you that have forgotten set theory, Section 6.1 has a refresher

Definition 1.3.3. *A collection of events $\mathscr{F}$ of Ω satisy following conditions:*

1. $\emptyset \in \mathscr{F}$
2. *If* $A_1, \ldots, \in \mathscr{F}$ *then* $\bigcup_{i=1}^{\infty} A_i \in \mathscr{F}$
3. *If* $A \in \mathscr{F}$ *then* $A^c \in \mathscr{F}$

Example 1.3.4. *Consider flipping a single coin, letting H denote a head and T denoting a tail then: $\Omega = \{H, T\}$ and $\mathscr{F} = \{\Omega, \emptyset, \{H\}, \{T\}\}$.*

Thus, to summarize, so far $(\Omega, \mathscr{F})$ is sample space and collection of events. The former is the collection of all possible outcomes and the latter is collection of sets on Ω (the events) that follow Definition 1.3.3. Our next objective is define a way of assigning a likelihood to each event.

Exercise 1.3.5. *You are given De-Morgan's Laws:*

$$\Big(\bigcup_i A_i\Big)^c = \bigcap_i A_i^c \quad \Big(\bigcap_i A_i\Big)^c = \bigcup A_i^c.$$

Show that if $A, B \in \mathscr{F}$, $A \cap B$ and $A \setminus B$ are in $\mathscr{F}$ (just use the rules of σ-fields).

1.3.3 Probability

We now introduce a way to assign a likelihood to an event, via probability. One possible interpretation is the following: suppose my experiment is repeated many times, then the probability of any event is the limit of the ratio of times the event occurs over the number of events. We note that this is not the only interpretation of probability, but we do not diverge into a discussion of the philosophy of probability. Formally, we introduce the notion of a probability measure:

Definition 1.3.6. *A probability measure $\mathbb{P}$ on $(\Omega, \mathscr{F})$ is a function $\mathbb{P} : \mathscr{F} \to [0, 1]$ which satisfies:*

1. $\mathbb{P}(\Omega) = 1$ *and* $\mathbb{P}(\emptyset) = 0$
2. *For* $A_1, A_2, \ldots$ *disjoint* ($\forall i \neq j$, $A_i \cap A_j = \emptyset$) *members of* $\mathscr{F}$ *then*

$$\mathbb{P}\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

The triple $(\Omega, \mathscr{F}, \mathbb{P})$, comprising a set Ω, a collection of events $\mathscr{F}$ of subsets of Ω and a probability measure $\mathbb{P}$, is called a probability space (or probability triple).

The idea is to associate the probability space with an experiment:

Example 1.3.7. *Consider flipping a coin as in Example 1.3.4, with $\Omega = \{H, T\}$ and $\mathscr{F} = \{\Omega, \emptyset, \{H\}, \{T\}\}$. Then we can set:*

$$\mathbb{P}(\{H\}) = p \quad \mathbb{P}(\{T\}) = 1 - p \quad p \in [0, 1].$$

If $p = 1/2$ we say that coin is fair.

Example 1.3.8. *Consider rolling a 6-sided die, then $\Omega = \{1, \ldots, 6\}$ and $\mathscr{F} = \{0, 1\}^{\Omega}$ (the power set of Ω the set of all subsets of Ω). Define a probability measure on $(\Omega, \mathscr{F})$ as:*

$$\mathbb{P}(A) = \sum_{i \in A} p_i \quad \forall A \in \mathscr{F}$$

where $\forall i \in \{1, \ldots, 6\}$, $0 \leq p_i \leq 1$ and $\sum_{i=1}^{6} p_i = 1$. If $p_i = 1/6$, $\forall i \in \{1, \ldots, 6\}$ then:

$$\mathbb{P}(A) = \frac{Card(A)}{6}$$

where $Card(A)$ is the cardinality of A (the number of elements in the set A).

We now consider a sequence of results on probability spaces.

Lemma 1.3.9. *We have the following properties on probability space $(\Omega, \mathscr{F}, \mathbb{P})$:*

1. *For any* $A \in \mathscr{F}$, $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
2. *For any* $A, B \in \mathscr{F}$, *if* $A \subseteq B$ *then,* $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$.
3. *For any* $A, B \in \mathscr{F}$, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
4. *(inclusion-exclusion formula) For any* $A_1, \ldots, A_n \in \mathscr{F}$ *then:*

$$\mathbb{P}\Big(\bigcup_{i=1}^n A_i\Big) = \sum_i \mathbb{P}(A_i) - \sum_{i<j} \mathbb{P}(A_i \cap A_j) + \sum_{i<j<k} \mathbb{P}(A_i \cap A_j \cap A_k) - \cdots + (-1)^{n+1}\mathbb{P}\Big(\bigcap_{i=1}^n A_i\Big).$$

where, for example, $\sum_{i<j} = \sum_{j=1}^n \sum_{i=1}^{j-1}$ *etc.*

Proof. For (1), $A \cup A^c = \Omega$, so $1 = \mathbb{P}(\Omega) = \mathbb{P}(A \cup A^c) = \mathbb{P}(A) + \mathbb{P}(A^c)$ $(A \cap A^c = \emptyset)$. Thus, one concludes by rearranging.
For (2), as $B = A \cup (B \setminus A)$ and $A \cap (B \setminus A) = \emptyset$, (recall Exercise 1.3.5) $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A)$. The inequality is clear from the definition of a probability.
For (3), $A \cup B = A \cup (B \setminus A)$, which is a disjoint union, and $B \setminus A = (B \setminus (A \cap B))$ hence

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) = \mathbb{P}(A) + \mathbb{P}(B \setminus (A \cap B)).$$

Now $(A \cap B) \subseteq B$ so by (2)

$$\mathbb{P}(B \setminus (A \cap B)) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

and thus

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

For (4), this can be proved by induction and is an exercise. □

To conclude the Section, we introduce two terms:

- An event $A \in \mathscr{F}$ is *null* if $\mathbb{P}(A) = 0$.
- An event $A \in \mathscr{F}$ occurs *almost surely* if $\mathbb{P}(A) = 1$.

Null events are not impossible. We will see this when considering *random variables* which take values in the real line.

1.4 Conditional Probability

We now move onto the notion of conditional probability. It takes the concept of a probability of an event, given another event is known to have occurred. For example, what is the probability that it rains today, given that it rained yesterday? We formalize the notion of conditional probability:

Definition 1.4.1. *Consider probability space* $(\Omega, \mathscr{F}, \mathbb{P})$ *and let* $A, B \in \mathscr{F}$ *with* $\mathbb{P}(B) > 0$. *Then the* ***conditional probability*** *that* A *occurs given* B *occurs is defined to be:*

$$\mathbb{P}(A|B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Example 1.4.2. *A family has two children of different ages. What is the probability that both children are boys, given that at least one is a boy? The older and younger child may be either boys or girl so the sample space is:*

$$\Omega = \{GG, BB, GB, BG\}$$

and we assume that all outcomes are equally likely $\mathbb{P}(GG) = \mathbb{P}(BB) = \mathbb{P}(GB) = \mathbb{P}(BG) = 1/4$ *(the uniform distribution). We are interested in:*

$$\begin{aligned}
\mathbb{P}(BB|GB \cup BB \cup BG) &= \frac{\mathbb{P}(BB \cap (GB \cup BB \cup BG))}{\mathbb{P}(GB \cup BB \cup BG)} \\
&= \frac{\mathbb{P}(BB)}{\mathbb{P}(GB \cup BB \cup BG)} \\
&= \frac{1/4}{3/4} = \frac{1}{3}.
\end{aligned}$$

One can also ask the question: what is the probability that in a family of two children, where the younger of two is a boy, that both are boys? We want:

$$\begin{aligned}\mathbb{P}(BB|BG \cup BB) &= \frac{\mathbb{P}(BB \cap (BG \cup BB))}{\mathbb{P}(BG \cup BB)} \\ &= \frac{\mathbb{P}(BB)}{\mathbb{P}(BG \cup BB)} \\ &= \frac{1/4}{1/2} = \frac{1}{2}.\end{aligned}$$

1.4.1 Bayes Theorem

An important result in conditional probability is *Bayes theorem.*

Theorem 1.4.3. *Consider probability space* $(\Omega, \mathscr{F}, \mathbb{P})$ *and let* $A, B \in \mathscr{F}$ *with* $\mathbb{P}(A), \mathbb{P}(B) > 0$. *Then we have:*

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(A|B)\mathbb{P}(B)}{\mathbb{P}(A)}.$$

Proof. By definition

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}.$$

As $\mathbb{P}(B) > 0$, $\mathbb{P}(A \cap B) = \mathbb{P}(A|B)\mathbb{P}(B)$ and hence:

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(A|B)\mathbb{P}(B)}{\mathbb{P}(A)}.$$

□

1.4.2 Theorem of Total Probability

We will use Bayes Theorem after introducing the following result (the *Theorem of Total Probability.* We begin with a preliminary definition:

Definition 1.4.4. *A family of sets* $B_1, \ldots, B_n$ *is called a* partition *of* Ω *if:*

$$\forall i \neq j \;\; B_i \cap B_j = \emptyset \quad \textit{and} \quad \bigcup_{i=1}^{n} B_i = \Omega.$$

Lemma 1.4.5. *Consider probability space* $(\Omega, \mathscr{F}, \mathbb{P})$ *and let for any fixed* $n \geq 2$, $B_1, \ldots, B_n \in \mathscr{F}$ *be a partition of* Ω, *with* $\mathbb{P}(B_i) > 0, \forall i \in \{1, \ldots, n\}$. *Then, for any* $A \in \mathscr{F}$:

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A|B_i)\mathbb{P}(B_i).$$

Proof. We give the proof for $n = 2$, the other cases being very similar. We have $A = (A \cap B_1) \cup (A \cap B_2)$ (recall $B_1 \cup B_2 = \Omega$, and $B_2 = B_1^c$), so

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A \cap B_1) + \mathbb{P}(A \cap B_2) \\ &= \mathbb{P}(A|B_1)\mathbb{P}(B_1) + \mathbb{P}(A|B_2)\mathbb{P}(B_2).\end{aligned}$$

□

Example 1.4.6. *Consider two factories which manufacture a product. If the product comes from factory I, it is defective with probability 1/5 and if it is from factory II, it is defective with probability 1/20. It is twice as likely that a product comes from factory I. What is the probability that a given product operates properly (i.e. is not defective)? Let A denote the event of 'operates properly' and B denote the event 'the product is made in factory I'. Then:*

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^c)\mathbb{P}(B^c) \\ &= \frac{4}{5}\frac{2}{3} + \frac{19}{20}\frac{1}{3} = \frac{51}{60}.\end{aligned}$$

If a product is defective, what is the probability that it came from factory I? This is $\mathbb{P}(B|A^c)$*; by Bayes Theorem*

$$\begin{aligned}\mathbb{P}(B|A^c) &= \frac{\mathbb{P}(A^c|B)\mathbb{P}(B)}{\mathbb{P}(A^c)} \\ &= \frac{(1/5)(2/3)}{9/60} = \frac{8}{9}.\end{aligned}$$

Example 1.4.7. *The Monty Hall Game Show. In a game show, a contestent selects one of three doors; behind one of the doors, there is a prize, and behind the other two there are no prizes. After the contestant selects a door, the game show host opens one of the remaining doors, and reveals that there is no prize behind it. The host then asks the contestant whether they want to SWITCH their choice to the unopened door, or STICK to their original choice.*

Is it probabilistically advantageous for the contestant to SWITCH doors, or is the probability of winning the same whether they STICK or SWITCH? We will assume that the host selects a door to open from those available*, with equal probability.*

Without loss of generality, let events A, B, C *correspond to the prize being behind the selected, opened and remaining door respectively and let* H_B *denote the event that the host opens door* B*. We want to compare* $\mathbb{P}(A|H_B)$ *(STICK) with* $\mathbb{P}(C|H_B)$ *(SWITCH). Now* $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = 1/3$ *and we are told that* $\mathbb{P}(H_B|A) = 1/2, \mathbb{P}(H_B|B) = 0, \mathbb{P}(H_B|C) = 1$*. Then via Bayes theorem:*

$$\begin{aligned}\mathbb{P}(A|H_B) &= \frac{\mathbb{P}(H_B|A)\mathbb{P}(A)}{\mathbb{P}(H_B)} = \frac{\mathbb{P}(H_B|A)\mathbb{P}(A)}{\mathbb{P}(H_B|A)\mathbb{P}(A) + \mathbb{P}(H_B|B)\mathbb{P}(B) + \mathbb{P}(H_B|C)\mathbb{P}(C)} \\ &= \frac{1/2 \times 1/3}{1/2 \times 1/3 + 0 \times 1/3 + 1 \times 1/3} = 1/3.\end{aligned}$$

Similarly, $\mathbb{P}(C|H_B) = 2/3$*. Thus one should SWITCH.*

Example 1.4.8. *Patients are recruited onto the two arms (0 - Control, 1-Treatment) of a clinical trial. The probability that an adverse outcome occurs on the control arm is* p_0*, and is* p_1 *for the treatment arm. Patients are allocated alternately onto the two arms in the sequence* $010101\ldots$*, and their outcomes are independent. What is the probability that the first adverse event occurs on the control arm?*

Define

- *sample space* Ω *to be all possible infinite sequences of patients outcomes*
- *event* E_1 *- first patient (allocated onto the control arm) suffers an adverse outcome*
- *event* E_2 *- first patient (allocated onto the control arm) does not suffer an adverse outcome, but the second patient (allocated onto the treatment arm) does suffer an adverse outcome*
- *event* E_0 *- neither of the first two patients suffer adverse outcomes*
- *event* F *- first adverse event occurs on the control arm*

We want $\mathbb{P}(F)$*. Now the events* E_1*,* E_2 *and* E_0 *partition* Ω*, so, by the Theorem of total probability,*

$$\mathbb{P}(F) = \mathbb{P}(F|E_1)\mathbb{P}(E_1) + \mathbb{P}(F|E_2)\mathbb{P}(E_2) + \mathbb{P}(F|E_0)\mathbb{P}(E_0).$$

Now

$$\mathbb{P}(E_1) = p_0 \quad \mathbb{P}(E_2) = (1-p_0)p_1 \quad \mathbb{P}(E_0) = (1-p_0)(1-p_1)$$

and $\mathbb{P}(F|E_1) = 1, \mathbb{P}(F|E_2) = 0$*. Finally, as after two non-adverse outcomes, the allocation process effectively re-starts, so* $\mathbb{P}(F|E_0) = \mathbb{P}(F)$*. Hence*

$$\mathbb{P}(F) = (1 \times p_0) + (0 \times (1-p_0)p_1) + (\mathbb{P}(F)(1-p_0)(1-p_1)) = p + 0 + (1-p_0)(1-p_1)\mathbb{P}(F)$$

which can be re-arranged to give

$$\mathbb{P}(F) = \frac{p_0}{p_0 + p_1 - p_0 p_1}.$$

Example 1.4.9. *While watching a game of English Premier League football in a shop, you observe someone who is clearly supporting Manchester United in the game. What is the probability that they were actually born within 25 miles of Manchester? Assume that the probability that a randomly selected person in a typical local bar environment is born within 25 miles of Manchester is* $1/20$ *, and that the chance that a person born within 25 miles of Manchester actually supports United is* $7/10$*. Assume also that the probability that a person not born within 25 miles of Manchester supports United with probability* $1/10$*.*

Define

- B *- event that the person is born within 25 miles of Manchester*
- U *- event that the person supports United.*

We want $\mathbb{P}(B|U)$*. By Bayes Theorem,*

$$\begin{aligned}\mathbb{P}(B|U) &= \frac{\mathbb{P}(U|B)\mathbb{P}(B)}{\mathbb{P}(U)} = \frac{\mathbb{P}(U|B)\mathbb{P}(B)}{\mathbb{P}(U|B)\mathbb{P}(B)+\mathbb{P}(U|B^c)\mathbb{P}(B^c)} \\ &= \frac{\frac{7}{10}\frac{1}{20}}{\frac{7}{10}\frac{1}{20}+\frac{1}{10}\frac{19}{20}} \\ &= \frac{7}{26}.\end{aligned}$$

Example 1.4.10. *It is believed that a sought after treasure will be in a certain sea area with probability 2/5. A search in that area will detect the treasure with probability 9/10 if it is there (you may assume that the probability of not detecting when the treasure is not there is 1). What is the probability of the treasure being in the area given that the area is searched and no treasure is found?*

Define

- A *the event that the treasure is in the area*
- D *the event that the treasure is detected.*

We want $\mathbb{P}(A|D^c)$*. By Bayes theorem*

$$\begin{aligned}\mathbb{P}(A|D^c) &= \frac{\mathbb{P}(D^c|A)\mathbb{P}(A)}{\mathbb{P}(D^c)} = \frac{\mathbb{P}(D^c|A)\mathbb{P}(A)}{\mathbb{P}(D^c|A)\mathbb{P}(A)+\mathbb{P}(D^c|A^c)\mathbb{P}(A^c)} \\ &= \frac{\frac{1}{10}\frac{4}{10}}{\frac{1}{10}\frac{4}{10}+1\times\frac{6}{10}} \\ &= \frac{1}{16}.\end{aligned}$$

Reading 1.4.11. *We will now consider Simpson's paradox. This is covered in [1, pp. 76-79]. Please open the relevant pages in the book.*

1.5 Independence

The idea of independence is roughly that the probability of an event A is 'unaffected' by the occurrance of a (non-null) event B; that is, $\mathbb{P}(A) = \mathbb{P}(A|B)$, when $\mathbb{P}(B) > 0$. More formally:

Definition 1.5.1. *Consider probability space* $(\Omega, \mathscr{F}, \mathbb{P})$ *and let* $A, B \in \mathscr{F}$*.* A *and* B *are* ***independent*** *if*

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

More generally, a family of $\mathscr{F}-$*sets* $A_1, \ldots, A_n$ $(\infty > n \geq 2)$ *are independent if*

$$\mathbb{P}\Big(\bigcap_{i=1}^{n} A_i\Big) = \prod_{i=1}^{n} \mathbb{P}(A_i).$$

Two important concepts include *pairwise independence*:

$$\mathbb{P}(A_i \cap A_j) = \mathbb{P}(A_i)\mathbb{P}(A_j) \quad \forall i \neq j.$$

This does not necessarily mean that $A_1, \ldots, A_n$ are independent events. Another important concept is *conditional independence*; given an event $C \in \mathscr{F}$, with $\mathbb{P}(C) > 0$, $A, B \in \mathscr{F}$ are conditionally independent events if

$$\mathbb{P}(A \cap B|C) = \mathbb{P}(A|C)\mathbb{P}(B|C).$$

This can be extended to families of sets A_i given C.

Example 1.5.2. *Consider a pack of playing cards, and one chooses a card completely at random (i.e. no card counting etc). One can assume that the probability of choosing a suit (e.g. spade) is independent of its rank. So, for example:*

$$\mathbb{P}(\textit{spade king}) = \mathbb{P}(\textit{spade})\mathbb{P}(\textit{king}) = \frac{13}{52}\frac{4}{52} = \frac{1}{52}.$$

1.5.1 Gambler's Ruin

Consider a gambler that starts with a fortune i and in a gambling transaction they will either gain 1 unit with a probability p or lose a unit with probability $1-p$ and each transaction is independent of all the others. The gambler stops when they reach N units (wins) or when they have zero units (ruin).

The total fortune of the gambler depends only on what they currently have and whether they win or lose the gambling transaction. So if we denote by P_i the probability that they win, starting at a fortune of i, this is equal to, for $i \in \{1, \ldots, N-1\}$

$$P_{i+1}p + P_{i-1}(1-p)$$

where $P_0 = 0$, $P_N = 1$. This simply says that the probability that they win, starting at a fortune of i is equal to the probability they win one gambling transaction and then win starting at $i+1$ added to the probability that the lose the one gambling transaction and then win starting at $i-1$. This can be justified by the theorem of total probability. Set A_i the event that the gambler wins starting at i and B the probability that the gambler wins a transaction:

$$\begin{aligned}
\mathbb{P}(A_i) &= \mathbb{P}(A_i|B)\mathbb{P}(B) + \mathbb{P}(A_i|B^c)\mathbb{P}(B^c) \\
&= \mathbb{P}(A_i|B)p + \mathbb{P}(A_i|B^c)(1-p) \\
&= P_{i+1}p + P_{i-1}(1-p).
\end{aligned}$$

The first equality follows by the theorem of total probability and the second by using the definition of winning and losing the gambling transaction. Now consider $\mathbb{P}(A_i|B)$ which is the that the gambler wins starting at i given that they won the first gambling transaction. This, by our construction, is the same as if the gambler started at a fortune of $i+1$. A similar statement follows for $\mathbb{P}(A_i|B^c)$. Simple algebra yields

$$P_{i+1} - P_i = \frac{1-p}{p}\left(P_i - P_{i-1}\right).$$

Then consider $i = 1$ then $P_2 - P_1 = (1-p)/pP_1$ and then $i = 2$, $P_3 - P_2 = (1-p)/p(P_2 - P_1) = (1-p)^2/p^2P_1$; hence iterating gives for $i \in \{1, \ldots, N-1\}$

$$P_{i+1} - P_i = \left(\frac{1-p}{p}\right)^i P_1.$$

Now, we have the telscoping sum representation:

$$\begin{aligned}
P_{i+1} - P_1 &= \sum_{k=1}^{i}\{P_{k+1} - P_k\} \\
&= P_1\sum_{k=1}^{i}\left(\frac{1-p}{p}\right)^k.
\end{aligned}$$

Therefore

$$P_{i+1} = P_1\sum_{k=0}^{i}\left(\frac{1-p}{p}\right)^k$$

and using the gometric sum formula:

$$P_{i+1} = P_1 \begin{cases} (i+1) & \text{if } p = 1-p \\ \frac{1-\left(\frac{1-p}{p}\right)^{i+1}}{1-\left(\frac{1-p}{p}\right)} & \text{o/w} \end{cases}$$

Then setting $i = N-1$ and using $P_N = 1$ we have after re-arranging for P_1, for $i \in \{0, \ldots, N\}$

$$P_i = \begin{cases} \frac{i}{N} & \text{if } p = 1-p \\ \frac{1-\left(\frac{1-p}{p}\right)^{i}}{1-\left(\frac{1-p}{p}\right)^{N}} & \text{o/w.} \end{cases}$$

Chapter 2

Discrete Random Variables

2.1 Introduction

This Chapter introduces the notion of a random variable (Section 2.2). This is essentially just a function on the sample space that produces a number and, in particular, we focus upon the case where the collection of numbers associated to the random variable form a countable set; the scenario of discrete random variables. We then develop the theory of discrete random variables in the guise of probability mass functions (Section 2.3) which provide a charaterization of the probability (measure). The notion of independence (Section 2.4) is revisted and some specific probability mass functions are introduced (Section 2.5). We conclude the Chapter with the notion of Expectation (Section 2.6) which is a summary of a random variable.

2.2 Random Variables and Distribution Functions

2.2.1 Random Variables

In general, we are often concerned by an experiment itself, but by an associated consequence of the outcome. For example, a gambler is often interested in his or hers profit or loss, rather that in the result that occurs. To deal with this issue, we introduce the notion of a random variable:

Definition 2.2.1. *A **random variable** is a function $X : \Omega \to \mathbb{R}$.*

One can think of $X(\omega)$ as a mapping from the sample space to the real line and continue onwards. In general we omit the argument ω and just write X, with possible realized values (numbers) written in lower-case x. The distinction between a random variable X and its realized value is very important and you should maintain this convention. To move from the random variable, back to our probability measure $\mathbb{P}$, we write events $\{\omega \in \Omega : X(\omega) \leq x\}$ as $\{X \leq x\}$, and hence we write $\mathbb{P}(\{X \leq x\})$ as $\mathbb{P}(X \leq x)$ to mean $\mathbb{P}(\{\omega \in \Omega : X(\omega) \leq x\})$ (recall $\{\omega \in \Omega : X(\omega) \leq x\} \in \mathscr{F}$ and so this makes sense). This leads us to the notion of a distribution function:

2.2.2 Distribution Functions

Definition 2.2.2. *The **distribution function** of a random variable X is the function $F : \mathbb{R} \to [0, 1]$ given by $F(x) = \mathbb{P}(X \leq x)$, $x \in \mathbb{R}$.*

Example 2.2.3. *Consider flipping a fair coin twice; then $\Omega = \{HH, TT, HT, TH\}$, with $\mathscr{F} = \{0, 1\}^{\Omega}$ and $\mathbb{P}(HH) = \mathbb{P}(TT) = \mathbb{P}(HT) = \mathbb{P}(TH) = 1/4$. Define a random variable X as the number of heads; so*

$$X(HH) = 2 \quad X(HT) = X(TH) = 1 \quad X(TT) = 0.$$

The associated distribution function is:

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{4} & \text{if } 0 \leq x < 1 \\ \frac{3}{4} & \text{if } 1 \leq x < 2 \\ 1 & \text{if } x \geq 2 \end{cases}$$

A distribution function has the following properties, which we state without proof. See [1, Theorem 3.6.3] for the proof. The lemma *characterizes* a distribution function: a function F is a distribution function for some random variable if and only if it satisfies the three properties in the following lemma.

Lemma 2.2.4. *A distribution function F has the following properties:*

1. $\lim_{x\to-\infty} F(x) = 0$, $\lim_{x\to\infty} F(x) = 1$,
2. *If $x < y$ then $F(x) \leq F(y)$,*
3. *F is right continuous; $\lim_{\delta\downarrow 0} F(x+\delta) = F(x)$ for each $x \in \mathbb{R}$.*

Example 2.2.5. *Consider flipping a coin as in Example 1.3.4, which lands heads with probability $p \in (0,1)$. Let $X : \Omega \to \mathbb{R}$ be given by*

$$X(H) = 1 \quad X(T) = 0.$$

The associated distribution function is

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1-p & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

X is said to have a Bernoulli distribution.

We end the Section with another lemma, which we do not prove.

Lemma 2.2.6. *A distribution function F of random variable X, has the following properties:*

1. *$\mathbb{P}(X > x) = 1 - F(x)$ for any fixed $x \in \mathbb{R}$,*
2. *$\mathbb{P}(x < X \leq y) = F(y) - F(x)$ for any fixed $x, y \in \mathbb{R}$ with $x < y$,*
3. *$\mathbb{P}(X = x) = F(x) - \lim_{y\uparrow x} F(y)$ for any fixed $x \in \mathbb{R}$.*

2.3 Probability Mass Functions

We now move onto an important class of random variable called discrete random variables.

Definition 2.3.1. *A random variable is said to be discrete if it takes values in some countable subset $\mathsf{X} = \{x_1, x_2, \dots\}$ of $\mathbb{R}$.*

A discrete random variable takes values only at countable values and hence its distribution is a jump function, in that it shifts between different values when they change. An important concept is the probability mass function (PMF):

Definition 2.3.2. *The **probability mass function** of a discrete random variable X, is the function $f : \mathsf{X} \to [0,1]$ defined by $f(x) = \mathbb{P}(X = x)$.*

Remark 2.3.3. ***We generally use sans-serif notation X, Z to denote the supports of a random variable. That is the range of values for which its PMF (or PDF - as will be defined much later on) is (potentially) non-zero. Note however, the PMF may be defined on say $\mathbb{Z}$ or $\mathbb{R}$ but is taken as (potentially) non-zero only at those points in X - it is always zero outside X. We call this the* support *of a random variable.***

We remark that

$$F(x) = \sum_{y\leq x} f(y) \quad f(x) = F(x) - \lim_{y\uparrow x} F(y)$$

which provides and association between the distribution function and the PMF. We have the following result, whose proof follows easily from the above definitions and results.

Lemma 2.3.4. *A PMF satisfies:*

1. *the set of x such that $f(x) \neq 0$ is countable,*

2. $\sum_{x \in \mathsf{X}} f(x) = 1$.

Example 2.3.5. *A coin is flipped n times and the probability one obtains a head is $p \in (0,1)$; $\Omega = \{H, T\}^n$. Let X denote the number of heads, which takes values in the set $\mathsf{X} = \{0, 1, \dots, n\}$ and is thus a discrete random variable. Consider $x \in \mathsf{X}$, exactly $\binom{n}{x}$ points in Ω give us x heads and each point occurs with probability $p^x(1-p)^{n-x}$; hence*

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x} \quad x \in \mathsf{X}.$$

*The random variable X is said to have a **Binomial distribution** and this is denoted $X \sim \mathcal{B}(n, p)$. Note that*

$$\binom{n}{x} = \frac{n!}{(n-x)!x!}$$

with $x! = x \times (x-1) \times \cdots \times 1$ and $0! = 1$.

Example 2.3.6. *Let $\lambda > 0$ be given. A random variable X that takes values on $\mathsf{X} = \{0, 1, 2, \dots, \}$ is said have a **Poisson distribution** with parameter λ (denoted $X \sim \mathcal{P}(\lambda)$) if its PMF is:*

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!} \quad x \in \mathsf{X}.$$

Reading 2.3.7. *Additional reading to enhance your understanding can be found in [1, Section 3.2, pp. 106-111].*

2.4 Independence

We now extend the notion of independence of events to the domain of random variables. Recall that events A and B are independent if the joint probability is equal to the product (A does not affect B).

Definition 2.4.1. *Discrete random variables X and Y are independent if the events $\{X = x\}$ and $\{Y = y\}$ are independent for each $(x, y) \in \mathsf{X} \times \mathsf{Y}$.*

To understand this idea, let $\mathsf{X} = \{x_1, x_2, \dots\}$, $\mathsf{Y} = \{y_1, y_2, \dots\}$, then X and Y are independent if and only if the events $A_i = \{X = x_i\}$, $B_j = \{Y = y_j\}$ are independent for every possible pair i, j.

Example 2.4.2. *Consider flipping a coin once, which lands tail with probability $p \in (0,1)$. Let X be the number of heads seen and Y the number of tails. Then:*

$$\mathbb{P}(X = Y = 1) = 0$$

and

$$\mathbb{P}(X = 1)\mathbb{P}(Y = 1) = (1-p)p \neq 0$$

so X and Y cannot be independent.

A useful result (which we do not prove) that is worth noting is the following.

Theorem 2.4.3. *If X and Y are independent random variables and $g : \mathsf{X} \to \mathbb{R}$, $h : \mathsf{Y} \to \mathbb{R}$, then the random variables $g(X)$ and $h(Y)$ are also independent.*

In full generality[1], consider a sequence of discrete random variables $X_1, X_2, \dots, X_n$, $X_i \in \mathsf{X}_i$; they are said to be independent if the events $\{X_1 = x_1\}, \dots, \{X_n = x_n\}$ are independent for every possible $(x_1, \dots, x_n) \in \mathsf{X}_1 \times \cdots \times \mathsf{X}_n$. That is:

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^{n} \mathbb{P}(X_i = x_i) \quad \forall (x_1, \dots, x_n) \in \mathsf{X}_1 \times \cdots \times \mathsf{X}_n.$$

[1] this point will not make too much sense, until we see Section 4.2

2.5 Some Probability Mass Functions

2.5.1 Discrete Uniform Distribution

Let X be a finite set of numbers and denote the cardinality as $\text{Card}(\mathsf{X})$. Then the discrete uniform distribution has a PMF:

$$f(x) = \frac{1}{\text{Card}(\mathsf{X})} \quad x \in \mathsf{X}.$$

That is, every element of X has equal probability of occuring. We use the notation $X \sim \mathcal{U}_{\mathsf{X}}$ to denote that X has a uniform distribution over the set X. The notion of a uniform distribution can be extended to any compact set X for instance if the support of the random variable X is a continuum (as will be the case in the next Chapter) and so, this notation will be maintained in that context also.

Let $\mathsf{A} \subseteq \mathsf{X}$, then one can easily deduce that:

$$\mathbb{P}(X \in \mathsf{A}) = \sum_{x \in \mathsf{A}} f(x) = \frac{\sum_{x \in \mathsf{A}}}{\text{Card}(\mathsf{X})} = \frac{\text{Card}(\mathsf{A})}{\text{Card}(\mathsf{X})}.$$

In words, this simply means that the chance that A occurs is the number of elements in A divided by the total number of elements in X.

2.5.2 Binomial Distribution

Recall the *binomial coefficient*: the number of ways, disregarding order, that x objects can be chosen from amongst n objects:

$$\binom{n}{x} = \frac{n!}{(n-x)!x!}.$$

Let $n \geq 1$ be fixed and finite integer and let $\mathsf{X} = \{0, 1, \ldots, n\}$. Then a random variable X has a binomial distribution, if its PMF is:

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x} \quad x \in \mathsf{X},\ p \in (0,1).$$

Such a PMF is suitable for a random variable that counts the 'successes' in a collection of n independent and identical *Bernoulli trials* (an experiment where there are only two possible outcomes), where the probability of success is $p \in (0,1)$.

An example of which, can be found in the notes; flipping a coin n times which lands heads with probability p and counting the number of heads which lands. In this example, the Bernoulli trial is that of flipping the coin; there are only two possible outcomes, head or tail. The success is a 'heads' landing and one would expect that these consecutive Bernoulli trials are independent.

We remark that $f(x)$ is indeed a valid PMF. First

$$f(x) \geq 0 \quad \forall x \in \mathsf{X}.$$

This is because $1 > p > 0$ and so for any $x \in \mathsf{X}$, $p^x(1-p)^{n-x} > 0$. The binomial coefficient is always positive in our scenario. Second,

$$\sum_{x=0}^{n} f(x) = \sum_{x=0}^{n} \binom{n}{x} p^x (1-p)^{n-x} = 1.$$

Here we use the fact that for any fixed $a, b \in (-\infty, \infty)$ and poitive integer n,

$$(a+b)^n = \sum_{x=0}^{n} \binom{n}{x} a^x b^{n-x}.$$

In our setting $a = p$ and $b = 1-p$ and for any $n \geq 1$, $(1)^n = 1$. Thus, our PMF is such that $1 \geq f(x) \geq 0$ for each $x \in \mathsf{X}$.

The distribution function is for $x \in \mathsf{X}$

$$\mathbb{P}(X \leq x) = F(x) = \sum_{y=0}^{x} \binom{n}{y} p^y (1-p)^{n-y}.$$

This latter summation does not always have an analytic expression. One can use the distribution function to answer various questions of interest. For example; if one flips a coin 10 times and the probability of heads is p, what is the probability of:

- 1 head
- at least 9 heads

For the first question we have

$$\mathbb{P}(X = 1) = f(1) = F(1) - F(0) = \binom{10}{1} p(1-p)^9.$$

For the second question, this is

$$\mathbb{P}(X \geq 9) = f(9) + f(10) = F(10) - F(8) = \binom{10}{9} p^9(1-p) + \binom{10}{10} p^{10}.$$

2.5.3 Hypergeometric Distribution

To start this discussion, we begin with Vandermonde's indentity:

$$\binom{m+n}{k} = \sum_{j=0}^{k} \binom{m}{j}\binom{n}{k-j} \tag{2.5.1}$$

with $(m, n, k) \in \mathbb{N}$. See for instance [1, Example 1.5.3] for a justification.

We now consider a box containing r red balls and g green balls, where r, g are strictly positive. The balls are identical except for the color. We will take out $1 \leq n \leq r + g$ balls out of the box *without* replacement and without looking; so that the chance of picking any ball out of the box is equally likely amongst those in the box. Let us consider the probability of obtaining k red balls and hence $n - k$ green balls where $0 \leq k \leq n$. Based upon our understanding of counting events the number of ways to obtain k red balls and $n - k$ green balls from our n balls taken out of the box is:

$$\binom{r}{k}\binom{g}{n-k}.$$

The total number of ways to draw n balls from $r + g$ total balls is simply

$$\binom{r+g}{n}.$$

Hence, denoting by X the random variable associated to the number of red balls drawn out of n, the probability of drawing $X = k$ red balls from the total of n drawn is:

$$\mathbb{P}(X = k) = \frac{\binom{r}{k}\binom{g}{n-k}}{\binom{r+g}{n}}.$$

By (2.5.1):

$$\sum_{k=0}^{n} \binom{r}{k}\binom{g}{n-k} = \binom{r+g}{n}$$

so this is a well-defined probability. Note that if k exceeds r we use the convention that $\binom{r}{k} = 0$, so this event has zero probability.

To conclude let $(r, g) \in \mathbb{N}^2$ and suppose that $n \in \{1, \ldots, r + g\}$ is fixed. We say that a random variable $X \in \mathsf{X} = \{0, \ldots, n\}$ is said to have a hypergeometric distribution, denoted $X \sim \mathcal{H}y(r, g, n)$ if the PMF is

$$f(x) = \frac{\binom{r}{x}\binom{g}{n-x}}{\binom{r+g}{n}} \quad x \in \mathsf{X}.$$

Note that the important distinction with the binomial distribution is in the former, one considers a sequence of independent Bernoulli trials. In the hypergeometric distribution, indeed one can consider the drawing of a red ball a success and a green ball a failure (a Bernoulli trial), but critically as there is no replacement, there is a dependence on what has happened in the past. The latter occurs because we do not replace the balls in the box.

2.5.4 Poisson Distribution

Let X be the set of non-negative integers: $\mathsf{X} = \{0, 1, \dots\}$. Then a random variable $X \in \mathsf{X}$ is said to have a Poisson distribution of parameter $\infty > \lambda > 0$ if the PMF is

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!} \quad x \in \mathsf{X}.$$

In general, the Poisson distribution is often used for random variables associated to counts (for example, counts which can grow indefinitely). This is a PMF as $f(x) \geq 0 \ \forall x \in \mathsf{X}$: $\lambda^x/x! > 0$ and $e^{-\lambda} > 0$. In addition

$$\begin{aligned} \sum_{x \in \mathsf{X}} f(x) &= \sum_{x=0}^{\infty} \frac{\lambda^x e^{-\lambda}}{x!} \\ &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} \\ &= e^{-\lambda} e^{\lambda} \\ &= 1. \end{aligned}$$

Here we have used the Taylor series for the exponential function.

The distribution function is for $x \in \mathsf{X}$

$$\mathbb{P}(X \leq x) = F(x) = \sum_{y=0}^{x} \frac{\lambda^y e^{-\lambda}}{y!}$$

This latter summation does not always have an analytic expression. One can use the distribution function or the PMF to answer questions of interest:

- what is the probability that X is 1?
- what is the probability that X is positive?

For the first question:

$$\mathbb{P}(X = 1) = f(1) = F(1) - F(0) = \lambda e^{-\lambda}.$$

For the second question:

$$\mathbb{P}(X \geq 1) = 1 - \mathbb{P}(X = 0) = 1 - f(0) = 1 - e^{-\lambda}.$$

Proposition 2.5.1. *Suppose that $X \sim \mathcal{B}(n, p)$ and denote*

$$\mathbb{P}_n(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Suppose that as $n \to \infty$, $p \to 0$, $\lambda = np$ is fixed. Then we have that for any fixed $k \in \{0, 1 \dots, n\}$

$$\lim_{n \to \infty, p \to 0} \mathbb{P}_n(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}.$$

Proof. We have that

$$\begin{aligned} \mathbb{P}_n(X = k) &= \binom{n}{k} p^k (1-p)^{n-k} \\ &= \frac{n(n-1) \times \cdots \times (n-k+1)}{k!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\ &= \frac{\lambda^k}{k!} \frac{n(n-1) \times \cdots \times (n-k+1)}{n^k} \left(1 - \frac{\lambda}{n}\right)^n (1-p)^{-k}. \end{aligned}$$

Then clearly (the second limit can be obtained by using the Binomial expansion) we have:

$$\begin{aligned} \lim_{n \to \infty, p \to 0} \frac{n(n-1) \times \cdots \times (n-k+1)}{n^k} &= 1 \\ \lim_{n \to \infty, p \to 0} \left(1 - \frac{\lambda}{n}\right)^n &= e^{-\lambda} \\ \lim_{n \to \infty, p \to 0} (1-p)^{-k} = 1. \end{aligned}$$

Therefore we can conclude the result. $\square$

Remark 2.5.2. *The result simply says that for a large number of Bernoulli trials, such that the probability of success is small, but the product np does not go to zero, one can approximate a Binomial probability with a Poisson one.*

2.5.5 Geometric Distribution

Consider a sequence of independent and identical Bernoulli trials with success probability $p \in (0,1)$. Define the random variable X as the number of trials until we see a success, and we include the successful trial (for example, if I flip a coin which shows heads (a success) with probability p, then X is the number of flips to obtain a head, including the successful flip). We know that:

$$X \in \mathsf{X} = \{1, 2, \dots, \}$$

that is, X is a positive integer. Now, what is the probability that X takes the value x? Well, suppose $X = 1$, then we must have:

$$\mathbb{P}(X = 1) = p.$$

This is because, we have only one Bernoulli trial, and it is a success. Suppose, now $X = 2$; then:

$$\mathbb{P}(X = 2) = (1 - p)p.$$

This is because we have two Bernoulli trials, and the first is a failure and the second a success. Similarly

$$\mathbb{P}(X = 3) = (1 - p)^2 p.$$

Thus, it follows that:

$$\mathbb{P}(X = x) = f(x) = (1 - p)^{x-1} p \quad x \in \mathsf{X} = \{1, 2, \dots, \}.$$

Any random variable with the above PMF is said to have a geometric distribution and we write $X \sim \mathcal{G}e(p)$.

The distribution function, for $x \in \mathsf{X}$:

$$F(x) = \sum_{y=1}^{x} (1 - p)^{y-1} p = p \sum_{y=1}^{x} (1 - p)^{y-1} = p \frac{1 - (1 - p)^x}{p} = 1 - (1 - p)^x.$$

Calculating the expectation is somewhat tedious:

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{x=1}^{\infty} x(1 - p)^{x-1} p \\
&= p \sum_{x=1}^{\infty} \frac{d}{dp} \Big[- (1 - p)^x \Big] \\
&= p \frac{d}{dp} \Big[- \sum_{x=1}^{\infty} (1 - p)^x \Big] \\
&= p \frac{d}{dp} \Big[- \frac{(1 - p)}{p} \Big] \\
&= \frac{1}{p}.
\end{aligned}$$

2.5.6 Negative Binomial Distribution

Consider again a sequence of independent and identical Bernoulli trials with success probability $p \in (0,1)$. Define the random variable X as the number of trials until we see $r \geq 1$ successes, and we include the rth successful trial (for example, if I flip a coin which shows heads (a success) with probability p, then X is the number of flips to obtain r heads, including the rth successful flip). We know that

$$X \in \mathsf{X} = \{r, r + 1, \dots\}.$$

That is, to achieve r successes, the minimum number of trials that we can have is r. Then as $p \in (0,1)$, we do not know when the trials will stop. Now what is the probability that X takes the

value x? Well, for $r = 1$ we have already obtained the solution (that is the Geometric distribution). Let us suppose that $r = 2$. Then we have, for $x \in \{2, 3, \dots\}$

$$\mathbb{P}(X = x) = (x-1)(1-p)^{x-2}p^2.$$

The logic is as follows: we have to have two successes and hence $x - 2$ failures, which accounts for the $(1-p)^{x-2}p^2$ part, then we know that the last successful trial is at x, so the first successful trial must lie in one of the first $x - 1$ trials; this is why we multiply by $x - 1$ (remember the trials are identical). Now suppose that $r = 3$, Then we have, for $x \in \{3, 4, \dots\}$

$$\mathbb{P}(X = x) = \binom{x-1}{2}(1-p)^{x-3}p^3.$$

The logic is as follows: we have to have three successes and hence $x - 3$ failures, which accounts for the $(1-p)^{x-3}p^3$ part, then we know that the last successful trial is at x, so the first and second successful trials must lie in one of the first $x-1$ trials; this is why we multiply by $\binom{x-1}{2}$ which is the number of ways of picking two out of $x - 1$ when the order does not matter. Then, following this reasoning, we have for any $r \geq 1$

$$\mathbb{P}(X = x) = f(x) = \binom{x-1}{r-1}(1-p)^{x-r}p^r \quad x \in \mathsf{X} = \{r, r+1, \dots\}.$$

A random variable with the above PMF is said to have a negative binomial distribution with parameters r, p, denoted $X \sim \mathcal{N}e(r, p)$.

2.6 Expectation

Throughout your statistical training, you may have encountered the notion of an average or mean value of data. In this Section we consider the idea of the average value of a random variable, which is called the *expected value*.

Definition 2.6.1. *The expected value of a discrete random variable X on X, with PMF f, denoted $\mathbb{E}[X]$ is defined as*

$$\mathbb{E}[X] := \sum_{x \in \mathsf{X}} x f(x)$$

whenever the sum on the R.H.S. is absolutely convergent.

Example 2.6.2. *Recall the Poisson random variable from Example 2.3.6, $X \sim \mathcal{P}(\lambda)$. The expected value is:*

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{x=0}^{\infty} x \frac{\lambda^x e^{-\lambda}}{x!} \\
&= \lambda e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!} \\
&= \lambda e^{-\lambda} e^{\lambda} \\
&= \lambda
\end{aligned}$$

where we have used the Taylor series expansion for the exponential function on the third line.

We now look at the idea of an expectation of a function of a random variable:

Lemma 2.6.3. *Given a discrete random variable X on X, with PMF f and $g : \mathsf{X} \to \mathbb{R}$:*

$$\mathbb{E}[g(X)] = \sum_{x \in \mathsf{X}} g(x) f(x)$$

whenever the sum on the R.H.S. is absolutely convergent.

Example 2.6.4. *Returning to Example 2.6.2, we have*

$$\begin{aligned}
\mathbb{E}[X^2] &= \sum_{x=0}^{\infty} x^2 \frac{\lambda^x e^{-\lambda}}{x!} \\
&= \lambda e^{-\lambda} \sum_{x=1}^{\infty} x \frac{\lambda^{(x-1)}}{(x-1)!} \\
&= \lambda e^{-\lambda} \sum_{x=1}^{\infty} \frac{d}{d\lambda}(\lambda^x) \frac{1}{(x-1)!} \\
&= \lambda e^{-\lambda} \frac{d}{d\lambda}\Big[\lambda\Big(\sum_{x=1}^{\infty}(\lambda^{x-1})\frac{1}{(x-1)!}\Big)\Big] \\
&= \lambda e^{-\lambda} \frac{d}{d\lambda}(\lambda e^{\lambda}) \\
&= \lambda^2 + \lambda
\end{aligned}$$

where we have assumed that it is legitimate to swap differentiation and summation (it turns out that this is true here).

Example 2.6.5. *Consider the indicator function of a set* A*:* $\mathbb{I}_{\mathsf{A}}(x)$*. This function is 1 if* $x \in \mathsf{A}$ *and zero otherwise. Now suppose we have a random variable* X *with support* X *and that* $\mathsf{A} \subseteq \mathsf{X}$*. Then we have that*

$$\begin{aligned}
\mathbb{E}[\mathbb{I}_A(X)] &= \sum_{x \in \mathsf{X}} \mathbb{I}_{\mathsf{A}}(x) f(x) \\
&= \sum_{x \in \mathsf{A}} f(x) \\
&= \mathbb{P}(X \in \mathsf{A}).
\end{aligned}$$

That is to say, that the probability that a random variable lies in a set A *can be written as an expectation of the indicator of* A*, at least for a discrete random variable (it turns out that this is indeed true in general).*

An important concept is the moment generating function:.

Definition 2.6.6. *For a discrete random variable* X *the* ***moment generating function*** *(MGF) is*

$$M(t) = \mathbb{E}[e^{Xt}] = \sum_{x \in \mathsf{X}} e^{xt} f(x) \quad t \in \mathsf{T}$$

where T *is the set of* t *for which* $\sum_{\mathsf{X}} e^{xt} f(x) < \infty$*.*

Exercise 2.6.7. *Show that*

$$\mathbb{E}[X] = M'(0) \quad \mathbb{E}[X^2] = M^{(2)}(0)$$

when the right-hand derivatives exist.

The moment generating function is thus a simple way to obtain moments, if it is simple to differentiate $M(t)$. Note also, that it can be proven that the MGF *uniquely* characterizes a distribution.

Example 2.6.8. *Let* $X \sim \mathcal{P}(\lambda)$ *then:*

$$\begin{aligned}
\mathbb{E}[e^{Xt}] &= \sum_{x=0}^{\infty} e^{xt} \frac{\lambda^x}{x!} e^{-\lambda} \\
&= \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!} e^{-\lambda} \\
&= \exp\{\lambda(e^t - 1)\}.
\end{aligned}$$

Then

$$M'(0) = \lambda.$$

Example 2.6.9. *The moment generating function of $X \sim \mathcal{N}e(r,p)$ can be calulated as follows:*

$$\begin{aligned} M(t) &= \mathbb{E}[e^{Xt}] \\ &= \sum_{x=r}^{\infty} \binom{x-1}{r-1} (1-p)^{x-r} p^r e^{xt} \\ &= \sum_{x=r}^{\infty} \binom{x-1}{r-1} ((1-p)e^t)^{x-r} (pe^t)^r \\ &= \Big(\frac{pe^t}{1-(1-p)e^t}\Big)^r \sum_{x=r}^{\infty} \binom{x-1}{r-1} ((1-p)e^t)^{x-r} (1-(1-p)e^t)^r \\ &= \Big(\frac{pe^t}{1-(1-p)e^t}\Big)^r \end{aligned}$$

if $t < \log(1/(1-p))$ i.e.

$$\mathsf{T} = \{t \in \mathbb{R} : t < \log(1/(1-p))\}$$

and we have used the fact that the last summation is 1, as we are summing a $\mathcal{N}e(r, 1-(1-p)e^t)$ random variable. Now

$$M'(t) = r\Big(\frac{pe^t}{1-(1-p)e^t}\Big)^{r-1}\Big(\frac{pe^t}{1-(1-p)e^t} + \frac{p(1-p)e^{2t}}{(1-(1-p)e^t)^2}\Big).$$

So

$$\mathbb{E}[X] = M'(0) = \frac{r}{p}.$$

Reading 2.6.10. *[1, Section 6.4] has some coverage of MGFs and further understanding can be obtained by reading of the book.*

An important special case of functions of random variables are:

Definition 2.6.11. *Given a discrete random variable X on X, with PMF f and $k \in \mathbb{Z}^+$, the kth moment of X is*

$$\mathbb{E}[X^k]$$

and the kth central moment of X is

$$\mathbb{E}[(X - \mathbb{E}[X])^k].$$

Of particular importance, is the idea of the **variance**:

$$\mathbb{V}\mathrm{ar}[X] := \mathbb{E}[(X - \mathbb{E}[X])^2].$$

Now, we have the following calculations:

$$\begin{aligned} \mathbb{E}[(X-\mathbb{E}[X])^2] &= \sum_{x\in\mathsf{X}} (x-\mathbb{E}[X])^2 f(x) \\ &= \sum_{x\in\mathsf{X}} (x^2 - 2x\mathbb{E}[X] + \mathbb{E}[X]^2) f(x) \\ &= \sum_{x\in\mathsf{X}} x^2 f(x) - 2\mathbb{E}[X] \sum_{x\in\mathsf{X}} x f(x) + \mathbb{E}[X]^2 \sum_{x\in\mathsf{X}} f(x) \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X]^2 + \mathbb{E}[X]^2 = \mathbb{E}[X^2] - \mathbb{E}[X]^2. \end{aligned}$$

Hence we have shown that

$$\mathbb{V}\mathrm{ar}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2. \tag{2.6.1}$$

Note a very important point: for any absolutely convergent sum $\sum a_n$, if $a_n \geq 0$ for each n, then $\sum a_n \geq 0$. Now as $\mathbb{V}\mathrm{ar}[X] := \mathbb{E}[(X-\mathbb{E}[X])^2] = \sum_{x\in\mathsf{X}} (x-\mathbb{E}[X])^2 f(x)$ and clearly

$$(x-\mathbb{E}[X])^2 f(x) \geq 0 \quad \forall x \in \mathsf{X}$$

it follows that

Variances cannot be negative[2].

Example 2.6.12. *Returning to Examples 2.6.2 and 2.6.4, when $X \sim \mathcal{P}(\lambda)$, we have shown that*

$$\mathbb{E}[X^2] = \lambda^2 + \lambda \quad \mathbb{E}[X] = \lambda.$$

Hence on using (2.6.1):

$$\mathbb{V}\mathrm{ar}[X] = \lambda.$$

That is, for a Poisson random variable, its mean and variance are equal.

Exercise 2.6.13. *Compute the mean and variance for a Binomial distribution $\mathcal{B}(n,p)$ random variable. [Hint: consider the differentiation, w.r.t. x, of the equality*

$$\sum_{k=0}^{n} \binom{n}{k} x^k = (1+x)^n.$$

].

We now state a Theorem, whose proof we do not give. In general, it is simple to establish, once the concept of a joint distribution has been introduced; we have not done this, so we simply state the result.

Theorem 2.6.14. *The expectation operator has the following properties:*

1. *if $X \geq 0$, $\mathbb{E}[X] \geq 0$*
2. *if $a, b \in \mathbb{R}$ then $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$*
3. *if $X = c \in \mathbb{R}$ always, then $\mathbb{E}[X] = c$.*

An important result that we will use later on and whose proof is omitted is as follows.

Lemma 2.6.15. *If X and Y are independent then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.*

A notion of dependence (linear dependence) is correlation. This will be discussed in details, later on.

Definition 2.6.16. *X and Y are uncorrelated if $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.*

It is important to remark that random variables that are independent are uncorrelated. However, uncorrelated variables are not necessarily independent; we will explore this idea later on.

We end the section with some useful properties of the variance operator.

Theorem 2.6.17. *For random variables X and Y*

1. *For $a \in \mathbb{R}$, $\mathbb{V}\mathrm{ar}[aX] = a^2\mathbb{V}\mathrm{ar}[X]$,*
2. *For X, Y uncorrelated $\mathbb{V}\mathrm{ar}[X+Y] = \mathbb{V}\mathrm{ar}[X] + \mathbb{V}\mathrm{ar}[Y]$.*

Proof. For 1. we have:

$$\begin{aligned} \mathbb{V}\mathrm{ar}[aX] &= \mathbb{E}[(aX)^2) - \mathbb{E}[aX]^2 \\ &= a^2\mathbb{E}[X^2] - a^2\mathbb{E}[X]^2 = a^2\mathbb{V}\mathrm{ar}[X] \end{aligned}$$

where we have used Theorem 2.6.14 2. in the second line.

Now for 2. we have:

$$\begin{aligned} \mathbb{V}\mathrm{ar}[(X+Y)] &= \mathbb{E}[(X+Y-\mathbb{E}[X+Y])^2) \\ &= \mathbb{E}[X^2 + Y^2 + 2XY + \mathbb{E}[X+Y]^2 - 2(X+Y)\mathbb{E}[X+Y]] \\ &= \mathbb{E}[X^2 + Y^2 + 2XY + \mathbb{E}[X]^2 + \mathbb{E}[Y]^2 + 2\mathbb{E}[XY]] - 2\mathbb{E}[X+Y]^2 \\ &= \mathbb{E}[X^2] + \mathbb{E}[Y^2] + 4\mathbb{E}[X]\mathbb{E}[Y] + \mathbb{E}[X]^2 + \mathbb{E}[Y]^2 - 2(\mathbb{E}[X]^2 + \mathbb{E}[Y]^2 + 2\mathbb{E}[X]\mathbb{E}[Y]) \\ &= \mathbb{E}[X^2] + \mathbb{E}[Y^2] - \mathbb{E}[X]^2 - \mathbb{E}[Y]^2 = \mathbb{V}\mathrm{ar}[X] + \mathbb{V}\mathrm{ar}[Y] \end{aligned}$$

where we have repeatedly used Theorem 2.6.14 2.. □

Reading 2.6.18. *[1, Chapter 4] gives a rather complete introduction to expectations.*

[2] This is very important and you will not be told again

Chapter 3

Continuous Random Variables

3.1 Introduction

In the previous Capter, we considered random variables which produce real numbers within a countably finite or infinite set. However, there are of course a wide variety of applications where one sees numerical outcomes of an experiment that can lie potentially anywhere on the real-line. Thus we extend each of concepts that we saw for discrete random variables to continuous ones. In a rather informal (and incorrect) manner, one can simply think of the idea of replacing summation with integration; of course things will be more challenging than this, but one should keep this idea in the back of your mind.

This Chapter is structured as follows. In Section 3.2 we introduce the continuous analogue of a PMF; the probability density function. We consider independence and then expectation in Sections 3.3 and 3.4 respectively.

Reading 3.1.1. *[1, Chapter 5] gives an important complement to the work in this Chapter and should be read to help your understanding.*

3.2 Probability Density Functions

A random variable is said to be continuous if its distribution function $F(x) = \mathbb{P}(X \leq x)$, can be written as

$$F(x) = \int_{-\infty}^{x} f(u)du$$

where $f : \mathbb{R} \to [0, \infty)$, the R.H.S. is the usual Riemann integral and we will assume that the R.H.S. is differentiable.

Definition 3.2.1. *The function f is called the **probability density function** (PDF) of the continuous random variable X.*

We note that, under our assumptions, $f(x) = F'(x)$.

Example 3.2.2. *Let $\mathsf{X} = [a, b]$, where $-\infty < a < b < +\infty$. Then a random variable X is said to have a uniform distribution on X, denoted $X \sim \mathcal{U}_{\mathsf{X}}$ if the PDF can be written as, for $x \in \mathsf{X}$:*

$$f(x) = \frac{1}{b-a}.$$

Let $\mathsf{A} \subseteq \mathsf{X}$ then:

$$\mathbb{P}(X \in \mathsf{A}) = \frac{\int_{\mathsf{A}} dx}{b-a}.$$

That is, the probability that X lies in the set A is the 'volume' of the set A divided by the total 'volume' of the interval. The CDF is easily computed for any $x \in \mathsf{X}$

$$F(x) = \frac{1}{b-a}\int_{a}^{x} du = \frac{x-a}{b-a}.$$

Example 3.2.3. *One of the most commonly used PDFs is the Gaussian or normal distribution:*

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\{-\frac{1}{2\sigma^2}(x-\mu)^2\} \quad x \in \mathsf{X} = \mathbb{R}$$

where $\mu \in \mathbb{R}$, $\sigma^2 > 0$. *We use the notation* $X \sim \mathcal{N}(\mu, \sigma^2)$.

One of the key points associated to PDFs is as follows. The numerical value $f(x)$ *does not* represent the probability that X takes the value x. The technical explanation goes far beyond the mathematical level of this course, but perhaps an intuitive reason is simply that there are uncountably infinite points in X (so assigning a probability to each point is seemingly impossible). In general, one assigns probability to sets of 'non-zero width'. For example let $A = [a, b]$, $-\infty < a < b < \infty$, then one might expect:

$$\mathbb{P}(X \in A) = \int_A f(x)dx.$$

Indeed this holds true, but we are deliberately vague about this. We give the following result, which is not proved and should be taken as true.

Lemma 3.2.4. *If* X *has a PDF* f, *then*

1. $\int_{\mathsf{X}} f(x)dx = 1$.
2. $\mathbb{P}(X = x) = 0$ *for each* $x \in \mathsf{X}$.
3. $\mathbb{P}(X \in [a, b]) = \int_a^b f(x)dx$, $-\infty \leq a < b \leq \infty$.

Example 3.2.5. *Returning to Example 3.2.3, we have*

$$\begin{aligned} \mathbb{P}(X \in \mathsf{X}) &= \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\{-\frac{1}{2\sigma^2}(x-\mu)^2\}dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-u^2/2}du = 1. \end{aligned}$$

Here, we have used the substitution $u = (x-\mu)/\sigma$ *to go to the second line and the fact that* $\int_{-\infty}^{\infty} e^{-u^2/2}du = \sqrt{2\pi}$.

3.2.1 Some Remarks

Consider a **discrete** random variable X with support X and PMF $f(x)$. Then we have learnt that

$$\mathbb{P}(X = x) = f(x) \quad x \in \mathsf{X}.$$

Now for a **continuous** random variable Y with support Y (lets say that Y has 'nothing to do' with X, for example associated to a different experiment) and PDF $g(y)$. Then

$$\mathbb{P}(Y = y) = 0 \quad y \in \mathsf{Y}$$

that is, in general, one has $\mathbb{P}(Y = y)$*is not normally equal to* $g(y)$.

For X (the discrete random variable) we had learnt that the distribution function $F(x)$ is

$$F(x) = \sum_{\{u \leq x \cap u \in \mathsf{X}\}} f(u).$$

Similarly, for Y we have for any $y \in \mathsf{Y}$

$$G(y) = \int_{\{u \leq y \cap u \in \mathsf{Y}\}} g(u)du.$$

If one thinks as $g(y)$ as a continuous function, the distribution function represents the *area under the curve, to the left of* y. More generally we had for X:

$$\mathbb{P}(X \in A) = \sum_{x \in A} f(x).$$

Now for our continuous random variable

$$\mathbb{P}(Y \in B) = \int_B g(y)dy.$$

In essence, all we are doing is replacing summation with integration. Note that all the rules we have are repeated; e.g. for B_1 and B_2 two disjoint sets

$$\mathbb{P}(Y \in B_1 \cup B_2) = \int_{B_1 \cup B_2} g(y)dy = \int_{B_1} g(y)dy + \int_{B_2} g(y)dy = \mathbb{P}(Y \in B_1) + \mathbb{P}(Y \in B_2).$$

We note also, if $\mathsf{Y} = [a, b]$, $-\infty < a < b < \infty$

$$G(a) = 0 \quad G(b) = 1.$$

If instead $\mathsf{Y} = [0, \infty)$:

$$G(0) = 0 \quad \lim_{u \to \infty} G(u) = 1.$$

Finally if $\mathsf{Y} = (-\infty, \infty)$

$$\lim_{u \to -\infty} G(u) = 0 \quad \lim_{u \to \infty} G(u) = 1$$

Example

Suppose that $X \sim \mathcal{N}(\mu, \sigma^2)$, that is the PDF of X is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\Big\{ -\frac{1}{2\sigma^2}(x-\mu)^2 \Big\} \quad x \in \mathsf{X} = \mathbb{R}.$$

Let us find the distribution function.

First suppose that $\mu = 0$ and $\sigma^2 = 1$; in this situation, we call X a standard normal random variable. Now **define**

$$\mathbb{P}(X \le x) = \Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} \exp\Big\{ -\frac{1}{2}u^2 \Big\} du.$$

In general, the integral on the RHS cannot be computed in a closed form, but by now there are very accurate approximations on a computer.

Second, let us consider the general case. We have

$$\begin{aligned} \mathbb{P}(X \le x) &= \int_{-\infty}^{x} \frac{1}{\sigma\sqrt{2\pi}} \exp\Big\{ -\frac{1}{2\sigma^2}(u-\mu)^2 \Big\} du \\ &= \int_{-\infty}^{\frac{x-\mu}{\sigma}} \frac{1}{\sqrt{2\pi}} \exp\Big\{ -\frac{1}{2}v^2 \Big\} dv \\ &= \Phi\Big(\frac{x-\mu}{\sigma}\Big). \end{aligned}$$

We made the substitution $v = (u-\mu)/\sigma$ to go to the second line. This substituion is called *standardization* for normal random variables.

Given a computer program to calculate $\Phi(x)$ we can then compute a variety of probabilities associated to normal random variables. For example, if we want to calculate the probability that $X \in [a, b]$ for some $-\infty < a < b < \infty$, we have

$$\begin{aligned} \mathbb{P}(X \in [a,b]) &= \mathbb{P}(X \le b) - \mathbb{P}(X \le a) \\ &= \Phi\Big(\frac{b-\mu}{\sigma}\Big) - \Phi\Big(\frac{a-\mu}{\sigma}\Big). \end{aligned}$$

To obtain the first equality, we remark that

$$\mathbb{P}(X \in (-\infty, b]) = \mathbb{P}(X \in (-\infty, a) \cup [a, b]) = \mathbb{P}(X \in (-\infty, a)) + \mathbb{P}(X \in [a, b])$$

so we have

$$\mathbb{P}(X \in (-\infty, b]) = \mathbb{P}(X \in (-\infty, a)) + \mathbb{P}(X \in [a, b]) \Rightarrow \mathbb{P}(X \in [a, b]) = \mathbb{P}(X \in (-\infty, b]) - \mathbb{P}(X \in (-\infty, a)).$$

We then note $\mathbb{P}(X \in (-\infty, b]) = \mathbb{P}(X \le b)$ and $\mathbb{P}(X \in (-\infty, a)) = \mathbb{P}(X \in (-\infty, a])$ (note that $\mathbb{P}(X = a) = 0$). If this explanation is confusing;

we are just saying the area under the curve between $[a, b]$ (for a continuous function) is the difference between all the area to the left of b take away all the area to the left of a.

Similarly for $-\infty < a < b < c < d < \infty$:

$$\mathbb{P}(X \in [a,b] \cup [c,d]) = \mathbb{P}(X \in [a,b]) + \mathbb{P}(X \in [c,d]) = \Phi\Big(\frac{b-\mu}{\sigma}\Big) - \Phi\Big(\frac{a-\mu}{\sigma}\Big) + \Phi\Big(\frac{d-\mu}{\sigma}\Big) - \Phi\Big(\frac{c-\mu}{\sigma}\Big).$$

3.3 Independence

To define an idea of independence for continuous random variables, we cannot use the one for discrete random variables (recall Definition 2.4.1); the sets $\{X = x\}$ and $\{Y = y\}$ have zero probability and are hence trivially independent. Thus we use a new definition for independence:

Definition 3.3.1. *The random variables X and Y are independent if*

$$\{X \leq x\} \quad \{Y \leq y\}$$

are independent events for each $x, y \in \mathbb{R}$.

As for PMFs one can consider independence through PDFs; however, as we are yet to define the notion of joint PDFs, we will leave this idea for now. We note that one can show that for any real valued functions h and g (at least within some technical conditions which are assumed in this course) that if X and Y are independent, so are the random variables $h(X)$ and $g(Y)$. We do not prove why.

3.4 Expectation

As for discrete random variables one can consider the idea of the average of a random variable. This is simply brought about by replacing summations with integrations:

Definition 3.4.1. *The expectation of a continuous random variable X with PDF f is given by*

$$\mathbb{E}[X] = \int_{\mathsf{X}} x f(x) dx$$

whenever the integral exists.

Example 3.4.2. *Consider the exponential density:*

$$f(x) = \lambda e^{-\lambda x} \quad x \in \mathsf{X} = [0, \infty) \quad \lambda > 0.$$

We use the notation $X \sim \mathcal{E}(\lambda)$. Then

$$\mathbb{E}[X] = \int_0^\infty x\lambda e^{-\lambda x} dx = [-xe^{-\lambda x}]_0^\infty + \int_0^\infty e^{-\lambda x} dx = [-\frac{1}{\lambda} e^{-\lambda x}]_0^\infty = \frac{1}{\lambda}$$

where we have used integration by parts.

Example 3.4.3. *Let us return to Example 3.2.3. Then*

$$\begin{aligned}
\mathbb{E}[X] &= \int_{-\infty}^\infty x \frac{1}{\sigma\sqrt{2\pi}} \exp\{-\frac{1}{2\sigma^2}(x-\mu)^2\} dx \\
&= \int_{-\infty}^\infty (u\sigma + \mu) \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du \\
&= \sigma \int_{-\infty}^\infty u \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du + \mu \\
&= \frac{\sigma}{\sqrt{2\pi}} [-e^{-u^2/2}]_{-\infty}^\infty + \mu \\
&= \mu.
\end{aligned}$$

Here, we have used the substitution $u = (x - \mu)/\sigma$ to go to the second line and the fact that

$$\int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du = 1$$

to go to the third.

Example 3.4.4. *Consider the gamma density:*

$$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} \quad x \in \mathsf{X} = [0, \infty) \quad \lambda, \alpha > 0$$

where $\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt$. *We use the notation* $X \sim \mathcal{G}(\alpha, \lambda)$ *and note that if* $X \sim \mathcal{G}(1, \lambda)$ *then* $X \sim \mathcal{E}(\lambda)$. *Now*

$$\begin{aligned}
\mathbb{E}[X] &= \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty x^\alpha e^{-\lambda x} dx \\
&= \frac{\lambda^\alpha}{\Gamma(\alpha)} \frac{1}{\lambda^{\alpha+1}} \int_0^\infty u^\alpha e^{-u} du \\
&= \frac{1}{\lambda \Gamma(\alpha)} \Gamma(\alpha+1) = \frac{\alpha}{\lambda}.
\end{aligned}$$

Where we have used the substitution $u = \lambda x$ *to go the second line and* $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$ *on the final line (from herein you may use that identity without proof).*

We now state a useful technical result

Theorem 3.4.5. *If* X *and* $g(x)$ *(*$g : \mathbb{R} \to \mathbb{R}$*) are continuous random variables*

$$\mathbb{E}[g(X)] = \int_{\mathsf{X}} g(x) f(x) dx.$$

We remark that all the extensions to expectations discussed in Section 2.6 can be extended to the continuous case. In particular Definitions 2.6.11 and 2.6.16 can be imported to the continuous case. In addition the results: Theorems 2.6.14 and 2.6.17 and Lemma 2.6.15 all can be extended. It is assumed that this is the case from herein.

Example 3.4.6. *Let us return to Example 3.4.2*

$$\begin{aligned}
\mathbb{E}[X^2] &= \int_0^\infty x^2 \lambda e^{-\lambda x} dx \\
&= [-x^2 e^{-\lambda x}]_0^\infty + \frac{2}{\lambda} \int_0^\infty x \lambda e^{-\lambda x} dx \\
&= \frac{2}{\lambda^2}.
\end{aligned}$$

Thus, for an exponential random variable:

$$\mathbb{V}\mathrm{ar}[X] = \frac{1}{\lambda^2}.$$

Example 3.4.7. *Let us return to Example 3.2.3. Then*

$$\begin{aligned}
\mathbb{E}[X^2] &= \int_{-\infty}^\infty x^2 \frac{1}{\sigma\sqrt{2\pi}} \exp\{-\frac{1}{2\sigma^2}(x-\mu)^2\} dx \\
&= \int_{-\infty}^\infty (u\sigma + \mu)^2 \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du \\
&= \sigma^2 \int_{-\infty}^\infty u^2 \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du + \mu^2 + 2\sigma\mu \int_{-\infty}^\infty u \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du \\
&= \frac{\sigma^2}{\sqrt{2\pi}} \{[-ue^{-u^2/2}]_{-\infty}^\infty + \int_{-\infty}^\infty e^{-u^2/2} du\} + \mu^2 \\
&= \frac{\sigma^2}{\sqrt{2\pi}} \sqrt{2\pi} + \mu^2.
\end{aligned}$$

Here we have used $\int_{-\infty}^\infty u \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du = 0$ *and* $\int_{-\infty}^\infty e^{-u^2/2} du = \sqrt{2\pi}$. *Thus, for a normal random variable:*

$$\mathbb{V}\mathrm{ar}[X] = \sigma^2.$$

Example 3.4.8. *Let us return to Example 3.4.4 Now*

$$\begin{aligned}\mathbb{E}[X^2] &= \frac{\lambda^\alpha}{\Gamma(\alpha)}\int_0^\infty x^{\alpha+1}e^{-\lambda x}dx \\ &= \frac{\lambda^\alpha}{\Gamma(\alpha)}\frac{1}{\lambda^{\alpha+2}}\int_0^\infty u^{\alpha+1}e^{-u}du \\ &= \frac{1}{\lambda^2\Gamma(\alpha)}\Gamma(\alpha+2) = \frac{\alpha(\alpha+1)}{\lambda^2}\end{aligned}$$

where we have used $\Gamma(\alpha+2) = (\alpha+1)\alpha\Gamma(\alpha)$*. Thus, for a gamma random variable:*

$$\mathbb{V}\mathrm{ar}[X] = \frac{\alpha}{\lambda^2}.$$

Example 3.4.9. *Let* $X \sim \mathcal{N}(0,1)$*, let us compute* $\mathbb{E}[|X|]$*. We have*

$$\begin{aligned}\mathbb{E}[|X|] &= \int_{-\infty}^\infty |x|\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}dx \\ &= -\int_{-\infty}^0 x\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}dx + \int_0^\infty x\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}dx \\ &= \frac{1}{\sqrt{2\pi}}\left\{\left[e^{-\frac{x^2}{2}}\right]_{-\infty}^0 + \left[-e^{-\frac{x^2}{2}}\right]_0^\infty\right\} \\ &= \sqrt{\frac{2}{\pi}}.\end{aligned}$$

Example 3.4.10. *Consider the indicator function of a set* A*:* $\mathbb{I}_{\mathsf{A}}(x)$*. This function is 1 if* $x \in \mathsf{A}$ *and zero otherwise. Now suppose we have a random variable* X *with support* X *and that* $\mathsf{A} \subseteq \mathsf{X}$*. Then we have that*

$$\begin{aligned}\mathbb{E}[\mathbb{I}_A(X)] &= \int_{\mathsf{X}} \mathbb{I}_{\mathsf{A}}(x)f(x)dx \\ &= \int_{\mathsf{A}} f(x)dx \\ &= \mathbb{P}(X \in \mathsf{A}).\end{aligned}$$

That is to say, that the probability that a random variable lies in a set A *can be written as an expectation of the indicator of* A*, at least for a continuous random variable; see Example 2.6.5 for the discrete case.*

To end this section we consider an important concept; the MGF for continuous random variables.

Definition 3.4.11. *For a continuous random variable* X *the* ***moment generating function*** *(MGF) is*

$$M(t) = \mathbb{E}[e^{Xt}] = \int_{\mathsf{X}} e^{xt}f(x)dx \quad t \in \mathsf{T}$$

where T *is the set of* t *for which* $\int_{\mathsf{X}} e^{xt}f(x)dx < \infty$*.*

Exercise 3.4.12. *Show that*

$$\mathbb{E}[X] = M'(0) \quad \mathbb{E}[X^2] = M^{(2)}(0)$$

when the right-hand derivatives exist.

As for discrete random variables, the moment generating function is a simple way to obtain moments, if it is simple to differentiate $M(t)$. Note also, that it can be proven that the MGF *uniquely* characterizes a distribution.

Example 3.4.13. *Suppose* $X \sim \mathcal{E}(\lambda)$ *then, supposing* $\lambda > t$

$$\begin{aligned}M(t) &= \int_0^\infty \lambda e^{-x(\lambda-t)}dx \\ &= \left[-\frac{\lambda}{(\lambda-t)}e^{-x(\lambda-t)}\right]_0^\infty \\ &= \frac{\lambda}{(\lambda-t)}.\end{aligned}$$

Clearly

$$M'(t) = \frac{\lambda}{(\lambda - t)^2}$$

and thus $\mathbb{E}[X] = 1/\lambda$.

Example 3.4.14. *Suppose* $X \sim \mathcal{N}(\mu, \sigma^2)$ *then*

$$\begin{aligned} M(t) &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\{-\frac{1}{2\sigma^2}(x-\mu)^2 + xt\} dx \\ &= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\{-\frac{1}{2\sigma^2}[(x-(\mu+t\sigma^2))^2 - (\mu+t\sigma^2)^2 + \mu^2]\} dx \\ &= \frac{1}{\sigma\sqrt{2\pi}} \exp\{\mu t + \frac{\sigma^2 t^2}{2}\} \int_{-\infty}^{\infty} \exp\{-\frac{1}{2\sigma^2}(x-(\mu+t\sigma^2))^2\} dx \\ &= \exp\{\mu t + \frac{\sigma^2 t^2}{2}\} \end{aligned}$$

where we have used a change-of-variables $u = (x - (\mu + t\sigma^2))/\sigma$ *to deal with the integral.*

Chapter 4

Joint and Multivariate Distributions

4.1 Introduction

The idea of defining probability densities and mass functions over many random variables is now developed. We begin with the discrete case in Section 4.2 and continuous case in Section 4.3. In Section 4.4 we describe the idea that a sequence of i.i.d. continuous random variables have a symmetric distribution. Then in Section 4.5 the concept of order statistics is introduced. In Section 4.6 we show the link between the exponential distribution and a stochastic process, called the Poisson process. Finally in Section 4.7 we give a definition for the multivariate normal distribution.

4.2 Discrete Case

As we have seen, there is a need to define distributions on more than one random variable. We will do that in this Section. We start with the following definition (of which one can easily generalize to a collection of $n \geq 1$ discrete random variables):

Definition 4.2.1. *The joint distribution function $F : \mathbb{R}^2 \to [0,1]$ of X, Y where X and Y are discrete random variables, is given by*

$$F(x,y) = \mathbb{P}(X \leq x \cap Y \leq y).$$

Their joint mass function $f : \mathbb{R}^2 \to [0,1]$ is given by

$$f(x,y) = \mathbb{P}(X = x \cap Y = y).$$

In general, it may be that the random variables are defined on a space Z which may not be decomposable into a cartesian product $\mathsf{X} \times \mathsf{Y}$. In such scenarios we write the joint support as simply Z and omit X and Y, in this scenario, we will use the notation $\sum_x$ or $\sum_y$ to denote sums over the supports induced by Z. This concept will be clarified during a reading of the subsequent text.

Of particular importance, are the marginal PMFs of X and Y:

$$f(x) = \sum_y f(x,y) \quad f(y) = \sum_x f(x,y).$$

Note that, clearly

$$\sum_x f(x) = 1 = \sum_y f(y).$$

	$y=-1$	$y=0$	$y=2$	$f(x)$
$x=1$	$\frac{1}{18}$	$\frac{3}{18}$	$\frac{2}{18}$	$\frac{6}{18}$
$x=2$	$\frac{2}{18}$	0	$\frac{3}{18}$	$\frac{5}{18}$
$x=3$	0	$\frac{4}{18}$	$\frac{3}{18}$	$\frac{7}{18}$
$f(y)$	$\frac{3}{18}$	$\frac{7}{18}$	$\frac{8}{18}$	

Table 4.1: The Joint Mass Function in Example 4.2.3. The row totals are the marginal mass function on X (and sum to 1) and respectively the column totals are the marginal mass function on Y.

Example 4.2.2. *Consider Theorem 2.6.14 2. and for simplicity suppose* $\mathsf{Z} = \mathsf{X} \times \mathsf{Y}$. *Now, we have*

$$\begin{aligned}
\mathbb{E}[aX + bY] &= \sum_{x\in\mathsf{X}}\sum_{y\in\mathsf{Y}}(ax+by)f(x,y) \\
&= \sum_{x\in\mathsf{X}} ax \sum_{y\in\mathsf{Y}} f(x,y) + \sum_{y\in\mathsf{Y}} by \sum_{x\in\mathsf{X}} f(x,y) \\
&= a\sum_{x\in\mathsf{X}} xf(x) + b\sum_{y\in\mathsf{Y}} yf(y) \\
&= a\mathbb{E}[X] + b\mathbb{E}[Y].
\end{aligned}$$

Example 4.2.3. *Suppose* $\mathsf{X} = \{1,2,3\}$ *and* $\mathsf{Y} = \{-1,0,2\}$ *then an example of a joint PMF can be found in Table 4.2.3. From the table, we have:*

$$\mathbb{E}[XY] = \sum_{x\in\mathsf{X}}\sum_{y\in\mathsf{Y}} xyf(x,y) = \frac{29}{18}$$

(just sum the 9 values in the table, multiplying each time by the product of the associated x and y. Similarly

$$\mathbb{E}[X] = \frac{6}{18} + 2\frac{5}{18} + 3\frac{7}{18} = \frac{37}{18}$$

and

$$\mathbb{E}[Y] = -1\frac{3}{18} + 0\frac{7}{18} + 2\frac{8}{18} = \frac{13}{18}.$$

We now formalize independence in a result, which provides us a more direct way to ascertain if two random variables X and Y are independent.

Lemma 4.2.4. *The discrete random variables X and Y are independent if an only if*

$$f(x,y) = f(x)f(y) \quad \forall x,y \in \mathbb{R}.$$

More generally, X and Y are independent if and only if $f(x,y)$ factorizes into the product $g(x)h(y)$ with g a function only of x and h a function only of y.

Example 4.2.5. *Consider the joint PMF:*

$$f(x,y) = \frac{\lambda^x e^{-\lambda}}{x!}\frac{\lambda^y e^{-\lambda}}{y!} \quad \mathsf{X} = \mathsf{Y} = \{0,1,\ldots,\}.$$

Now clearly

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!} \; x \in \mathsf{X} \quad f(y) = \frac{\lambda^y e^{-\lambda}}{y!} \; y \in \mathsf{Y}.$$

It is also clear that via Lemma 4.2.4 the random variables X and Y are independent and identically distributed (and Poisson distributed).

Example 4.2.6. *Consider $X \sim \mathcal{P}(\lambda_1)$ and independently $Y \sim \mathcal{P}(\lambda_2)$ what is the distribution of $Z = X + Y$? We have that*

$$f(x,z) = f_X(x)f_Y(z-x)$$

for $(x,z) \in \mathsf{Z} = \{(x,z) \in \{0,1\dots\}^2 : x \le z\}$ *and where the subscripts on the R.H.S. denote the PMFs of* X *and* Y *respectively. Then for* $z \in \{0,1,\dots\}$

$$\begin{aligned} f(z) &= \sum_{x=0}^{z} f_X(x) f_Y(z-x) \\ &= \sum_{x=0}^{z} \frac{\lambda_1^x}{x!} e^{-\lambda_1} \frac{\lambda_2^{z-x}}{(z-x)!} e^{-\lambda_2} \\ &= \frac{1}{z!} e^{-(\lambda_1+\lambda_2)} \sum_{x=0}^{z} \binom{z}{x} \lambda_1^x \lambda_2^{z-x} \\ &= \frac{(\lambda_1+\lambda_2)^z}{z!} e^{-(\lambda_1+\lambda_2)}. \end{aligned}$$

That is, $Z \sim \mathcal{P}(\lambda_1+\lambda_2)$.

As in the case of a single variable, we are interested in the expectation of a functional of two random variables (strictly, in the proof of Theorem 2.6.17 we have already used the following result):

Lemma 4.2.7. $\mathbb{E}[g(X,Y)] = \sum_{(x,y)\in \mathsf{X}\times Y} g(x,y) f(x,y)$.

Recall that in the previous Section, we mentioned a notion of dependence called correlation. In order to formally define this concept, we introduce first the **covariance** and then correlation.

Definition 4.2.8. *The covariance between* X *and* Y *is*

$$\mathbb{C}\mathrm{ov}[X,Y] := \mathbb{E}[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])].$$

The correlation between X *and* Y *is*

$$\rho(X,Y) = \frac{\mathbb{C}\mathrm{ov}[X,Y]}{\sqrt{\mathbb{V}\mathrm{ar}[X]\mathbb{V}\mathrm{ar}[Y]}}.$$

Exercise 4.2.9. *Show that*

$$\mathbb{E}[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Thus, independent random variables have zero correlation.

Example 4.2.10. *Returning to Example 4.2.3, we have:*

$$\mathbb{V}\mathrm{ar}[X] = 233/324 \quad \mathbb{V}\mathrm{ar}[Y] = 461/324$$

and

$$\mathbb{C}\mathrm{ov}[X,Y] = 29/18 - (37/18)(13/18) = 41/324.$$

Thus

$$\rho(X,Y) = 41/\sqrt{107413}.$$

Some useful facts about correlations that we do not prove:

- $|\rho(X,Y)| \le 1$. Random variables with correlation 1 or -1 are said to be perfectly positively or negatively correlated.
- The correlation coefficient is 1 iff Y increases linearly with X and -1 iff Y decreases linearly as X increases.

When we consider *continuous random variables* an example of uncorrelated random variables that are not independent, will be given.

4.2.1 An Extended Example

Suppose that $(X, Y) \in \mathsf{Z} = \mathsf{X} \times \mathsf{Y}$ are jointly defined discrete random variables, with PMF

$$f(x, y) = c(x + y)\frac{a^{x+y}}{x!y!} \quad \mathsf{X} = \mathsf{Y} = \{0, 1, \dots\}$$

where $a > 0$ is a known finite constant, but $c > 0$ is unknown. In this example we will answer the following questions:

1. Find c in terms of a.
2. Find the marginal PMFs of X and Y; are X and Y independent?
3. Calculate: $\mathbb{E}[X]$, $\mathbb{E}[Y]$, $\mathbb{V}\mathrm{ar}[X]$, $\mathbb{V}\mathrm{ar}[Y]$.
4. Calculate: $\rho(X, Y)$.

Solution to (1)

We know that $f(x, y)$ is a PMF. This means it must be non-negative on Z and sum to 1. We already know that it is non-negative. Thus, we have that

$$c = \Big(\sum_{(x,y)\in\mathsf{Z}} (x + y)\frac{a^{x+y}}{x!y!} \Big)^{-1}.$$

This then means that $\sum_{(x,y)\in\mathsf{Z}} f(x, y) = 1$. Therefore, we need to compute the sum.

$$\begin{aligned}
\sum_{(x,y)\in\mathsf{Z}} (x + y)\frac{a^{x+y}}{x!y!} &= \sum_{x=0}^{\infty}\sum_{y=0}^{\infty}(x + y)\frac{a^{x+y}}{x!y!} \\
&= \sum_{x=0}^{\infty}\frac{a^x}{x!}\Big\{x\sum_{y=0}^{\infty}\frac{a^y}{y!} + \sum_{y=1}^{\infty}\frac{a^y}{(y-1)!}\Big\} \\
&= \sum_{x=0}^{\infty}\frac{a^x}{x!}\Big\{xe^a + a\sum_{y=1}^{\infty}\frac{a^{y-1}}{(y-1)!}\Big\} \\
&= \sum_{x=0}^{\infty}\frac{a^x}{x!}(xe^a + ae^a) \\
&= e^a\Big\{\sum_{x=1}^{\infty}\frac{a^x}{(x-1)!} + a\sum_{x=0}^{\infty}\frac{a^x}{x!}\Big\} \\
&= e^a(ae^a + ae^a) \\
&= 2ae^{2a}.
\end{aligned}$$

Hence $c = e^{-2a}/2a$.

Solution to (2)

We have that, for $x \in \mathsf{X}$

$$\begin{aligned}
f(x) &= \sum_{y\in\mathsf{Y}} f(x, y) \\
&= \sum_{y=0}^{\infty}\frac{e^{-2a}}{2a}(x + y)\frac{a^{x+y}}{x!y!} \\
&= \frac{e^{-2a}}{2a}\frac{a^x}{x!}\Big\{x\sum_{y=0}^{\infty}\frac{a^y}{y!} + \sum_{y=1}^{\infty}\frac{a^y}{(y-1)!}\Big\} \\
&= \frac{e^{-2a}}{2a}\frac{a^x}{x!}\Big\{xe^a + ae^a\Big\} \\
&= \frac{e^{-a}}{2a}\frac{a^x}{x!}(x + a) \quad x \in \mathsf{X}.
\end{aligned}$$

One can repeat almost the same calculations to deduce that

$$f(y) = \frac{e^{-a}}{2a}\frac{a^y}{y!}(y+a) \quad x \in \mathsf{Y}.$$

Clearly, we do not have, for every $(x, y) \in \mathsf{Z}$

$$f(x, y) = f(x)f(y)$$

and X and Y are **not** independent. Note however X and Y have the same marginal PMF's.

Solution to (3)

As noted in the solution to (2), X and Y have the same marginal PMF's, so to compute the expectation and variance, we need only consider the properties for one of the marginal PMFs. That is;

$$\mathbb{E}[X] = \mathbb{E}[Y] \quad \mathbb{V}\mathrm{ar}[X] = \mathbb{V}\mathrm{ar}[Y].$$

We begin with $\mathbb{E}[X]$:

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{x\in\mathsf{X}} xf(x) \\
&= \sum_{x=0}^{\infty} x\frac{e^{-a}}{2a}\frac{a^x}{x!}(x+a) \\
&= \sum_{x=1}^{\infty} \frac{e^{-a}}{2a}\frac{a^x}{(x-1)!}(x+a) \\
&= \frac{e^{-a}}{2a}\Big\{\sum_{x=1}^{\infty} x\frac{a^x}{(x-1)!} + a\sum_{x=1}^{\infty}\frac{a^x}{(x-1)!}\Big\} \\
&= \frac{e^{-a}}{2a}\Big\{\sum_{x=1}^{\infty} x\frac{a^x}{(x-1)!} + a^2e^a\Big\}. \qquad (4.2.1)
\end{aligned}$$

Now the remaining summation deserves a little attention, so let us focus upon it. We have

$$\begin{aligned}
\sum_{x=1}^{\infty} x\frac{a^x}{(x-1)!} &= a\sum_{x=1}^{\infty}\Big[\frac{d}{da}a^x\Big]\frac{1}{(x-1)!} \\
&= a\frac{d}{da}\Big[\sum_{x=1}^{\infty}\frac{a^x}{(x-1)!}\Big] \\
&= a\frac{d}{da}[ae^a] \\
&= a(a+1)e^a. \qquad (4.2.2)
\end{aligned}$$

So, returning to (4.2.1), we have

$$\begin{aligned}
\mathbb{E}[X] &= \frac{e^{-a}}{2a}\Big\{a(a+1)e^a + a^2e^a\Big\} \\
&= \frac{2a+1}{2}.
\end{aligned}$$

Now, to compute the variance, we recall that

$$\mathbb{V}\mathrm{ar}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

So, we compute the second moment of X:

$$\begin{aligned}
\mathbb{E}[X^2] &= \sum_{x\in\mathsf{X}} x^2 f(x) \\
&= \sum_{x=0}^{\infty} x^2 \frac{e^{-a}}{2a}\frac{a^x}{x!}(x+a) \\
&= \frac{e^{-a}}{2a}\Big\{\sum_{x=1}^{\infty} x^2 \frac{a^x}{(x-1)!} + a\sum_{x=1}^{\infty} x\frac{a^x}{(x-1)!}\Big\} \\
&= \frac{e^{-a}}{2a}\Big\{\sum_{x=1}^{\infty} x^2 \frac{a^x}{(x-1)!} + aa(a+1)e^a\Big\} \qquad (4.2.3)
\end{aligned}$$

where we have applied (4.2.7) on the third line. Now, again, we focus on the remaining summation:

$$\begin{aligned}
\sum_{x=1}^{\infty} x^2 \frac{a^x}{(x-1)!} &= a^2\sum_{x=1}^{\infty}\Big\{\frac{d^2}{da^2}[a^x] + xa^{x-2}\Big\}\frac{1}{(x-1)!} \\
&= a^2\Big\{\frac{d^2}{da^2}\sum_{x=1}^{\infty}\frac{a^x}{(x-1)!} + \sum_{x=1}^{\infty}\frac{xa^{x-2}}{(x-1)!}\Big\} \\
&= a^2\Big\{\frac{d^2}{da^2}(ae^a) + \frac{1}{a^2}\sum_{x=1}^{\infty}\frac{xa^x}{(x-1)!}\Big\} \\
&= a^2\Big\{\frac{d}{da}((a+1)e^a) + \frac{a(a+1)e^a}{a^2}\Big\} \\
&= a^2\Big\{(a+2)e^a + \frac{a(a+1)e^a}{a^2}\Big\} \\
&= (a^2+3a+1)ae^a.
\end{aligned}$$

Here we have used (4.2.7) on line 3. Then returning to (4.2.3), we have

$$\begin{aligned}
\mathbb{E}[X^2] &= \frac{e^{-a}}{2a}\Big\{(a^2+3a+1)ae^a + a^2(a+1)e^a\Big\} \\
&= a^2+2a+\frac{1}{2}.
\end{aligned}$$

Thus, we have, on recalling $\mathbb{E}[X] = \frac{2a+1}{2}$

$$\begin{aligned}
\mathbb{V}\mathrm{ar}[X] &= a^2+2a+\frac{1}{2} - (\frac{2a+1}{2})^2 \\
&= a+\frac{1}{4}.
\end{aligned}$$

Solution to (4)

In order to calculate the correlation, we require the covariance. Recall

$$\mathbb{C}\mathrm{ov}[X,Y] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Thus, we need $\mathbb{E}[XY]$:

$$\begin{aligned}
\mathbb{E}[XY] &= \sum_{(x,y)\in\mathsf{Z}} xyf(x,y) \\
&= \frac{e^{-2a}}{2a}\sum_{x=0}^{\infty}\sum_{y=0}^{\infty} xy(x+y)\frac{a^{x+y}}{x!y!} \\
&= \frac{e^{-2a}}{2a}\sum_{x=1}^{\infty}\sum_{y=1}^{\infty} (x+y)\frac{a^{x+y}}{(x-1)!(y-1)!} \\
&= \frac{e^{-2a}}{2a}\sum_{x=1}^{\infty}\frac{a^x}{(x-1)!}\Big\{x\sum_{y=1}^{\infty}\frac{a^y}{(y-1)!}+\sum_{y=1}^{\infty}\frac{ya^y}{(y-1)!}\Big\} \\
&= \frac{e^{-2a}}{2a}\sum_{x=1}^{\infty}\frac{a^x}{(x-1)!}\{xae^a+a(a+1)e^a\} \\
&= \frac{e^{-2a}}{2a}\Big\{ae^a\sum_{x=1}^{\infty}\frac{xa^x}{(x-1)!}+a(a+1)e^a\sum_{x=1}^{\infty}\frac{a^x}{(x-1)!}\Big\} \\
&= \frac{e^{-2a}}{2a}\Big\{ae^a\times a(a+1)e^a+a(a+1)e^a\times ae^a\Big\} \\
&= a(a+1)
\end{aligned}$$

Here we have used (4.2.7) on lines 4 and 6. Then recalling $\mathbb{E}[Y]=\mathbb{E}[X]=\frac{2a+1}{2}$

$$\mathbb{C}\mathrm{ov}[X,Y]=a(a+1)-\frac{1}{4}(4a^2+4a+1)=-\frac{1}{4}.$$

Now, by definition

$$\rho(X,Y)=\frac{\mathbb{C}\mathrm{ov}[X,Y]}{\sqrt{\mathbb{V}\mathrm{ar}[X]\mathbb{V}\mathrm{ar}[Y]}}.$$

As, in this example, $\mathbb{V}\mathrm{ar}[X]=\mathbb{V}\mathrm{ar}[Y]$, we have

$$\rho(X,Y)=\frac{\mathbb{C}\mathrm{ov}[X,Y]}{\mathbb{V}\mathrm{ar}[X]},$$

so

$$\rho(X,Y)=\frac{-\frac{1}{4}}{a+\frac{1}{4}}=-\frac{1}{4a+1}.$$

Note $|\rho(X,Y)|\le 1$.

4.2.2 Conditional Distributions and Expectations

In Section 1.4 we discussed the idea of conditional probabilities associated to events. This idea can then be extended to discrete-valued random variables:

Definition 4.2.11. *The conditional distribution function of Y given X, written $F_{Y|x}(\cdot|x)$, is defined by*

$$F_{y|x}(y|x)=\mathbb{P}(Y\le y|X=x)$$

for any x with $\mathbb{P}(X=x)>0$. The conditional PMF of Y given $X=x$ is defined by

$$f(y|x)=\mathbb{P}(Y=y|X=x)$$

when x is such that $\mathbb{P}(X=x)>0$.

We remark that, in particular, the conditional mass function of Y given $X=x$ is:

$$f(y|x)=\frac{f(x,y)}{f(x)}=\frac{f(x,y)}{\sum_y f(x,y)}.$$

Example 4.2.12. *Let us return to Example 4.2.6. We now ask what is the conditional PMF for X given Z? We already have calculated that*

$$f(x,z) = f_X(x)f_Y(z-x)$$

for $(z,x) \in \{0,1,\dots\} \times \{0.\dots,z\}$. Therefore, to compute the conditonal PMF $X|Z$ we need the marginal of Z. However, in Example 4.2.6 we also obtained that $Z \sim \mathcal{P}(\lambda_1+\lambda_2)$. Therefore, for $x \in \{0,1,\dots,z\}$:

$$\begin{aligned} f(x|z) &= \frac{f(x,z)}{f(z)} \\ &= \frac{\lambda_1^x}{x!}e^{-\lambda_1}\frac{\lambda_2^{z-x}}{(z-x)!}e^{-\lambda_2} \Bigg/ \left\{\frac{(\lambda_1+\lambda_1)^z}{z!}e^{-(\lambda_1+\lambda_2)}\right\} \\ &= \binom{z}{x}\left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^x\left(\frac{\lambda_z}{\lambda_1+\lambda_2}\right)^{z-x}. \end{aligned}$$

That is, $X|Z=z \sim \mathcal{B}\left(z, \frac{\lambda_1}{\lambda_1+\lambda_2}\right)$.

Example 4.2.13. *Suppose that $Y \sim \mathcal{P}(\lambda)$ and $X|Y=y \sim \mathcal{B}(y,p)$. Find the conditional probability mass function of $Y|X=x$. Note that the random variables lie in the space $\mathsf{Z} = \{(x,y) : y \in \{0,1,\dots\}, x \in \{0,1,\dots,y\}\}$. We have*

$$f(y|x) = \frac{f(x,y)}{f(x)} = \frac{f(x|y)f(y)}{f(x)} \quad (x,y) \in \mathsf{Z}$$

which is a version of Bayes Theorem *for discrete random variables. Note that for $(x,y) \in \mathsf{Z}$ all the PMFs above are positive. Now for $x \in \{0,1,\dots\}$*

$$\begin{aligned} f(x) &= \sum_{y=x}^{\infty}\binom{y}{x}p^x(1-p)^{y-x}\frac{\lambda^y e^{-\lambda}}{y!} \\ &= (\lambda p)^x\frac{e^{-\lambda}}{x!}\sum_{y=x}^{\infty}\frac{((1-p)\lambda)^{y-x}}{(y-x)!} \\ &= \frac{(\lambda p)^x}{x!}e^{-\lambda p}. \end{aligned}$$

Thus, we have for $y \in \{x, x+1, \dots\}$:

$$f(y|x) = \frac{\binom{y}{x}p^x(1-p)^{y-x}\frac{\lambda^y e^{-\lambda}}{y!}}{\frac{(\lambda p)^x}{x!}e^{-\lambda p}}$$

which after some algebra, becomes:

$$f(y|x) = \frac{(\lambda(1-p))^{y-x}e^{-\lambda(1-p)}}{(y-x)!} \quad y \in \{x, x+1, \dots\}.$$

Given the idea of a conditional distribution, we move onto the idea of a conditional expectation:

Definition 4.2.14. *The **conditional expectation** of a random variable Y, given $X=x$ is*

$$\mathbb{E}[Y|X=x] = \sum_y yf(y|x)$$

given that the conditional PMF is well-defined. We generally write $\mathbb{E}[Y|X]$ or $\mathbb{E}[Y|x]$.

An important result associated to conditional expectations is as follows:

Theorem 4.2.15. *The conditional expectation satisfies:*

$$\mathbb{E}[\mathbb{E}[Y|X]] = \mathbb{E}[Y]$$

assuming the expectations all exist.

Proof. We have

$$\begin{aligned}
\mathbb{E}[Y] &= \sum_y y f(y) \\
&= \sum_{(x,y)\in \mathsf{Z}} y f(x,y) \\
&= \sum_{(x,y)\in \mathsf{Z}} y f(y|x) f(x) \\
&= \sum_x [\sum_y y f(y|x)] f(x) \\
&= \mathbb{E}[\mathbb{E}[Y|X]].
\end{aligned}$$

□

Example 4.2.16. *Let us return to Example 4.2.13. Find* $\mathbb{E}[X|Y]$, $\mathbb{E}[Y|X]$ *and* $\mathbb{E}[X]$. *From Exercise 2.6.13, you should have derived that if* $Z \sim \mathcal{B}(n,p)$ *then* $\mathbb{E}[Z] = np$. *Then*

$$\mathbb{E}[X|Y=y] = yp.$$

Thus

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X|Y=y]] = p\mathbb{E}[Y].$$

As $Y \sim \mathcal{P}(\lambda)$

$$\mathbb{E}[X] = p\lambda.$$

From Example 4.2.13,

$$f(y|x) = \frac{(\lambda(1-p))^{y-x} e^{-\lambda(1-p)}}{(y-x)!} \quad y \in \{x, x+1, \dots\}.$$

Thus

$$\mathbb{E}[Y|X=x] = \sum_{y=x}^{\infty} y \frac{(\lambda(1-p))^{y-x} e^{-\lambda(1-p)}}{(y-x)!}.$$

Setting $u = y - x$ *in the summation, one yields*

$$\begin{aligned}
\mathbb{E}[Y|X=x] &= \sum_{u=0}^{\infty} (u+x) \frac{(\lambda(1-p))^u e^{-\lambda(1-p)}}{u!} \\
&= \sum_{u=0}^{\infty} u \frac{(\lambda(1-p))^u e^{-\lambda(1-p)}}{u!} + x \\
&= \lambda(1-p) + x.
\end{aligned}$$

Here, we have used the fact that $U \sim \mathcal{P}(\lambda(1-p))$.

We end the Section with a result which can be very useful in practice.

Theorem 4.2.17. *We have, for any* $g : \mathbb{R} \to \mathbb{R}$

$$\mathbb{E}[\mathbb{E}[Y|X] g(X)] = \mathbb{E}[Y g(X)]$$

assuming the expectations all exist.

Proof. We have

$$\begin{aligned}
\mathbb{E}[Y g(X)] &= \sum_{(x,y)\in \mathsf{Z}} y g(x) f(x,y) \\
&= \sum_{(x,y)\in \mathsf{Z}} y g(x) f(y|x) f(x) \\
&= \sum_x g(x) [\sum_y y f(y|x)] f(x) \\
&= \mathbb{E}[\mathbb{E}[Y|X] g(X)].
\end{aligned}$$

□

Example 4.2.18. *Let $X_1, X_2, \dots$ be a possibly infinite sequence of independent and identically distributed discrete random variables, $X_i \in \mathsf{X}$, with mean $\mu = \mathbb{E}[X_1]$. Indepedently suppose $N \in \{1, 2, \dots\}$ is a random variable with finite mean. Find*

$$\mathbb{E}\Big[\sum_{i=1}^{N} X_i\Big].$$

This particular problem can be dealt with using conditional expectations. We begin by conditioning on $N = n$ for some fixed $n \in \{1, \dots, n\}$. This is useful as suppose we were only interested in

$$\mathbb{E}\Big[\sum_{i=1}^{n} X_i\Big].$$

with n known. Then

$$\begin{aligned}
\mathbb{E}\Big[\sum_{i=1}^{n} X_i\Big] &= \mathbb{E}[X_1] + \mathbb{E}\Big[\sum_{i=2}^{n} X_i\Big] \\
&= \mathbb{E}[X_1] + \mathbb{E}[X_2] + \mathbb{E}\Big[\sum_{i=3}^{n} X_i\Big] \\
&= \sum_{i=1}^{n} \mathbb{E}[X_i].
\end{aligned}$$

We know that $\mathbb{E}[X_1] = \mu$ and since the random variables are identically distributed $\mathbb{E}[X_i] = \mu$ for any $i \in \{2, 3, \dots\}$. Hence

$$\mathbb{E}\Big[\sum_{i=1}^{n} X_i\Big] = n\mu.$$

Thus, we have

$$\mathbb{E}\Big[\sum_{i=1}^{N} X_i \Big| N = n\Big] = \sum_{i=1}^{n} \mathbb{E}[X_i] = n\mu.$$

So, it then follows that

$$\begin{aligned}
\mathbb{E}\Big[\sum_{i=1}^{N} X_i\Big] &= \mathbb{E}\Big[\mathbb{E}\Big[\sum_{i=1}^{N} X_i \Big| N = n\Big]\Big] \\
&= \mathbb{E}[N\mu] \\
&= \mu\mathbb{E}[N].
\end{aligned}$$

Example 4.2.19. *Suppose that $Y \sim \mathcal{N}e(n, p)$ for some known $p \in (0, 1)$ and fixed $n > 1$. In addition that $X|Y = y \sim \mathcal{B}(y, p)$. Then, we have that the joint PMF is*

$$f(x, y) = f(x|y)f(y) = \binom{y}{x} p^x (1-p)^{y-x} \binom{y-1}{n-1} p^n (1-p)^{y-n} \quad (x, y) \in \mathsf{Z}$$

where $\mathsf{Z} = \{(x, y) : y \in \{n, n+1, \dots\}, x \in \{0, \dots, y\}\}$. Find the moment generating function of X.

This problem can be solved by considering the conditional *moment generating function of $X|Y$. We have*

$$\begin{aligned}
M_X(t) &= \mathbb{E}[e^{Xt}] \\
&= \mathbb{E}[\mathbb{E}[e^{Xt}|Y = y]].
\end{aligned}$$

Now, as $X|Y = y$ is a binomial distribution, we know from the tables in the notes (just set $n = y$ in the table) that

$$M_{X|y}(t) = ((1-p) + pe^t)^y.$$

Thus, we have that

$$\begin{aligned}
M_X(t) &= \mathbb{E}[\mathbb{E}[e^{Xt}|Y = y]] \\
&= \mathbb{E}[M_{X|Y}(t)] \\
&= \mathbb{E}[((1-p) + pe^t)^Y].
\end{aligned}$$

Now, we can deal with the right hand side using the defintion of an expectation of a function of a random variable (we will set $\alpha = (1-p) + pe^t$ in the forthcoming calculations):

$$\begin{aligned}
\mathbb{E}[((1-p)+pe^t)^Y] &= \sum_{y=n}^{\infty}((1-p)+pe^t)^y \binom{y-1}{n-1} p^n (1-p)^{y-n} \\
&= \frac{1}{(1-p)^n} \sum_{y=n}^{\infty} \binom{y-1}{n-1} p^n [(1-p)\alpha]^y \\
&= (\alpha p)^n \sum_{y=n}^{\infty} \binom{y-1}{n-1} [(1-p)\alpha]^{y-n}.
\end{aligned}$$

Now if t is such that

$$(1-p)\alpha < 1$$

that is

$$\mathsf{T} = \Big\{ t : t < \log\Big(\frac{1}{p}\Big(\frac{1}{1-p} - (1-p)\Big)\Big)\Big\}$$

(rearrange the condition $(1-p)\alpha < 1$) then we can proceed as follows. Set $\rho = (1-p)\alpha$, then we have

$$\begin{aligned}
\mathbb{E}[((1-p)+pe^t)^Y] &= (\alpha p)^n \sum_{y=n}^{\infty} \binom{y-1}{n-1} \rho^{y-n} \\
&= \Big(\frac{\alpha p}{1-\rho}\Big)^n \sum_{y=n}^{\infty} \binom{y-1}{n-1} (1-\rho)^n \rho^{y-n} \\
&= \Big(\frac{\alpha p}{1-\rho}\Big)^n.
\end{aligned}$$

To go to the last line, we remark that the summation is that of a PMF of a $\mathcal{N}e(n, 1-\rho)$ random variable, so it is 1. Then, recalling that $\alpha = (1-p) + pe^t$ and $\rho = (1-p)\alpha$, we have

$$M_X(t) = \Big(\frac{((1-p)+pe^t)p}{1-(1-p)[(1-p)+pe^t]}\Big)^n \quad t \in \mathsf{T}.$$

4.2.3 Further Facts about Conditional Expectations

Just as for ordinary expectations, we have the following properties of conditional expectations. For X, Y, Z jointly defined discrete random variables and a, b finite real constants, we have:

- $\mathbb{E}[a|X] = a$.
- $\mathbb{E}[aX + bY|Z] = a\mathbb{E}[X|Z] + b\mathbb{E}[Y|Z]$.
- If $X \geq 0$ then $\mathbb{E}[X|Y] \geq 0$.
- If X and Y are independent then $\mathbb{E}[X|Y] = \mathbb{E}[X]$.

In particular, one can also define the conditional variance:

$$\mathbb{V}\mathrm{ar}[X|Y] = \mathbb{E}[(X - \mathbb{E}[X|Y])^2|Y] = \mathbb{E}[X^2|Y] - \mathbb{E}[X|Y]^2.$$

Example 4.2.20. *We return to Example 4.2.13: recall $Y \sim \mathcal{P}(\lambda)$, $X|Y = y \sim \mathcal{B}(n, p)$. We had shown that $X \sim \mathcal{P}(\lambda p)$. We will compute:*

$$\mathbb{E}[X], \mathbb{E}[X|Y], \mathbb{E}[Y|X], \mathbb{E}[X^2|Y], \mathbb{V}ar[X|Y].$$

From lectures we know that if $Z \sim \mathcal{P}(\phi)$ then $\mathbb{E}[Z] = \phi$. So $\mathbb{E}[X] = \lambda p$. Again, from an exercise in lectures, if $Z \sim \mathcal{B}(n, p)$ then $\mathbb{E}[Z] = np$. So $\mathbb{E}[X|Y = y] = yp$. For $\mathbb{E}[Y|X]$, we had computed it in lectures; $\mathbb{E}[Y|X = x] = \lambda(1-p) + x$.

Now, for $\mathbb{E}[X^2|Y]$, we have

$$\begin{aligned}\mathbb{E}[X^2|Y=y] &= \sum_x x^2 f(x|y)\\ &= \sum_{x=0}^{y} x^2 \binom{y}{x} p^x (1-p)^{y-x}\\ &= (1-p)^y \sum_{x=0}^{y} x^2 \binom{y}{x} \Big(\frac{p}{1-p}\Big)^x.\end{aligned}$$

Now set $u = p/(1-p)$ and note that

$$x^2 u^{x-2} = \frac{d^2}{du^2}[u^x] + x u^{x-2}.$$

Thus:

$$\mathbb{E}[X^2|Y=y] = (1-p)^y u^2 \sum_{x=0}^{y} \binom{y}{x} \Big[\frac{d^2}{du^2}[u^x] + x u^{x-2}\Big]. \tag{4.2.4}$$

Now

$$(1-p)^y u^2 \sum_{x=0}^{y} \binom{y}{x} x u^{x-2} = yp \tag{4.2.5}$$

this is because it is the expectation of a binomial random variable with parameters y, p. So we focus on computing

$$\begin{aligned}\sum_{x=0}^{y} \binom{y}{x} \frac{d^2}{du^2}[u^x] &= \frac{d^2}{du^2}\Big[\sum_{x=0}^{y} \binom{y}{x} u^x\Big]\\ &= \frac{d^2}{du^2}(1+u)^y\\ &= y(y-1)(1+u)^{y-2}.\end{aligned}$$

Returning to (4.2.4), noting (4.2.5) and recalling $u = p/(1-p)$, we have

$$\begin{aligned}\mathbb{E}[X^2|Y=y] &= (1-p)^y u^2 y(y-1)(1+u)^{y-2} + yp\\ &= (1-p)^y \Big(\frac{p}{1-p}\Big)^2 y(y-1)\Big(\frac{1}{1-p}\Big)^{y-2} + yp\\ &= p^2 y(y-1) + yp.\end{aligned}$$

Then

$$\mathbb{V}ar[X|Y=y] = \mathbb{E}[X^2|Y=y] - \mathbb{E}[X|Y=y]^2 = p^2 y(y-1) + yp - (yp)^2 = yp(1-p)$$

Example 4.2.21. *A factory has produced n robots and, independently, each robot is faulty with probability $p \in (0,1)$. For each robot a test is applied which (independently for each robot) detects a fault if present, with probability $\delta \in (0,1)$. Let X be the number of faulty robots and Y the number detected as faulty. Show that*

$$\mathbb{E}[X|Y=y] = \frac{1}{1-p\delta}\Big\{np(1-\delta) + (1-p)y\Big\}.$$

From the description of the problem, we can deduce that

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x} \quad x \in \{0, \ldots, n\}.$$

That is, there are n robots and independently and indentically, each one is faulty with probability p; this is a description of a Binomial random variable. Now conditionally on knowing X, Y must also be Binomial; i.e. $Y|X = x \sim \mathcal{B}(x, \delta)$. That is, if x robots are faulty, independently and identically, the fault is detected with probability δ. So we know that

$$f(x,y) = f(y|x)f(x) = \binom{x}{y} \delta^y (1-\delta)^{x-y} \binom{n}{x} p^x (1-p)^{n-x} \quad (x,y) \in \mathsf{Z}$$

where $\mathsf{Z} = \{(x,y) : x \in \{0,\dots,n\}, y \in \{0,\dots,x\}\}$. *There are many ways we could proceed to solve this question, but I will use the definitions of conditional expectations, instead of using tricks such as the MGFs. We first note that via Bayes rule:*

$$f(x|y) = \frac{f(x,y)}{f(y)}.$$

So to calculate $\mathbb{E}[X|Y=y]$, *we must first know this conditional PMF.*

We have that, for $y \in \{0,\dots,n\}$

$$\begin{aligned} f(y) &= \sum_x f(x,y) \\ &= \sum_{x=y}^{n} \binom{x}{y} \delta^y (1-\delta)^{x-y} \binom{n}{x} p^x (1-p)^{n-x} \\ &= \Big(\frac{\delta}{1-\delta}\Big)^y (1-p)^n \sum_{x=y}^{n} \binom{x}{y}\binom{n}{x} \Big(\frac{p(1-\delta)}{1-p}\Big)^x \end{aligned}$$

Now, let $\alpha = \frac{p(1-\delta)}{1-p}$ *and set* $u = x - y$ *in the summation; then we have*

$$\begin{aligned} f(y) &= \Big(\frac{\delta}{1-\delta}\Big)^y (1-p)^n \sum_{u=0}^{n-y} \binom{u+y}{y}\binom{n}{u+y} \alpha^{u+y} \\ &= \Big(\frac{\delta}{1-\delta}\Big)^y (1-p)^n \alpha^y \sum_{u=0}^{n-y} \binom{n-y}{u}\binom{n}{y} \alpha^{u} \\ &= \Big(\frac{\delta}{1-\delta}\Big)^y (1-p)^n \alpha^y \binom{n}{y} \sum_{u=0}^{n-y} \binom{n-y}{u} \alpha^{u} \\ &= \Big(\frac{\delta}{1-\delta}\Big)^y (1-p)^n \alpha^y \binom{n}{y} (1+\alpha)^{n-y}. \end{aligned}$$

Here, we have used the definition of the Binomial coefficient

$$\binom{n}{x} = \frac{n!}{(n-x)!x!}$$

to go from the first to second line and the binomial series expansion to go from the fourth to fifth line. Now, recalling the definition of α, *we have*

$$\begin{aligned} f(y) &= \binom{n}{y}\Big(\frac{\delta}{1-\delta}\Big)^y (1-p)^n \Big(\frac{p(1-\delta)}{1-p}\Big)^y \Big(\frac{1-\delta p}{1-p}\Big)^{n-y} \\ &= \binom{n}{y} (\delta p)^y (1-\delta p)^{n-y} \quad y \in \{0,\dots,n\}. \end{aligned}$$

So marginally $Y \sim \mathcal{B}(n, \delta p)$.

Now, we can compute the desired conditional expectation, for $y \in \{0,\dots,n\}$:

$$\begin{aligned} \mathbb{E}[X|Y=y] &= \sum_x x f(x|y) \\ &= \frac{1}{f(y)} \sum_{x=y}^{n} x \binom{x}{y} \delta^y (1-\delta)^{x-y} \binom{n}{x} p^x (1-p)^{n-x} \\ &= \frac{1}{f(y)} \Big(\frac{\delta}{1-\delta}\Big)^y (1-p)^n \sum_{x=y}^{n} x \binom{x}{y}\binom{n}{x} \Big(\frac{p(1-\delta)}{1-p}\Big)^x \\ &= \frac{1}{f(y)} \Big(\frac{\delta}{1-\delta}\Big)^y (1-p)^n \sum_{u=0}^{n-y} (u+y) \binom{u+y}{y}\binom{n}{u+y} \alpha^{u+y}. \end{aligned} \tag{4.2.6}$$

To go to the final line, we have set $u = x - y$ *in the summation and recalled the definition of* α. *At this stage, we will focus on the summation:*

$$\sum_{u=0}^{n-y} u \binom{u+y}{y}\binom{n}{u+y} \alpha^u$$

as we know how to do the summation

$$\sum_{u=0}^{n-y} y\binom{u+y}{y}\binom{n}{u+y}\alpha^u = \binom{n}{y}y(1+\alpha)^{n-y} \tag{4.2.7}$$

as this was part of the above calculations. We have

$$\begin{aligned}
\sum_{u=0}^{n-y} u\binom{u+y}{y}\binom{n}{u+y}\alpha^u &= \sum_{u=0}^{n-y} u\binom{n-y}{u}\binom{n}{y}\alpha^u \\
&= \alpha\binom{n}{y}\sum_{u=0}^{n-y}\binom{n-y}{u}u\alpha^{u-1} \\
&= \alpha\binom{n}{y}\sum_{u=0}^{n-y}\binom{n-y}{u}\frac{d}{d\alpha}(\alpha^u) \\
&= \alpha\binom{n}{y}\frac{d}{d\alpha}\sum_{u=0}^{n-y}\binom{n-y}{u}\alpha^u \\
&= \alpha\binom{n}{y}\frac{d}{d\alpha}[(1+\alpha)^{n-y}] \\
&= \alpha\binom{n}{y}(n-y)(1+\alpha)^{n-y-1}.
\end{aligned}$$

Returning to (4.2.6) and combining (4.2.7) with the above result, we have

$$\begin{aligned}
\mathbb{E}[X|Y=y] &= \frac{1}{f(y)}\Big(\frac{\delta}{1-\delta}\Big)^y(1-p)^n\alpha^y\binom{n}{y}\Big[y(1+\alpha)^{n-y}+\alpha(n-y)(1+\alpha)^{n-y-1}\Big] \\
&= \frac{1}{f(y)}\Big(\frac{\delta}{1-\delta}\Big)^y(1-p)^n\alpha^y\binom{n}{y}(1+\alpha)^{n-y-1}[y+\alpha n].
\end{aligned}$$

Now, recalling the definition of α and the form of $f(y)$, we have

$$\begin{aligned}
\mathbb{E}[X|Y=y] &= \Big(\frac{\delta}{1-\delta}\Big)^y(1-p)^n\Big(\frac{p(1-\delta)}{1-p}\Big)^y\binom{n}{y}\Big(\frac{1-\delta p}{1-p}\Big)^{n-y-1}\times \\
&\quad \Big[y+\frac{p(1-\delta)}{1-p}n\Big]\frac{1}{\binom{n}{y}(\delta p)^y(1-\delta p)^{n-y}} \\
&= \frac{(1-p)(\delta p)^y(1-\delta p)^{n-y-1}(y+\frac{np(1-\delta)}{1-p})}{(\delta p)^y(1-\delta p)^{n-y}} \\
&= \frac{1-p}{1-\delta p}\Big(y+\frac{np(1-\delta)}{1-p}\Big) \\
&= \frac{1}{1-p\delta}\Big\{np(1-\delta)+(1-p)y\Big\}
\end{aligned}$$

which was the desired result.

Reading 4.2.22. *We will now consider the multinomial distribution as described in [1, pp. 332-334]. Please go to these pages in the book and we will expand upon the discussion there.*

4.3 Continuous Case

Just as for discrete random variables, one can consider the idea of joint distributions for continuous random variables.

Definition 4.3.1. *The **joint distribution function** of X and Y is the function $F : \mathbb{R}^2 \to [0,1]$ given by*

$$F(x,y) = \mathbb{P}(X \le x, Y \le y).$$

Again, as for discrete random variables, one requires a PDF:

Definition 4.3.2. *The random variables are jointly continuous with joint PDF* $f : \mathbb{R}^2 \to [0, \infty)$ *if*

$$F(x, y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f(u, v) du dv$$

for each $(x, y) \in \mathbb{R}^2$.

In this course, we will assume (generally) that

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y).$$

Note that, $\mathbb{P}(X = x, Y = y) = 0$ and for sufficiently well defined sets $A \subseteq \mathsf{Z} \subseteq \mathbb{R}^2$

$$\mathbb{P}((X, Y) \in A) = \int_A f(x, y) dx dy$$

note that the R.H.S. is a double integral, but we will only write one integral sign in such contexts.

As for discrete random variables, we can introduce the idea of marginal PDFs. Here, we take a little longer to consider these ideas:

Definition 4.3.3. *The* ***marginal distribution functions*** *of* X *and* Y *are*

$$F(x) = \lim_{y \to \infty} F(x, y) \quad F(y) = \lim_{x \to \infty} F(x, y).$$

As

$$F(x) = \int_{-\infty}^{x} \int_{-\infty}^{\infty} f(u, y) dy du \quad F(y) = \int_{-\infty}^{y} \int_{-\infty}^{\infty} f(x, u) dx du$$

the ***marginal density functions*** *of* X *and* Y

$$f(x) = \int_{-\infty}^{\infty} f(x, y) dy \quad f(y) = \int_{-\infty}^{\infty} f(x, y) dx.$$

We remark that expectation is much the same for joint distributions (as for Section 3.4) of continuous random variables:

$$\mathbb{E}[g(X, Y)] = \int_{\mathsf{Z}} g(x, y) f(x, y) dx dy = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dx dy.$$

So as before if $g(x, y) = ah(x) + bg(y)$

$$\mathbb{E}[g(X, Y)] = a\mathbb{E}[h(X)] + b\mathbb{E}[g(Y)].$$

We now turn to independence; we state the following result with no proof. If you are unconvinced, simply take the below as a definition.

Theorem 4.3.4. *The random variables* X *and* Y *are independent if and only if*

$$F(x, y) = F(x)F(y)$$

or equivalently

$$f(x, y) = f(x)f(y).$$

Example 4.3.5. *Let* $\mathsf{Z} = (\mathbb{R}^+)^2 = \mathsf{X} \times \mathsf{Y}$ *and*

$$f(x, y) = \lambda^2 e^{-\lambda(x+y)} \quad (x, y) \in \mathsf{Z}, \lambda > 0.$$

Then one has

$$f(x) = \lambda e^{-\lambda x} \; x \in \mathsf{X} \quad f(y) = \lambda e^{-\lambda y} \; y \in \mathsf{Y}$$

In addition, what is the probability that $X > Y$?

$$\begin{aligned}
\mathbb{P}(X > Y) &= \int_0^\infty \int_0^x f(x,y)dydx \\
&= \int_0^\infty \int_0^x \lambda e^{-\lambda y} dy \lambda e^{-\lambda x} dx \\
&= \int_0^\infty \Big[-e^{-\lambda y} \Big]_0^x \lambda e^{-\lambda x} dx \\
&= \int_0^\infty (1 - e^{-\lambda x}) \lambda e^{-\lambda x} dx \\
&= \Big[-e^{-\lambda x} + \frac{1}{2} e^{-2\lambda x} \Big]_0^\infty \\
&= 1 - \frac{1}{2} = \frac{1}{2}.
\end{aligned}$$

Again, one has the idea of covariance and correlation for continuous valued random variables:

$$\begin{aligned}
\mathbb{C}\mathrm{ov}[X,Y] &= \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] \\
&= \int_{\mathsf{Z}} xyf(x,y)dxdy - \int xf(x)dx \int yf(y)dy \\
\rho(X,Y) &= \frac{\mathbb{C}\mathrm{ov}[X,Y]}{\sqrt{\mathbb{V}\mathrm{ar}[X]\mathbb{V}\mathrm{ar}[Y]}}
\end{aligned}$$

Example 4.3.6. *Let $\mathsf{Z} = \mathbb{R}^2$ and define*

$$f(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\{-\frac{1}{2(1-\rho^2)}(x^2 + y^2 - 2xy\rho)\} \quad (x,y) \in \mathsf{Z} \ \rho \in [-1,1].$$

Let us check if this is a PDF; clearly it is non-negative on Z. So

$$\begin{aligned}
\int_{\mathsf{Z}} f(x,y)dxdy &= \int_{-\infty}^\infty \int_{-\infty}^\infty \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\{-\frac{1}{2(1-\rho^2)}(x^2 + y^2 - 2xy\rho)\}dxdy \\
&= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^\infty \exp\{-\frac{1}{2(1-\rho^2)}(y^2 - y^2\rho^2)\} \int_{-\infty}^\infty \exp\{-\frac{1}{2(1-\rho^2)}(x - y\rho)^2\}dxdy \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty \exp\{-\frac{1}{2(1-\rho^2)}(y^2 - y^2\rho^2)\}dy \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty \exp\{-\frac{1}{2(1-\rho^2)}(1-\rho^2)y^2\}dy = 1.
\end{aligned}$$

Here we have completed the square in the integral in x and used the change of variable $u = (x - \rho y)/(1-\rho^2)^{1/2}$. Thus $f(x,y)$ defines a joint PDF. Moreover, it is called the standard bi-variate normal distribution. Now let us consider the marginal PDFs:

$$\begin{aligned}
f(x) &= \int_{-\infty}^\infty \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\{-\frac{1}{2(1-\rho^2)}(x^2 + y^2 - 2xy\rho)\}dy \quad x \in \mathbb{R} \\
&= \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\{-\frac{1}{2(1-\rho^2)}(x^2 - x^2\rho^2)\} \int_{-\infty}^\infty \exp\{-\frac{1}{2(1-\rho^2)}(y - x\rho)^2\}dy \\
&= \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.
\end{aligned}$$

That is $X \sim \mathcal{N}(0,1)$. By Similar arguments, $Y \sim \mathcal{N}(0,1)$. Now, let us consider:

$$\begin{aligned}
\mathbb{E}[XY] &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\{-\frac{1}{2(1-\rho^2)}(x^2+y^2-2xy\rho)\}dxdy \\
&= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} y \exp\{-\frac{1}{2(1-\rho^2)}(y^2-y^2\rho^2)\} \int_{-\infty}^{\infty} x \exp\{-\frac{1}{2(1-\rho^2)}(x-y\rho)^2\}dxdy \\
&= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} y \exp\{-\frac{1}{2(1-\rho^2)}(y^2-y^2\rho^2)\} \int_{-\infty}^{\infty} [u(1-\rho^2)^{1/2}+\rho y](1-\rho^2)^{1/2} e^{-u^2/2}dudy \\
&= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} y \exp\{-\frac{1}{2(1-\rho^2)}(y^2-y^2\rho^2)\}[(1-\rho^2)^{1/2}\{(1-\rho^2)^{1/2}\sqrt{2\pi}\mathbb{E}[X]+\rho y\sqrt{2\pi}\}]dy \\
&= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} y \exp\{-\frac{1}{2(1-\rho^2)}(y^2-y^2\rho^2)\}(1-\rho^2)^{1/2}\rho y\sqrt{2\pi}dy \\
&= \frac{\rho}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y^2 e^{-y^2/2}dy = \rho.
\end{aligned}$$

Here we have used the results in Examples 3.4.3 and 3.4.7 that the first and second (raw) moment of $\mathcal{N}(\mu,\sigma^2)$ are μ and $\mu^2+\sigma^2$. Therefore

$$\begin{aligned}
\mathbb{C}\mathrm{ov}[X,Y] &= \rho \\
\rho(X,Y) &= \frac{\rho}{\sqrt{1}} = \rho.
\end{aligned}$$

Note also that if $\rho = 0$ then:

$$f(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\{-\frac{1}{2}(x^2+y^2)\} \quad (x,y) \in \mathsf{Z}.$$

so as

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \ x \in \mathbb{R} \quad f(y) = \frac{1}{\sqrt{2\pi}} e^{-y^2/2} \ y \in \mathbb{R}$$

$$f(x,y) = f(x)f(y).$$

That is standard bi-variate normal random variables are independent if and only they are uncorrelated. Note that this does not always occur for other random variables.

We conclude this section with a mention of joint moment generating functions:

Definition 4.3.7. *For a continuous random variables X and Y the* ***joint moment generating function*** *is*

$$M(t_1,t_2) = \mathbb{E}[e^{Xt_1+Yt_2}] = \int_{\mathsf{Z}} e^{xt_1+yt_2} f(x,y)dxdy \quad (t_1,t_2) \in \mathsf{T}$$

where T is the set of t_1, t_2 for which $\int_{\mathsf{Z}} e^{xt_1+yt_2} f(x,y)dxdy < \infty$.

Exercise 4.3.8. *Show that*

$$\mathbb{E}[X] = \frac{\partial}{\partial t_1} M(t_1,t_2)\Big|_{t_1=t_2=0}, \quad \mathbb{E}[Y] = \frac{\partial}{\partial t_2} M(t_1,t_2)\Big|_{t_1=t_2=0}, \quad \mathbb{E}[XY] = \frac{\partial^2}{\partial t_1 \partial t_2} M(t_1,t_2)\Big|_{t_1=t_2=0}$$

when the derivatives exist.

4.3.1 Conditional Distributions and Expectations

The idea of conditional distributions and expectations can be further extended to continuous random variables. However, we recall that the event $\{X = x\}$ has zero probability, so how can we condition on it? It turns out that we cannot have a rigorous answer to this question (i.e. at the level of mathematics in this course), so some faith must be extended towards the following definitions.

Definition 4.3.9. *The* ***conditional distributions function*** *of Y given $X = x$ is the function:*

$$F(y|x) = \int_{-\infty}^{y} \frac{f(x,v)}{f(x)} dv$$

for any x such that $f(x) > 0$.

Definition 4.3.10. *The* ***conditional density function*** *of $F(y|x)$, written $f(y|x)$ is given by:*

$$f(y|x) = \frac{f(x,y)}{f(x)}$$

for any x such that $f(x) > 0$.

We remark that clearly

$$f(y|x) = \frac{f(x,y)}{\int_{-\infty}^{\infty} f(x,y)dy}.$$

Example 4.3.11. *Suppose that $Y|X = x \sim \mathcal{N}(x,1)$ and $X \sim \mathcal{N}(0,1)$, $\mathsf{Z} = \mathbb{R}^2$. Find the PDFs: $f(x,y)$ and $f(x|y)$. We have*

$$f(x,y) = f(y|x)f(x)$$

so

$$f(x,y) = \frac{1}{2\pi}\exp\{-\frac{1}{2}((y-x)^2 + x^2)\} \quad (x,y) \in \mathsf{Z}.$$

Now

$$f(x|y) = \frac{f(x,y)}{f(y)} = \frac{f(y|x)f(x)}{f(y)}$$

which is Bayes theorem *for continuous random-variables. Now for $y \in \mathbb{R}$*

$$\begin{aligned}
f(y) &= \int_{-\infty}^{\infty} \frac{1}{2\pi}\exp\{-\frac{1}{2}((y-x)^2 + x^2)\}dx \\
&= \frac{1}{2\pi}\exp\{-y^2/2\}\int_{-\infty}^{\infty}\exp\{-\frac{1}{2}(2x^2 - 2xy)\}dx \\
&= \frac{1}{2\pi}\exp\{-y^2/2\}\int_{-\infty}^{\infty}\exp\{-[(x-y/2)^2 - y^2/4]\}dx \\
&= \frac{1}{2\pi}\exp\{-y^2/4\}\int_{-\infty}^{\infty} e^{-u^2/2}\frac{1}{\sqrt{2}}du \\
&= \frac{1}{2\pi}\exp\{-y^2/4\}\frac{\sqrt{2\pi}}{\sqrt{2}} \\
&= \frac{1}{\sqrt{2}\sqrt{2\pi}}\exp\{-y^2/4\}.
\end{aligned}$$

So $Y \sim \mathcal{N}(0,2)$. Then:

$$f(x|y) = \frac{\frac{1}{2\pi}\exp\{-\frac{1}{2}((y-x)^2 + x^2)\}}{\frac{1}{\sqrt{2}\sqrt{2\pi}}\exp\{-y^2/4\}} \quad x \in \mathbb{R}.$$

After some calculations (exercise) one obtains:

$$f(x|y) = \sqrt{\frac{2}{2\pi}}\exp\{-(x - \frac{y}{2})^2\} \quad x \in \mathbb{R}$$

that is $X|Y = y \sim \mathcal{N}(y/2, 1/2)$.

Example 4.3.12. *Let X and Y have joint PDF:*

$$f(x,y) = \frac{1}{x} \quad (x,y) \in \mathsf{Z} = \{(x,y) : 0 \le y \le x \le 1\}.$$

Then for $x \in [0,1]$

$$f(x) = \int_0^x \frac{1}{x}dy = 1.$$

In addition, $(x,y) \in \mathsf{Z}$:

$$f(y|x) = \frac{1/x}{1} = \frac{1}{x}$$

that is if $x \in [0,1]$, then conditional upon x $Y \sim \mathcal{U}_{[0,x]}$ (the uniform distribution on $[0,x]$). Now, suppose we are interested in $\mathbb{P}(X^2+Y^2 \leq 1|X=x)$. Let $x \geq 0$, and define

$$A(x) := \{y \in \mathbb{R} : 0 \leq y \leq x, x^2+y^2 \leq 1\}$$

that is, $A(x) = [0, x \wedge (1-x^2)^{1/2}]$, where $a \wedge b = \min\{a,b\}$. Thus ($x \in [0,1]$)

$$\begin{aligned}\mathbb{P}(X^2+Y^2 \leq 1|X=x) &= \int_{A(x)} f(y|x)dy \\ &= \int_0^{x\wedge(1-x^2)^{1/2}} \frac{1}{x}dy \\ &= \frac{x \wedge (1-x^2)^{1/2}}{x}.\end{aligned}$$

To compute $\mathbb{P}(X^2+Y^2 \leq 1)$, we note, by setting $A = \{(x,y) : (x,y) \in \mathsf{Z}, x^2+y^2 \leq 1\}$, we are interested in $\mathbb{P}(A)$, which is:

$$\begin{aligned}\mathbb{P}(X^2+Y^2 \leq 1) &= \int_A f(x,y)dydx \\ &= \int_0^1 \int_{A(x)} f(y|x)dyf(x)dx \\ &= \int_0^1 \frac{x \wedge (1-x^2)^{1/2}}{x}dx.\end{aligned}$$

Now $x^2 \leq (1-x^2)^{1/2}$ if

$$x \leq 1/\sqrt{2}$$

hence

$$\mathbb{P}(X^2+Y^2 \leq 1) = \int_0^{1/\sqrt{2}} dx + \int_{1/\sqrt{2}}^1 \frac{(1-x^2)^{1/2}}{x}dx. \tag{4.3.1}$$

Now we have, setting $x = \cos(\theta)$

$$\begin{aligned}\int_{1/\sqrt{2}}^1 \frac{(1-x^2)^{1/2}}{x}dx &= \int_0^{\pi/4} \sin(\theta)\tan(\theta)d\theta \\ &= \Big[\log(|\sec(\theta)+\tan(\theta)|) - \sin(\theta)\Big]_0^{\pi/4} \\ &= \log(1+\sqrt{2}) - \frac{1}{\sqrt{2}}\end{aligned}$$

where we have used integration tables to obtain the integral[1]*. Thus, returning to (4.3.1):*

$$\mathbb{P}(X^2+Y^2 \leq 1) = \frac{1}{\sqrt{2}} + \log(1+\sqrt{2}) - \frac{1}{\sqrt{2}} = \log(1+\sqrt{2}).$$

Given the previous discussion, the notion of a conditional expectation for continuous random variables:

$$\mathbb{E}[g(Y)|X=x] = \int g(y)f(y|x)dy.$$

As for discrete-valued random variables, we have the following result, which is again not proved.

Theorem 4.3.13. *Consider jointly continuous random variables X and Y with $g, h : \mathbb{R} \to \mathbb{R}$ with $h(X)$, $g(Y)$ continuous. Then:*

$$\mathbb{E}[h(X)g(Y)] = \mathbb{E}[\mathbb{E}[g(Y)|X]h(X)] = \int \Big(\int g(y)f(y|x)dy\Big)h(x)f(x)dx$$

whenever the expectations exist.

[1] you will not be expected to perform such an integration in the examination

Example 4.3.14. *Let us return to Example 4.3.11. Then*

$$\begin{aligned}\mathbb{E}[XY] &= \mathbb{E}[Y\mathbb{E}[X|Y]] \\ &= \mathbb{E}[Y(Y/2)] \\ &= \frac{1}{2}[2] = 1.\end{aligned}$$

Thus as $\mathbb{V}\mathrm{ar}[X] = 1$ *and* $\mathbb{V}\mathrm{ar}[Y] = 2$

$$\rho(X,Y) = \frac{1}{\sqrt{2}}.$$

Example 4.3.15. *Suppose* $Y|X = x \sim \mathcal{N}(x, \sigma_1^2)$ *and* $X \sim \mathcal{N}(\xi, \sigma_2^2)$*. To find the marginal distribution of* Y*, one can use MGFs and conditional expectations. We have:*

$$\begin{aligned}M(t) &= \mathbb{E}[e^{Yt}] \\ &= \mathbb{E}[\mathbb{E}[e^{Yt}|X]] \\ &= \mathbb{E}[e^{Xt+(\sigma_1^2t^2)/2}] \\ &= e^{(\sigma_1^2t^2)/2}\mathbb{E}[e^{Xt}] \\ &= e^{(\sigma_1^2t^2)/2}e^{\xi t+(\sigma_2^2t^2)/2} \\ &= \exp\{\xi t + (\sigma_1^2+\sigma_2^2)t^2/2\}.\end{aligned}$$

Here we have used Example 3.4.14 for the MGF of a normal distribution. Thus we can conclude that

$$Y \sim \mathcal{N}(\xi, \sigma_1^2 + \sigma_2^2).$$

4.3.2 Functions of Random Variables

Consider a random variable X and $g : \mathbb{R} \to \mathbb{R}$ a continuous and invertible function (continuity can be weakened, but let us use it for now). Suppose $Y = g(X)$; what is the distribution of Y? We have:

$$F(y) = \mathbb{P}(Y \le y) = \mathbb{P}(g(X) \le y) = \mathbb{P}(X \in g^{-1}((-\infty, y)) = \int_{g^{-1}(-\infty,y]} f(x)dx.$$

Note that for $A \subseteq \mathbb{R}$, then $g^{-1}A = \{x \in \mathbb{R} : g(x) \in A\}$.

Example 4.3.16. *Let* $X \sim \mathcal{N}(0,1)$ *and set*

$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du$$

to denote the standard normal CDF. Suppose $Y = X^2$*, then for* $y \ge 0$ *(note* $\mathsf{Y} = \mathbb{R}^+$*)*

$$\mathbb{P}(Y \le y) = \mathbb{P}(X^2 \le y) = \mathbb{P}(-\sqrt{y} \le X \le \sqrt{y}).$$

Now

$$\begin{aligned}\mathbb{P}(-\sqrt{y} \le X \le \sqrt{y}) &= \mathbb{P}(X \le \sqrt{y}) - \mathbb{P}(X \le -\sqrt{y}) \\ &= \Phi(\sqrt{y}) - \Phi(-\sqrt{y}) \\ &= \Phi(\sqrt{y}) - [1 - \Phi(\sqrt{y})] \\ &= 2\Phi(\sqrt{y}) - 1.\end{aligned}$$

Then

$$f(y) = \frac{d}{dy}F(y) = \frac{1}{\sqrt{y}}\Phi'(\sqrt{y}) = \frac{1}{\sqrt{2\pi y}}e^{-y/2} \quad y \in \mathsf{Y}$$

that is $Y \sim \mathcal{G}(1/2, 1/2)$*. This is also called the chi-squared distribution on one degree of freedom. We remark that* Y *and* X *are clearly not independent (if* X *changes so does* Y *and in a deterministic manner). Now*

$$\mathbb{E}[XY] = \mathbb{E}[X^3] = 0.$$

In addition $\mathbb{E}[X] = 0$ *and* $\mathbb{E}[Y] = 1$*. So* X *and* Y *are uncorrelated, but they are not independent.*

We now move onto a more general change of variable formula. Suppose X_1 and X_2 have a joint density on Z and we set

$$(Y_1, Y_2) = T(X_1, X_2)$$

where $T : \mathsf{Z} \to \mathsf{T}$ where T is differentiable and invertible and $\mathsf{T} \subseteq \mathbb{R}^2$. What is the joint density function of (Y_1, Y_2) on T? As T is invertible, we set

$$X_1 = T_1^{-1}(Y_1, Y_2) \quad X_2 = T_2^{-1}(Y_1, Y_2)$$

Then we define

$$J(y_1, y_2) := \begin{vmatrix} \frac{\partial T_1^{-1}}{\partial y_1} & \frac{\partial T_2^{-1}}{\partial y_1} \\ \frac{\partial T_1^{-1}}{\partial y_2} & \frac{\partial T_2^{-1}}{\partial y_2} \end{vmatrix} = \frac{\partial T_1^{-1}}{\partial y_1}\frac{\partial T_2^{-1}}{\partial y_2} - \frac{\partial T_2^{-1}}{\partial y_1}\frac{\partial T_1^{-1}}{\partial y_2}$$

to be the *Jacobian* of transformation. Then we have the following result, which simply follows from the change of variable rule for integration:

Theorem 4.3.17. *If (X_1, X_2) have joint density $f(x, y)$ on Z, then for $(Y_1, Y_2) = T(X_1, X_2)$, with T as described above, the joint density of (Y_1, Y_2), denoted g is:*

$$g(y_1, y_2) = f(T_1^{-1}(y_1, y_2), T_2^{-1}(y_1, y_2))|J(y_1, y_2)| \quad (y_1, y_2) \in \mathsf{T}.$$

Example 4.3.18. *Suppose $\mathsf{Z} = (\mathbb{R}^+)^2$ and*

$$f(x_1, x_2) = \lambda^2 e^{-\lambda(x_1+x_2)} \quad (x_1, x_2) \in \mathsf{Z}.$$

Let

$$(Y_1, Y_2) = (X_1 + X_2, X_1/X_2).$$

To find the joint density of (Y_1, Y_2) we first note that $\mathsf{T} = \mathsf{Z}$ and that

$$(X_1, X_2) = \Big(\frac{Y_1 Y_2}{1 + Y_2}, \frac{Y_1}{1 + Y_2}\Big).$$

Then the Jacobian is

$$J(y_1, y_2) := \begin{vmatrix} \frac{y_2}{1+y_2} & \frac{1}{1+y_2} \\ \frac{y_1}{(1+y_2)^2} & -\frac{y_1}{(1+y_2)^2} \end{vmatrix} = -\frac{y_1}{(1 + y_2)^2}.$$

Thus:

$$g(y_1, y_2) = \frac{y_1}{(1 + y_2)^2}\lambda^2 e^{-\lambda y_1} \quad (y_1, y_2) \in \mathsf{T}.$$

One can check that indeed Y_1 and Y_2 are independent and, that the marginal densities are:

$$g(y_1) = \lambda^2 y_1 e^{-\lambda y_1} \;\; y_1 \in \mathbb{R}^+ \quad g(y_2) = \frac{1}{(1 + y_2)^2} \;\; y_2 \in \mathbb{R}^+.$$

Example 4.3.19. *Suppose we have (X_1, X_2) as in the case of Example 4.3.18, except we have the mapping:*

$$(Y_1, Y_2) = (X_1, X_1 + X_2).$$

Now, clearly $Y_2 \geq Y_1$, so this transformation induces the support for (Y_1, Y_2) as:

$$\mathsf{T} = \{(y_1, y_2) : 0 \leq y_1 \leq y_2, y_2 \in \mathbb{R}^+\}.$$

Then

$$(X_1, X_2) = (Y_1, Y_2 - Y_1)$$

and clearly $J(y_1, y_2) = 1$; thus

$$g(y_1, y_2) = \lambda^2 e^{-\lambda y_2} \quad (y_1, y_2) \in \mathsf{T}.$$

Example 4.3.20. *Let X_1 and X_2 be independent $\mathcal{N}(0,1)$ random variables ($\mathsf{Z} = \mathbb{R}^2$) and let*

$$(Y_1, Y_2) = (X_1, \rho X_1 + \sqrt{1-\rho^2} X_2) \quad \rho \in [-1, 1]$$

Then $\mathsf{T} = \mathsf{Z}$ and

$$(X_1, X_2) = \left(Y_1, \frac{Y_2 - \rho Y_1}{\sqrt{1-\rho^2}}\right).$$

Then the Jacobian is

$$J(y_1, y_2) := \begin{vmatrix} 1 & -\frac{\rho}{\sqrt{1-\rho^2}} \\ 0 & \frac{1}{\sqrt{1-\rho^2}} \end{vmatrix} = \frac{1}{\sqrt{1-\rho^2}}.$$

Thus

$$\begin{aligned} g(y_1, y_2) &= \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\{-\frac{1}{2}(y_1^2 + (y_2 - \rho y_1)^2/(1-\rho^2)\} \quad (y_1, y_2) \in \mathsf{T} \\ &= \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\{-\frac{1}{2(1-\rho^2)}(y_1^2 + y_2^2 - 2y_1 y_2 \rho)\} \quad (y_1, y_2) \in \mathsf{T}. \end{aligned}$$

On inspection of Example 4.3.6, (Y_1, Y_2) follow a standard bivariate normal distribution.

Example 4.3.21. *Suppose that independently for $i \in \{1, 2\}$, $X_i \sim \mathcal{G}a(\alpha_i, 1)$, with $\alpha_i \in \mathbb{R}^+$. Suppose we set*

$$Y_1 = \frac{X_1}{X_1 + X_2}$$

what is the PDF of Y_1? First, we know that $Y_1 \in (0, 1)$, so the support of Y_1 is already determined. Second, consider th e change of variables:

$$\begin{aligned} Y_1 &= \frac{X_1}{X_1 + X_2} \\ Y_2 &= X_2 \end{aligned}$$

so that $\mathsf{T} = \{(y_1, y_2) : y_1 \in (0, 1), y_2 \in \mathbb{R}^+\}$. Then we have

$$(X_1, X_2) = \left(\frac{Y_1 Y_2}{(1 - Y_1)}, Y_2\right).$$

Then the Jacobian is

$$J(y_1, y_2) := \begin{vmatrix} \frac{Y_2}{(1-Y_1)^2} & \frac{Y_1}{(1-Y_1)} \\ 0 & 1 \end{vmatrix} = \frac{Y_2}{(1 - Y_1)^2}.$$

Thus

$$g(y_1, y_2) = \frac{1}{\Gamma(\alpha_1)\Gamma(\alpha_2)} y_1^{\alpha_1 - 1} y_2^{\alpha_1 + \alpha_2 - 1} \left(\frac{1}{(1 - y_1)}\right)^{\alpha_1 + 1} \exp\left\{-\frac{y_2}{(1 - y_1)}\right\} \quad (y_1, y_2) \in \mathsf{T}.$$

The marginal PDF of Y_1, for $y_1 \in (0, 1)$ is then

$$\begin{aligned} g(y_1) &= \frac{y_1^{\alpha_1 - 1}}{\Gamma(\alpha_1)\Gamma(\alpha_2)} \left(\frac{1}{(1 - y_1)}\right)^{\alpha_1 + 1} \int_0^\infty y_2^{\alpha_1 + \alpha_2 - 1} \exp\left\{-\frac{y_2}{(1 - y_1)}\right\} dy_2 \\ &= \frac{y_1^{\alpha_1 - 1}}{\Gamma(\alpha_1)\Gamma(\alpha_2)} \left(\frac{1}{(1 - y_1)}\right)^{\alpha_1 + 1} (1 - y_1)^{\alpha_1 + \alpha_2} \int_0^\infty u^{\alpha_1 + \alpha_2 - 1} \exp\{-u\} du \\ &= \frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)} y_1^{\alpha_1 - 1} (1 - y_1)^{\alpha_2 - 1}. \end{aligned}$$

That is, $Y_1 \sim \mathcal{B}e(\alpha_1, \alpha_2)$ which is called the Beta distribution.

Distributions associated to the Normal Distribution

In the following Subsection we investigate some distributions that are related to the normal distribution. They appear frequently in hypothesis testing. We note that, in a similar way to the way in which joint distributions are defined (Definition 4.3.2), one can easily extend to joint distributions of $n \geq 2$ variables (so that there is a joint distribution $F(x_1, \ldots, x_n)$ and density $f(x_1, \ldots, x_n)$).

We begin with the idea of the chi-square distribution on $n > 0$ degrees of freedom:

$$f(x) = \frac{1}{(2)^{n/2}\Gamma(n/2)} x^{n/2-1} e^{-x/2} \quad x \in \mathsf{X} = \mathbb{R}^+.$$

We write $X \sim \mathcal{X}_n^2$; you will notice that also $X \sim \mathcal{G}(n/2, 1/2)$, so that $\mathbb{E}[X] = n$. Note that from Table 6.2, we have that

$$M(t) = \frac{1}{(1-2t)^{n/2}} \quad \frac{1}{2} > t.$$

We have the following result:

Proposition 4.3.22. *Let $X_1, \ldots, X_n$ be independent and identically distributed (i.i.d.) standard normal random variables (i.e. $X_i \overset{i.i.d.}{\sim} \mathcal{N}(0,1)$). Let*

$$Z = X_1^2 + \cdots + X_n^2.$$

Then $Z \sim \mathcal{X}_n^2$.

Proof. We will use moment generating functions. We have

$$\begin{aligned} M(t) &= \mathbb{E}[e^{Zt}] \\ &= \mathbb{E}[e^{t\sum_{i=1}^n X_i^2}] \\ &= \prod_{i=1}^n \mathbb{E}[e^{tX_i^2}] \\ &= \mathbb{E}[e^{X_1^2 t}]^n. \end{aligned}$$

Here we have used the independent property on the third line and the fact that the random variables are identically distributed on the last line. Now

$$\begin{aligned} \mathbb{E}[e^{X_1^2 t}] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\{-\frac{1}{2}(1-2t)x_1^2\} dx_1 \quad \frac{1}{2} > t \\ &= \frac{1}{\sqrt{2\pi}(1-2t)^{1/2}} \int_{-\infty}^{\infty} \exp\{-\frac{1}{2}u^2\} du \\ &= \frac{1}{(1-2t)^{1/2}}. \end{aligned}$$

Hence

$$M(t) = \frac{1}{(1-2t)^{n/2}} \quad \frac{1}{2} > t$$

and we conclude the result. □

The standard $t-$distribution on $n-$degrees of freedom is ($n > 0$):

$$p(x) = \frac{\Gamma([n+1]/2)}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{x^2}{n}\right)^{-(n+1)/2} \quad x \in \mathsf{X} = \mathbb{R}.$$

We write $X \sim \mathcal{T}_n$. Then we have the following important result.

Proposition 4.3.23. *Let $X \sim \mathcal{N}(0,1)$ and independently $Y \sim \mathcal{X}_n^2$, $n > 0$. Let*

$$T = \frac{X}{\sqrt{\frac{Y}{n}}}.$$

Then $T \sim \mathcal{T}_n$.

Proof. We have

$$f(x,y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \frac{1}{2^{n/2}\Gamma(n/2)} y^{n/2-1} e^{-\frac{1}{2}y} \quad \mathsf{Z} = \mathbb{R} \times \mathbb{R}^+.$$

We will use Theorem 4.3.17, with the transformation defined by

$$\begin{aligned} T &= \frac{X}{\sqrt{\frac{Y}{n}}} \\ S &= Y \end{aligned}$$

and then marginalize out S. The Jacobian of transformation is:

$$J(t,s) := \begin{vmatrix} \sqrt{\frac{s}{n}} & 0 \\ \frac{t}{2\sqrt{sn}} & 1 \end{vmatrix} = \sqrt{\frac{s}{n}}.$$

Then

$$f(t,s) = \frac{1}{\sqrt{2n\pi}2^{n/2}\Gamma(n/2)} s^{n/2-1/2} \exp\{-\frac{s}{2}[t^2/n+1]\} \quad (t,s) \in \mathbb{R} \times \mathbb{R}^+.$$

Then we have for $t \in \mathbb{R}$

$$\begin{aligned} f(t) &= \frac{1}{\sqrt{2n\pi}2^{n/2}\Gamma(n/2)} \int_0^\infty s^{n/2-1/2} \exp\{-\frac{s}{2}[t^2/n+1]\} ds \\ &= \frac{1}{\sqrt{2n\pi}2^{n/2}\Gamma(n/2)} \int_0^\infty \frac{1}{(\frac{1}{2}+\frac{t^2}{2n})^{\frac{n+1}{2}}} u^{n/2-1/2} e^{-u} du \\ &= \frac{1}{\sqrt{2n\pi}2^{n/2}\Gamma(n/2)} \frac{1}{(\frac{1}{2}+\frac{t^2}{2n})^{\frac{n+1}{2}}} \Gamma([n+1]/2) \\ &= \frac{\Gamma([n+1]/2)}{\sqrt{n\pi}\Gamma(n/2)} \Big(1+\frac{t^2}{n}\Big)^{-(n+1)/2} \end{aligned}$$

and we conclude. □

The last distribution that we consider is the standard $F-$distribution on $d_1, d_2 > 0$ degrees of freedom:

$$f(x) = \frac{1}{B(\frac{d_1}{2}, \frac{d_2}{2})} \Big(\frac{d_1}{d_2}\Big)^{d_1/2} x^{d_1/2-1} \Big(1+\frac{d_1}{d_2}x\Big)^{-\frac{d_1+d_2}{2}} \quad x \in \mathsf{X} = \mathbb{R}^+$$

where

$$B(\frac{d_1}{2}, \frac{d_2}{2}) = \frac{\Gamma(\frac{d_1}{2})\Gamma(\frac{d_2}{2})}{\Gamma(\frac{d_1+d_2}{2})}.$$

We write $X \sim \mathcal{F}_{(d_1,d_2)}$. We have the following result.

Proposition 4.3.24. *Let $X \sim \mathcal{X}^2_{d_1}$ and independently $Y \sim \mathcal{X}^2_{d_2}$, $d_1, d_2 > 0$. Let*

$$F = \frac{X/d_1}{Y/d_2}.$$

Then $F \sim \mathcal{F}_{(d_1,d_2)}$.

Proof. We have

$$f(x,y) = \frac{1}{2^{d_1/2}\Gamma(d_1/2)} x^{d_1/2-1} e^{-\frac{1}{2}x} \frac{1}{2^{d_2/2}\Gamma(d_2/2)} y^{d_2/2-1} e^{-\frac{1}{2}y} \quad \mathsf{Z} = \mathbb{R}^+ \times \mathbb{R}^+.$$

We will use Theorem 4.3.17, with the transformation defined by

$$\begin{aligned} T &= \frac{X/d_1}{Y/d_2} \\ S &= Y \end{aligned}$$

and then marginalize out S. The Jacobian of transformation is:

$$J(t,s) := \begin{vmatrix} \frac{d_1 s}{d_2} & 0 \\ \frac{d_1 t}{d_2} & 1 \end{vmatrix} = \frac{d_1 s}{d_2}.$$

Then

$$f(t,s) = \frac{1}{2^{d_1/2+d_2/2}\Gamma(d_1/2)\Gamma(d_2/2)}\frac{d_1 s}{d_2}\Big(\frac{d_1 st}{d_2}\Big)^{d_1/2-1} e^{-\frac{d_1 st}{2d_2}} s^{d_2/2-1} e^{-s/2} \quad \mathsf{T} = (\mathbb{R}^+)^2.$$

To shorten the subsequent notations, let

$$g = \frac{1}{2^{d_1/2+d_2/2}\Gamma(d_1/2)\Gamma(d_2/2)}\Big(\frac{d_1}{d_2}\Big)^{d_1/2}.$$

Then, for $t \in \mathbb{R}^+$

$$\begin{aligned}
f(t) &= g t^{d_1/2-1}\int_0^\infty s^{\frac{d_1+d_2}{2}-1}\exp\{-s(\frac{1}{2}[1+\frac{d_1 t}{d_2}])\}ds \\
&= g t^{d_1/2-1}\int_0^\infty u^{(d_1+d_2)/2-1}e^{-u}(\frac{1}{2}[1+\frac{d_1 t}{d_2}])^{-(d_1+d_2)/2}du \\
&= g t^{d_1/2-1}(\frac{1}{2}[1+\frac{d_1 t}{d_2}])^{-(d_1+d_2)/2}\Gamma((d_1+d_2)/2) \\
&= g\Gamma((d_1+d_2)/2)2^{\frac{d_1+d_2}{2}} t^{d_1/2-1}\Big(1+\frac{d_1}{d_2}t\Big)^{-\frac{d_1+d_2}{2}} \\
&= \frac{1}{B(\frac{d_1}{2},\frac{d_2}{2})}\Big(\frac{d_1}{d_2}\Big)^{d_1/2} t^{d_1/2-1}\Big(1+\frac{d_1}{d_2}t\Big)^{-\frac{d_1+d_2}{2}}
\end{aligned}$$

and we conclude. □

4.4 Symmetry of I.I.D. Continuous Random Variables

Consider a sequence of i.i.d. continuous scalar random variables with PDF f and CDF F. Denote the sequence as $X_1, \ldots, X_n$ and write the collection of permutations of $\{1, \ldots, n\}$ as P; there are $n!$ such permutations. For $(X_1, \ldots, X_n)$ given we write $(X_{p(1)}, \ldots, X_{p(n)})$ with $p \in \mathsf{P}$ as the random variables $(X_1, \ldots, X_n)$ with a given permutation p of the indices $\{1, \ldots, n\}$.

We have the following result.

Proposition 4.4.1. *For any sequence of i.i.d. continuous scalar random variables we have that for every $p \in \mathsf{P}$:*

$$\mathbb{P}(X_{p(1)} < \cdots < X_{p(n)}) = \frac{1}{n!}.$$

Proof. Clearly the chance that any pair (X_i, X_j) are equal is zero for any $(i,j) \in \{1, \ldots, n\}^2$ with $i \neq j$ as X_i and X_j are continuous random variables. Then the probability of $\bigcup_{i \neq j}\{X_i = X_j\}$ is upper-bounded by the sum of the individual probabilities (i.e. for each $i \neq j$) and these probabilities are zero. Thus as the chances of ties are zero (i.e. all the random variables are distinct with probability 1) and since all of the random variables are independent and identically distributed it must be that

$$\mathbb{P}(X_{p(1)} < \cdots < X_{p(n)}) = \mathbb{P}(X_{q(1)} < \cdots < X_{q(n)})$$

for anty pair of of permutations $(p,q) \in \mathsf{P}^2$. As there are $n!$ such permutations, the result follows. □

4.5 Order Statistics

In following discussion, we consider a sequence of independent and identically distributed random variables $X_1, \ldots, X_n$. Such a sequence is referred to as a random sample.

An important type of summary of (univariate - assumed for the rest of the section) data are often associated, to the largest, smallest or mid-value of the data. For instance one can be concerned in the biggest loss for a trading transcation, or the highest level of rainfall in a region, or the 'average' (median) temperature over a month. Such statistics are often called order statistics.

Definition 4.5.1. *The order statistics of a random sample $X_1, \ldots, X_n$ are the sample values placed in ascending (increasing) order. They are denoted $X_{(1)}, \ldots, X_{(n)}$.*

In other words $X_{(1)} = \min_{i\in\{1,\dots,n\}} X_i$ and the $X_{(2)}$ is the second smallest etc, with $X_{(n)} = \max_{i\in\{1,\dots,n\}} X_i$. So we have $X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}$. Two interesting statistics based upon the order statistics are the range:

$$R_n = X_{(n)} - X_{(1)}$$

and the median

$$M_n = \begin{cases} X_{(n+1)/2} & \text{if } n \text{ is odd} \\ (X_{(n/2)} + X_{(n/2+1)})/2 & \text{if } n \text{ is even.} \end{cases}$$

The first statistic just indicates the difference in largest and smallest value and can be used as a measure of the scale of the data. The second is a popular alternative to the mean, as an average value. It can often be more robust to outlying values (than the mean), which is why it is often preferred in some scenarios.

Theorem 4.5.2. *Let $X_1, \dots, X_n$ be a random sample from a discrete distribution, with PMF $f(x)$, with $p_i = f(x_i)$ and $x_1 < x_2 < \cdots$. Define for $i \ge 1$*

$$P_i = \sum_{j=1}^{i} p_j$$

and $P_0 = 0$. Let $X_{(1)}, \dots, X_{(1)}$ be the order statistics, then for any $j \in \{1, \dots, n\}$ and $i \ge 1$

$$\mathbb{P}(X_{(j)} \le x_i) = \sum_{k=j}^{n} \binom{n}{k} P_i^k (1 - P_i)^{n-k}$$

and

$$\mathbb{P}(X_{(j)} = x_i) = \sum_{k=j}^{n} \binom{n}{k} [P_i^k (1 - P_i)^{n-k} - P_{i-1}^k (1 - P_{i-1})^{n-k}].$$

Proof. Fix i and let Y be the random variable that counts the number of $X_1, \dots, X_n$ that are less than or equal to x_i. For each of the $X_1, \dots, X_n$, call the event $\{X_j \le x_i\}$ a success and $\{X_j > x_i\}$ a failure. Then Y is the number of successes in n trials. The probability of success is the same value $P_i = \mathbb{P}(X_j \le x_i)$ for each trial due to the fact that $X_{1:n}$ are i.i.d.. The success or failure of the jth trial is independent of the outcome of any trial again for the same reason. Thus $Y \sim \mathcal{B}(n, P_i)$ where $\mathcal{B}(n, p)$ is the Binomial distribution with n trials and success probability $p \in (0, 1)$.

The event $\{X_{(j)} \le x_i\}$ is equivalent to the event $\{Y \ge j\}$; that is, at least j of the sample values are less than or equal to x_i. Then $\mathbb{P}(X_{(j)} \le x_i)$ expresses this probability:

$$\mathbb{P}(X_{(j)} \le x_i) = \mathbb{P}(Y \le j) = \sum_{k=j}^{n} \binom{n}{k} P_i^k (1 - P_i)^{n-k}.$$

For $i \ge 2$

$$\mathbb{P}(X_{(j)} = x_i) = \mathbb{P}(X_{(j)} \le x_i) - \mathbb{P}(X_{(j)} \le x_{i-1})$$

which verifies the desired equation. The case $i = 1$ follows from the defiinition of P_0. □

Theorem 4.5.3. *Let $X_1, \dots, X_n$ be a continuous random sample with CDF F and PDF f. The PDF of $X_{(j)}$ for $j \in \{1, \dots, n\}$ is*

$$f_{X_{(j)}}(x) = \frac{n!}{(j-1)!(n-j)!} f_X(x) F(x)^{j-1} [1 - F(x)]^{n-j}.$$

Proof. We will compute the CDF of $X_{(j)}$ and differentiate. Let Y be the random variable that counts the number of $X_{1:n}$ that are less than or equal to x. Defining success as the event $\{X_j \le x\}$, in a similar manner to the proof of Theorem 4.5.2, $Y \sim \mathcal{B}(n, F_x(x))$. Hence

$$F_{X_{(j)}}(x) = \mathbb{P}(Y \ge j) = \sum_{k=j}^{n} \binom{n}{k} [F(x)]^k [1 - F(x)]^{n-k}.$$

Then application of the chain rule yields

$$
\begin{aligned}
f_{X_{(j)}}(x) &= \sum_{k=j}^{n} \binom{n}{k} \Big\{ k[F(x)]^{k-1}[1-F(x)]^{n-k} f_X(x) - \\
&\quad (n-k)[F(x)]^{k}[1-F(x)]^{n-k-1} f(x) \Big\} \\
&= \binom{n}{j} j f(x)[F(x)]^{j-1}[1-F_X(x)]^{n-j} + \\
&\quad \sum_{k=j+1}^{n} \binom{n}{k} k[F(x)]^{k-1}[1-F(x)]^{n-k} f(x) \\
&\quad - \sum_{k=j}^{n-1} \binom{n}{k} (n-k)[F(x)]^{k}[1-F(x)]^{n-k-1} f(x) \\
&= \frac{n!}{(j-1)!(n-j)!} f_X(x) F(x)^{j-1}[1-F(x)]^{n-j} + \\
&\quad \sum_{k=j}^{n-1} \binom{n}{k+1} (k+1)[F(x)]^{k}[1-F(x)]^{n-k-1} f(x) \\
&\quad - \sum_{k=j}^{n-1} \binom{n}{k} (n-k)[F(x)]^{k}[1-F(x)]^{n-k-1} f(x).
\end{aligned}
$$

As

$$
\binom{n}{k+1}(k+1) = \binom{n}{k}(n-k)
$$

the last two sums cancel and we conclude. □

4.6 Connection Between the Exponential Distribution and Poisson Processes

We will now consider the link between the exponential distribution and a type of *stochastic process*. For the purposes of this course this is a random variable that is indexed by a time parameter $t \in [0, T]$, for some given $T > 0$. That is at $t \in [0, T]$ we have random variable, say N_t, that takes some random and (for this course) integer value. An example of such a process is given in Figure 4.1.

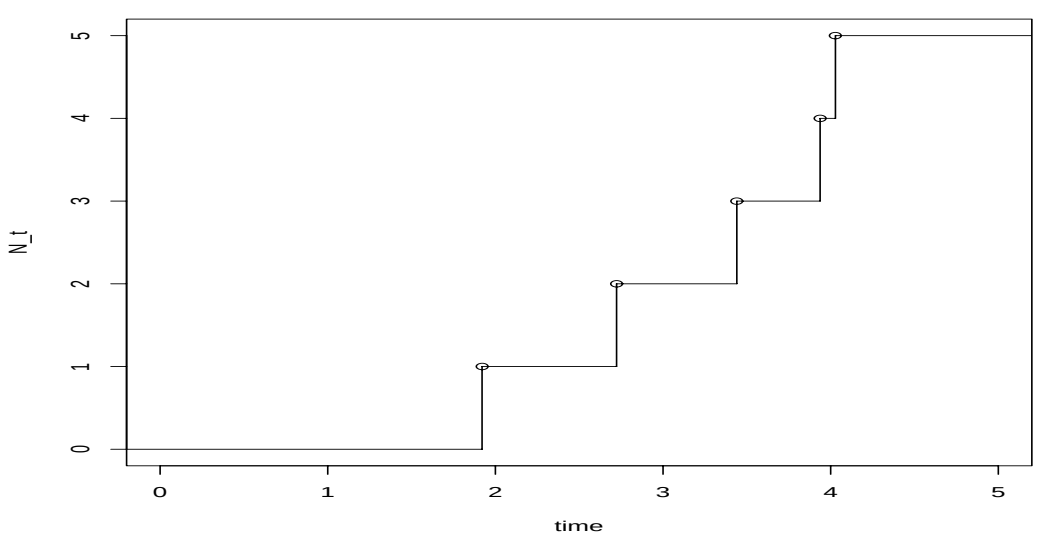

Figure 4.1: A Simulated Poisson Process on $[0, 5]$.

We begin by introducing the notion of a counting process. This will help to define the basic idea of the Poisson process.

Definition 4.6.1. *A stochastic process $\{N_t\}_{0\leq t\leq T}$ is said to be a counting process if N_t represents the total number of 'events' that have occurred up to time t. That is:*

1. $N_0 = 0$, $\forall t \in [0,T]$, $N_t \in \{0, 1, \dots\}$
2. *If* $0 \leq s < t \leq T$, $N_s \leq N_t$.

In the following definition, we consider a quantity called the increments. What this means is if $0 \leq s < t \leq T$ an increment is $N_t - N_s$ and in the context of a counting process is at least zero for any such s, t.

Definition 4.6.2. *A counting Process, $\{N_t\}_{0\leq t\leq T}$, is a Poisson Process of rate $\lambda > 0$ if*

1. $N_0 = 0$.
2. *The increments are independent. That is for any $0 \leq s_1 < s_2 < t_1 < t_2 \leq T$ the random variables $N_{s_2} - N_{s_1}$ and $N_{t_2} - N_{t_1}$ are independent.*
3. *For any $0 \leq s < t < T$, $k \in \mathbb{Z}_+$ we have*

$$\mathbb{P}(N_t - N_s = k) = \frac{(\lambda(t-s))^k e^{-\lambda(t-s)}}{k!}.$$

That is, the number of events in $[s,t]$ is a Poisson random variable, of mean $\lambda(t-s)$.

We have a process that counts events. A natural question is then: 'What is the time between events?'. To help answer this question, we derive the inter-arrival time distribution. That is, the distribution of the time to the next event.

The derivation is rather simple. Let $(X_1, \dots, X_n) := X_{1:n}$ be the inter-arrival times for the first n events. Now consider, for $t > 0$

$$\begin{aligned}\mathbb{P}(X_1 > t) &= \mathbb{P}(\text{no events in}[0,t]) \\ &= e^{-\lambda t}\end{aligned}$$

hence the density function is:

$$f_{X_1}(t) = \lambda e^{-\lambda t}$$

an exponential density function, of rate λ ($\mathcal{E}(\lambda)$). Consider the second inter-arrival time, for $t > s$

$$\begin{aligned}\mathbb{P}(X_2 > t | X_1 = s) &= \mathbb{P}\left(\text{no events in}[s, s+t] \middle| \text{one event in}[0,s]\right) \\ &= \mathbb{P}(N_{s+t} - N_s = 0 | N_s = 1) \\ &= \mathbb{P}(N_{s+t} - N_s = 0) \\ &= \mathbb{P}(N_t = 0) \\ &= e^{-\lambda t}.\end{aligned}$$

That is, independently of X_1, $X_2 \sim \mathcal{E}(\lambda)$. Here we have used the independent and stationary increment property of the Poisson process (also note that we consider inter-arrival times, so we consider the number of events on $[s, s+t]$). This construction can be repeated for any $n \geq 2$ (conditioning on $x_{1:n-1}$). Since

$$\mathbb{P}(X_{1:n}) = \mathbb{P}(X_n | x_{1:n-1}) \times \cdots \times \mathbb{P}(X_1)$$

we have established that:

Theorem 4.6.3. *Let $\{N_t\}_{0\leq t\leq T}$ be a Poisson process of rate $\lambda \in \mathbb{R}_+$. Then the inter-arrival times are independently and identically distributed exponential random variables, with rate parameter λ.*

4.7 Multivariate Normal Distribution

We saw in Example 4.3.6 the case where two random variables (X_1, X_2) are said to be jointly normally distributed (bi-variate normal). This can be extended to the case of multiple random variables, i.e. what is called a random vector $(X_1, \ldots, X_d)$ where each $X_i \in \mathbb{R}$, $i \in \{1, \ldots, d\}$.

Definition 4.7.1. *Consider a sequence of random variables $(X_1, \ldots, X_d)$, which is called a random vector. The random vector $(X_1, \ldots, X_d)$ is said to have a multivariate normal distribution if for any collection of constants $(t_1, \ldots, t_d)$ the random variable $Y = \sum_{j=1}^{d} t_j X_j$ is also normally distributed. We will exclude the case that **all** $t_i = 0$, $i \in \{1, \ldots, d\}$.*

Example 4.7.2. *To show that Definition 4.7.1 is consistent with Example 4.3.6 we use the moment generating function. Let*

$$\Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}.$$

Then it can be shown that if (X, Y) has a bivariate normal distribution with parameter ρ then the MGF for any fixed (t_1, t_2) is:

$$M(t_1, t_2) = \mathbb{E}[e^{t_1 X + t_2 Y}] = \exp\left\{\frac{1}{2} t^\top \Sigma t\right\}. \tag{4.7.1}$$

Now, to find the distribution of $t_1 X + t_2 Y$ we can use the MGF, which is by definition:

$$M(t_3) = \mathbb{E}[e^{t_1 t_3 X + t_2 t_3 Y}]$$

and then by using (4.7.1) this is equal to

$$M(t_3) = \exp\left\{\frac{1}{2} t^\top \tilde{\Sigma} t\right\}$$

where $\tilde{\Sigma} = t_3^2 \Sigma$. This MGF is that of $\mathcal{N}(0, t^\top \Sigma t)$ random variable, which establishes that the definition is consistent with Example 4.3.6.

Given Definition 4.7.1, several facts can be obtained as a direct consequence.

1. If $(X_1, \ldots, X_d)$ are multivariate normal, then any sub-vector, apart from the null one, is also (multivariate) normal. This clearly holds as one can make the requsite t_i equal to zero and by definition still yields a (multivariate) normal.

2. If $(X_1, \ldots, X_d)$ are multivariate normal and independently $(Y_1, \ldots, Y_m)$ are multivariate normal, then it follows that $(X_1, \ldots, X_d, Y_1, \ldots, Y_m)$ are multivariate normal. This follows from the fact that the sum of two independent normal random variables are also normal.

3. We can compute the MGF:

$$\begin{aligned} M(t_1, \ldots, t_n) &= \mathbb{E}\left[\exp\{\sum_{j=1}^{d} t_j X_j\}\right] \\ &= \exp\left\{\mathbb{E}[\sum_{j=1}^{d} t_j X_j] + \tfrac{1}{2}\mathbb{V}\mathrm{ar}[\sum_{j=1}^{d} t_j X_j]\right\} \\ &= \exp\left\{\sum_{j=1}^{d} t_j \mu_j + \tfrac{1}{2}\sum_{j=1}^{d} t_j^2 \mathbb{V}\mathrm{ar}[X_j] + \tfrac{1}{2}\sum_{i \neq j} t_i t_j \mathbb{C}\mathrm{ov}[X_i, X_j]\right\} \\ &= \exp\left\{t^\top \mu + \tfrac{1}{2} t^\top \Sigma t\right\} \end{aligned}$$

   where $\mu_j = \mathbb{E}[X_j]$, $\Sigma_{ij} = \mathbb{C}\mathrm{ov}[X_i, X_j]$. This establishes (4.7.1). In addition, if for each $(i, j) \in \{1, \ldots, d\}^2$ with $i \neq j$ we have $\mathbb{C}\mathrm{ov}[X_i, X_j] = 0$, that is, $\mathbb{C}\mathrm{orr}[X_i, X_j] = 0$ then the MGF is:

$$M(t_1, \ldots, t_n) = \exp\left\{t^\top \mu + \tfrac{1}{2}\sum_{j=1}^{d} t_j^2 \sigma_{ii}\right\}.$$

   This implies that each of the X_i are independent $\mathcal{N}(\mu_i, \sigma_{ii})$ random variables. In other words, if multivariate random variables are uncorrelated, they are also independent. We note that this result does not hold for every sequence of random variables.

Chapter 5

Inequalities and Modes of Convergence

5.1 Introduction

In the following Section we provide a brief introduction to some inequalities frequently used in probability (Section 5.2) and convergence of a sequence of random variables and in particular two modes of convergence:

- Convergence in Distribution (Section 5.3)
- Convergence in Probability (Section 5.4)

These properties are rather important in probability and mathematical statistics and provide a way to justify many statistical and numerical procedures. Section 5.5 gives the most classical result in probability: the central limit theorem.

In the case of convergence, the properties are rather loosely associated to the notion of a sequence of random variables $X_1, X_2, \ldots, X_n$ (there will be infinitely many of them typically) and we will be concerned with the idea of what happens to the distribution or some functional as $n \to \infty$. We will also present perhaps the most important result in probability: the central limit theorem. This provides a characterization of a random variable:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} X_i$$

with $X_1, \ldots, X_n$ independent and identically distributed with zero mean and unit variance. This particular result is very useful in hypothesis testing, which we shall see later on.

Reading 5.1.1. *[1, Chapter 10] provides a deeper introduction to this subject and should be read to improve your understanding.*

5.2 Probability Inequalities

We will consider bounds on tail probabilities. For a given random variable X and constant $\epsilon > 0$ we consider quantities of the type:

$$\mathbb{P}(|X| > \epsilon) \quad \mathbb{P}(|X - \mu| > \epsilon)$$

where $\mu = \mathbb{E}[X]$ (assumed to exist).

Example 5.2.1. *Let $X_1, \ldots, X_n$ be i.i.d. with finite, but unknown, mean μ. Consider*

$$\widehat{\mu}^n := \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Then we have that for any $n \in \mathbb{N}$:

$$\mathbb{E}[\widehat{\mu}^n] = \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}[X_i] = \mu.$$

We might expect that $\widehat{\mu}^n$ could be a good estimate of the unknown mean. A question we might then ask is, for any $\epsilon > 0$ given, what is the chance that $\widehat{\mu}^n$ is at least ϵ distance of μ in terms of the L_1-distance; i.e. what is $\mathbb{P}(|\widehat{\mu}^n - \mu| > \epsilon)$? We might hope that the larger n the smaller the probability and quantifiying how fast (as a function of n) would be important in understanding how many X_i are needed. We may not, in general, be able to compute $\mathbb{P}(|\widehat{\mu}^n - \mu| > \epsilon)$, but we may be able to upper-bound this quantity.

We begin with Markov's inequality.

Proposition 5.2.2. *Let X be a random variable and $\epsilon > 0$ then we have that*

$$\mathbb{P}(|X| > \epsilon) \leq \frac{\mathbb{E}[|X|]}{\epsilon}. \tag{5.2.1}$$

Proof. On the set $\mathsf{A} := \{x \in \mathbb{R} : |x| > \epsilon\}$ we have that $1 \leq \frac{|x|}{\epsilon}$. For any functions $\varphi_j : \mathsf{X} \to \mathbb{R}$, $j \in \{1, 2\}$ and any set A such that for every $x \in \mathsf{A}$ if $\varphi_1(x) \leq \varphi_2(x)$ we have

$$\mathbb{E}[\varphi_1(X)\mathbb{I}_{\mathsf{A}}(X)] \leq \mathbb{E}[\varphi_2(X)\mathbb{I}_{\mathsf{A}}(X)]$$

whenever the expectations exist, so it follows that

$$\mathbb{P}(|X| > \epsilon) = \mathbb{E}[\mathbb{I}_{\mathsf{A}}(X)] \leq \frac{1}{\epsilon}\mathbb{E}[|X|\mathbb{I}_{\mathsf{A}}(X)] \leq \frac{\mathbb{E}[|X|]}{\epsilon}$$

where the last inequality follows by the fact that for any $x \in \mathsf{X}$, $\mathbb{I}_{\mathsf{A}}(x) \leq 1$. □

Markov's inequality may not be particularly useful. For instance, assuming that the R.H.S. of (5.2.1) is finite it may not be less than 1 and hence not very useful. None-the-less we will see that it can be applied in a variety of scenarios for which the upper-bounds are meaningful.

We move onto an application of the Markov inequality, which is called Chebychev's inequality.

Proposition 5.2.3. *Let X be a random variable with finite mean μ and variance σ^2 and $\epsilon > 0$ then we have that*

$$\mathbb{P}(|X - \mu| > \epsilon) \leq \frac{\sigma^2}{\epsilon^2}. \tag{5.2.2}$$

Proof. As

$$\mathbb{P}(|X - \mu| > \epsilon) = \mathbb{P}(|X - \mu|^2 > \epsilon^2)$$

one can apply Markov's inequality to the R.H.S. of the above inequality and it follows that:

$$\mathbb{P}(|X - \mu| > \epsilon) \leq \frac{\mathbb{E}[|X - \mu|^2]}{\epsilon^2} = \frac{\sigma^2}{\epsilon^2}$$

which concludes the proof. □

Definition 5.2.4. *Recall the o and $\mathcal{O}$ notation from analysis. Let $(x_N)_{N\in\mathbb{N}}$ and $(y_N)_{N\in\mathbb{N}}$ be two* deterministic *sequences. Then we say that:*

$$x_N = o(y_N) \quad \text{if} \quad \lim_{N\to\infty} x_N/y_N = 0.$$

Additionally,

$$x_N = \mathcal{O}(y_N) \quad \text{if} \quad \exists M \in \mathbb{N}, C \in \mathbb{R} \text{ s.t. } |x_N/y_N| \leq C \ \forall N \geq M.$$

For instance if $x_N = N$ and $y_N = N^{1+\delta}$ for any $\delta > 0$ then $x_N = o(y_N)$.

Example 5.2.5. *Let us return to Example 5.2.1. By Chebychev's inequality we have that*

$$\mathbb{P}(|\widehat{\mu}^n - \mu| > \epsilon) \leq \frac{1}{\epsilon^2}\mathbb{E}[|\widehat{\mu}^n - \mu|^2].$$

Now

$$\mathbb{E}[|\widehat{\mu}^n - \mu|^2] = \mathbb{E}\left[\left(\frac{1}{n}\sum_{i=1}^{n}\{X_i - \mu\}\right)^2\right] = \frac{1}{n}\mathbb{V}\mathrm{ar}[X_1].$$

Then if $\mathbb{V}\mathrm{ar}[X_1] < +\infty$ we have established that

$$\mathbb{P}(|\widehat{\mu}^n - \mu| > \epsilon) \leq \frac{\mathbb{V}\mathrm{ar}[X_1]}{n\epsilon^2}$$

so the R.H.S. converges to zero at least as fast as $\mathcal{O}(n^{-1})$. We remark, however, there is nothing here to say that this rate is the fastest possible.

We finish this section with our final inequality, which is simply another application of Markov's inequality. It is called Chernoff's inequality.

Proposition 5.2.6. *Let X be a random variable and $\epsilon, t > 0$ then we have that*

$$\mathbb{P}(X > \epsilon) \leq \frac{\mathbb{E}[e^{tX}]}{e^{t\epsilon}}. \tag{5.2.3}$$

Proof. We have that as the exponential function is both invertible and monotonic increasing:

$$\mathbb{P}(X > \epsilon) = \mathbb{P}(e^{tX} > e^{t\epsilon}).$$

The proof is concluded by applying the Markov inequality to the R.H.S. of the above equation. □

Example 5.2.7. *We now consider an example to understand how sharp these bounds are. We consider $X \sim \mathcal{N}(0,1)$ and $\mathbb{P}(|X| > 3)$ and we know that this is exactly $2\Phi(-3)$ which is approximately 0.003.*

- *Using Markov's inequality we have*

  $$\mathbb{P}(|X| > 3) \leq \frac{\mathbb{E}[|X|]}{3}$$

  Recall from Example 3.4.9 that $\mathbb{E}[|X|] = \sqrt{\frac{2}{\pi}}$, so the upper-bound from Markov's inequality is approximately 0.27. This is a bound that is not very accurate.

- *Using Chebychev's inequality we have*

  $$\mathbb{P}(|X| > 3) \leq \frac{1}{3^2}$$

  which is a bound that is 0.111. This again is not very accurate.

- *Using Chernoff's inequality we have*

  $$\mathbb{P}(|X| > 3) = 2\mathbb{P}(X > 3) \leq \frac{e^{t^2/2}}{e^{3t}}.$$

  The upper-bound can be minimized in t, by using simple differential calculus (i.e. minimize any monotonic increasing function of it, by differentiating) and the optimal value is $t = 3$. The upper-bound in this case is 0.022. This is the most accurate upper-bound, but is still out by a factor of 10.

5.3 Convergence in Distribution

Consider a sequence of random variables $X_1, X_2, \ldots,$ with associated distribution functions $F_1, F_2, \ldots,$ we then have the following definition:

Definition 5.3.1. *We say that the sequence of distribution functions $F_1, F_2, \ldots$ converges to a distribution function F, if $\lim_{n\to\infty} F_n(x) = F(x)$ at each point where F is continuous. If X has distribution function F, we say that X_n converges in distribution to X and write $X_n \xrightarrow{d} X$.*

Example 5.3.2. *Consider a sequence of random variables $X_n \in \mathsf{X}_n = [0, n]$, $n \geq 1$, with*

$$F_n(x) = 1 - \left(1 - \frac{x}{n}\right)^n \quad x \in \mathsf{X}_n.$$

Then for any fixed $x \in \mathbb{R}_+$

$$\lim_{n\to\infty} F_n(x) = 1 - e^{-x}.$$

Now if $X \sim \mathcal{E}(1)$:

$$F(x) = \int_0^x e^{-u} du = 1 - e^{-x}.$$

So $X_n \xrightarrow{d} X$, $X \sim \mathcal{E}(1)$.

Example 5.3.3. *Consider a sequence of random variables $X_n \in \mathsf{X} = [0,1]$, $n \geq 1$, with*

$$F_n(x) = 1 - \frac{\sin(2n\pi x)}{2n\pi}.$$

Then for any fixed $x \in [0,1]$

$$\lim_{n\to\infty} F_n(x) = x.$$

Now if $X \sim \mathcal{U}_{[0,1]}$, $x \in [0,1]$:

$$F(x) = \int_0^x du = x.$$

So $X_n \xrightarrow{d} X$, $X \sim \mathcal{U}_{[0,1]}$.

5.3.1 Convergence in Distribution via Characteristic Functions

An method for determining convergence in distribution is via using characteristic functions. Let $i = \sqrt{-1}$, recall that the characteristic function for a random variable X, with support X is defined as

$$\phi(t) = \mathbb{E}[\exp\{iXt\}] = \int_{\mathsf{X}} e^{ixt} f(x)dx$$

where f is the PDF of X and t belongs to some domain T for which the function is well-defined. Note that for a given distribution, the characteristic function is unique. That is, no two different distributions have characteristic functions that are the same.

Example 5.3.4. *Suppose $X \sim \mathcal{N}(\mu, \sigma^2)$, then for $t \in \mathbb{R}$ fixed*

$$\phi(t) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\Big\{ixt - \frac{1}{2\sigma^2}(x-\mu)^2\Big\}dx.$$

As the exponent in the integrand is equal to

$$-\frac{1}{2\sigma^2}\Big((x - (\mu + it\sigma^2))^2 + t^2\sigma^4 - 2\mu it\sigma^2\Big)$$

we have

$$\phi(t) = \exp\{\mu it - t^2\sigma^2/2\} \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\Big\{-\frac{1}{2\sigma^2}(x - (\mu + it\sigma^2))^2\Big\}dx = \exp\{\mu it - t^2\sigma^2/2\}$$

as the integral is 1.

Theorem 5.3.5. *Let F_{X_n} and F_X be distribution functions the sequence $(X_n)_{n\in\mathbb{N}}$ and X respectively, with characteristic functions ϕ_n and ϕ respectively. A necessary and sufficient condition for $X_n \to_d X$ is that*

$$\lim_{n\to\infty} \phi_n(t) = \phi_X(t)$$

for each t.

An interesting corollary is as follows.

Corollary 5.3.6. *Suppose that $\lim_{n\to\infty} \phi_n(t) = g(t)$ for each t, with g continuous at zero. Then there exists a random variable X with CDF F_X, such that $X_n \to_d X$ and the characteristic function of X is g.*

Example 5.3.7. *Suppose that $X_n \sim \mathcal{N}(\mu + \frac{1}{n^2}, \sigma^2 + \frac{1}{n})$, then*

$$\phi_n(t) = \exp\{(\mu + \frac{1}{n^2})it - t^2(\sigma^2 + \frac{1}{n})/2\} \to \exp\{\mu it - t^2\sigma^2/2\}.$$

So if $X \sim \mathcal{N}(\mu, \sigma^2)$, $X_n \to_d X$. Note that for this example, the CDF of X_n

$$F_n(x) = \frac{1}{\sqrt{(\sigma^2 + \frac{1}{n})2\pi}} \int_{-\infty}^{x} \exp\Big\{-\frac{1}{2(\sigma^2 + \frac{1}{n})}(u - \mu - \frac{1}{n})^2\Big\}du$$

is more difficult to work with than the characteristic function. One could use the dominated convergence theorem for the CDF approach, but, our characteristic function method has only relied on standard limits from analysis.

5.4 Convergence in Probability

We now consider an alternative notion of convergence for a sequence of random variables $X_1, X_2, \ldots$

Definition 5.4.1. *We say that X_n converges in probability to a constant $c \in \mathbb{R}$ (written $X_n \xrightarrow{\mathbb{P}} c$) if for every $\epsilon > 0$*

$$\lim_{n\to\infty} \mathbb{P}(|X_n - c| > \epsilon) = 0.$$

Remark 5.4.2. *If $f : \mathbb{R} \to$ is continuous and $X_n \xrightarrow{\mathbb{P}} c$ then $f(X_n) \xrightarrow{\mathbb{P}} f(c)$.*

We note that convergence in probability can be extended to convergence to a random variable X, but we do not do this, as it is beyond the level of this course. We have the following useful result, which we do not prove.

Theorem 5.4.3. *Consider a sequence of random variables $X_1, X_2, \ldots,$ with associated distribution functions $F_1, F_2, \ldots$. If for every $x \in \mathsf{X}_n$*

$$\lim_{n\to\infty} F_n(x) = c$$

then $X_n \xrightarrow{\mathbb{P}} c$.

Example 5.4.4. *Consider a sequence of random variables $X_n \in \mathsf{X} = \mathbb{R}^+$, $n \geq 1$, with*

$$F_n(x) = \left(\frac{x}{1+x}\right)^n.$$

Then for any fixed $x \in \mathsf{X}$

$$\lim_{n\to\infty} F_n(x) = 0.$$

Thus $X_n \xrightarrow{\mathbb{P}} 0$.

We finish the Section with a rather important result. It is called the weak law of large numbers:

Theorem 5.4.5. *Let $X_1, X_2, \ldots$ be a sequence of independent and identically distributed random variables with $\mathbb{E}[|X_1|] < \infty$. Then*

$$\frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{\mathbb{P}} \mathbb{E}[X_1].$$

Proof. First we consider the case that $\mathbb{E}[X_1^2] < \infty$, then the result follows by Chebychev's inequality as established in Example 5.2.5.

Second, for the general case we will use the Marcinkiewicz-Zygmund (M-Z) inequality which states that

$$\mathbb{E}\left[\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mathbb{E}[X_1]\right|\right] \leq \frac{C}{\sqrt{n}}$$

where C is a constant that does not depend upon n. Note that in classical text-books the power, p, of $\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mathbb{E}[X_1]\right|^p$ excludes the case $p = 1$, but there is a version for $p = 1$ also. Now fix $\epsilon > 0$, one can apply Markov's inequality followed by the M-Z inequality:

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mathbb{E}[X_1]\right| > \epsilon\right) \leq \frac{1}{\epsilon}\mathbb{E}\left[\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \mathbb{E}[X_1]\right|\right] \leq \frac{C}{\epsilon\sqrt{n}}$$

and this completes the proof. □

Note that this result extends to functions; i.e. if $g : \mathbb{R} \to \mathbb{R}$ with $\mathbb{E}[|g(X_1)|] < \infty$ then

$$\frac{1}{n}\sum_{i=1}^{n} g(X_i) \xrightarrow{\mathbb{P}} \mathbb{E}[g(X_1)].$$

Example 5.4.6. *Consider an integral*

$$I = \int_{\mathbb{R}} g(x)dx$$

where we will suppose that $g(x) \neq 0$. *Suppose we cannot calculate* I. *Consider any non-zero pdf* $f(x)$ *on* $\mathbb{R}$. *Then*

$$I = \int_{\mathbb{R}} \frac{g(x)}{f(x)} f(x)dx$$

and suppose that

$$\int_{\mathbb{R}} \left|\frac{g(x)}{f(x)}\right| f(x)dx < \infty.$$

Then by the weak law of large numbers, if $X_1, \ldots, X_n$ *are i.i.d. with pdf* f *then*

$$\frac{1}{n}\sum_{i=1}^{n} \frac{g(X_i)}{f(X_i)} \xrightarrow{\mathbb{P}} I.$$

This provides a justification for a numerical method (called the Monte Carlo method) to approximate integrals.

5.5 Central Limit Theorem

The central limit theorem (CLT) is perhaps one of the most celebrated results in all of statistics and applied probability. It provides a reasoning for the universality of the normal distribution. It also provides, for us, a very important result on the asymptotic efficiency of statistics such as S_N studied in the previous section. We consider the case of i.i.d. and independent random variables.

Theorem 5.5.1. *Let* (X_i) *be a sequence of i.i.d. random variables such that* $\mathbb{E}[X_1^2] < +\infty$. *If* $\sigma^2 = \mathbb{V}\mathrm{ar}[X_1]$ *then:*

$$\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}(X_i - \mathbb{E}[X_1])\right) \to_d \mathcal{N}(0, \sigma^2)$$

or equivalently $\frac{1}{\sigma}\sqrt{n}\Big(\frac{1}{n}\sum_{i=1}^{n}(X_i - \mathbb{E}[X_1])\Big) \to_d \mathcal{N}(0,1)$.

In order to prove this Theorem, we state the following result without proof. For a random variable X with finite variance σ^2 and zero mean, we have for $t \to 0$

$$\phi_X(t) = 1 - \sigma^2 t^2 \frac{1}{2} + o(t^2) \tag{5.5.1}$$

where we will use a subscript X to denote the characteristic function associated to a given random variable X.

Proof. We give the proof of the second statement, the first being a simple by-product. Set

$$\begin{aligned} Y &= X_1 - \mathbb{E}[X_1] \\ Z_n &= \frac{1}{\sigma}\sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n}(X_i - \mathbb{E}[X_1])\right). \end{aligned}$$

Then we have

$$\begin{aligned} \phi_{Z_n}(t) &= \mathbb{E}\left[\exp\left\{it\frac{1}{\sigma}\sqrt{n}\Big(\frac{1}{n}\sum_{j=1}^{N}(X_j - \mathbb{E}[X_1])\Big)\right\}\right] \\ &= \prod_{j=1}^{n}\mathbb{E}\left[\exp\left\{it\Big(\frac{1}{\sigma\sqrt{n}}(X_j - \mathbb{E}[X_1])\Big)\right\}\right] \\ &= \mathbb{E}\left[\exp\left\{it\Big(\frac{1}{\sigma\sqrt{n}}(X_1 - \mathbb{E}[X_1])\Big)\right\}\right]^n \\ &= \phi_Y\left(\frac{t}{\sigma\sqrt{n}}\right)^n. \end{aligned}$$

So, by (5.5.1) as $n \to \infty$

$$\phi_{Z_n}(t) = \Big(1 - \sigma^2 t^2 \frac{1}{2\sigma^2 n} + o\Big(\frac{1}{N}\Big)\Big)^n \to \exp\{-t^2/2\}.$$

As $\exp\{-t^2/2\}$ is the characteristic function of a $\mathcal{N}(0,1)$ random variable, by Theorem 5.3.5, we can conclude the result. □

The Theorem deals with the behaviour of distribution of suitably centred and normalized sums of i.i.d. random variables. The normalization $\sqrt{n}$ is essentially the correct rate; if one uses $n^{1/2+\delta}$ for any $\delta > 0$ one will get convergence in probability to zero. If the term is slower than $\sqrt{n}$, then there is no meaningful limit. So this provides a rate of convergence of averages, in some sense. The Theorem essentially means that *whatever* the distribution of the i.i.d. random variables, so long as the conditions of the Theorem are satisfied, then the statistic is normally distributed. In other words, if one considers

$$\sqrt{n}\Big(\frac{1}{n}\sum_{i=1}^{n}(X_i - \mathbb{E}[X_1])\Big)$$

and then is able to obtain multiple valiues of this statistic, for n large, then the distribution of these statistics is normal with zero mean and variance σ^2. To better understand this we can consider

$$E_n := \mathbb{E}\Big[\Big(\sqrt{n}\Big(\frac{1}{n}\sum_{i=1}^{n}(X_i - \mathbb{E}[X_1])\Big)\Big)^2\Big]$$

then under some additional conditions one would have that

$$\lim_{n\to\infty} E_n = \mathbb{E}[Z^2] = \sigma^2$$

where $Z \sim \mathcal{N}(0, \sigma^2)$. So essentially σ^2 reflects the large sample (asymptotic) accuracy of $\frac{1}{n}\sum_{i=1}^{n} X_i$ as an estimator of $\mathbb{E}[X_1]$ and $n^{-1/2}$ represents how fast the estimator converges.

Example 5.5.2. *Let $X_1, \dots$ be a sequence of i.i.d $\mathcal{E}(1)$ random variables. We know that $\mathbb{E}[X_1] = \mathbb{V}\mathrm{ar}[X_1] = 1$ and hence that $S_n := \frac{1}{n}\sum_{i=1}^{n} X_i$ will converge to 1. However, we might like to know how many samples we need so that our error is at most (say) 10 % with high probability. That is, what should we take N as, so that $\mathbb{P}(0.9 \le S_n \le 1.1) \ge 0.95$?*

We will use the CLT. For simplicity of the subsequent notations, set for $x \in \mathbb{R}$

$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du$$

and recall that $\Phi(-x) = 1 - \Phi(x)$ for $x > 0$. Then we have

$$\begin{aligned} \mathbb{P}(0.9 \le S_n \le 1.1) &= \mathbb{P}(-0.1 \le S_n - 1 \le 0.1) \\ &= \mathbb{P}(-0.1\sqrt{n} \le \sqrt{n}(S_n - 1) \le 0.1\sqrt{n}). \end{aligned}$$

By the CLT $\mathbb{P}(-0.1\sqrt{n} \le \sqrt{n}(S_n - 1) \le 0.1\sqrt{n}) \approx \Phi(0.1\sqrt{n}) - \Phi(-0.1\sqrt{n})$ so we want to find n so that

$$\Phi(0.1\sqrt{n}) - \Phi(-0.1\sqrt{n}) \ge 0.95.$$

Thus as $\Phi(-0.1\sqrt{n}) = 1 - \Phi(0.1\sqrt{n})$ we want to find n such that

$$2\Phi(0.1\sqrt{n}) - 1 \ge 0.95.$$

Simple calculations yield that $n > 384$.

Chapter 6

Miscellaneous Results

In the following Chapter we quote some results of use in the course. The following Sections give general facts which should be taken as true. You can easily find the theory behind each of these results in any undergraduate text in maths or statistics.

6.1 Set Theory

Recall that $A \cup B$ means all the elements in A or all the elements of B. E.g. $A = \{1, 2\}$, $B = \{3, 4\}$ then $A \cup B = \{1, 2, 3, 4\}$. In addition that $A \cap B$ means all the elements in A *and* B. E.g. $A = \{1, 2, 3\}$, $B = \{3, 4\}$ then $A \cap B = \{3\}$. Finally recall A^c is the complement of the set A, that is everything in Ω that is not in A, e.g. $\Omega = \{1, 2, 3\}$ and $A = \{1, 2\}$, then $A^c = \{3\}$.

Let A, B, C be sets in some state-space Ω. Then the following holds true:

$$\begin{array}{rll} A \cup B = B \cup A & A \cap B = B \cap A & \text{ASSOCIATIVITY} \\ A \cup (B \cup C) = (A \cup B) \cup (A \cup C) & A \cap (B \cap C) = (A \cap B) \cap (A \cap C) & \text{DISTRIBUTIVITY.} \end{array}$$

Note that the second rule holds when mixing intersection and union, e.g.:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

We also have De-Morgan's laws:

$$(A \cup B)^c = A^c \cap B^c \quad (A \cap B)^c = A^c \cup B^c.$$

A neat trick for probability:

$$A = (A \cap B) \cup (A \cap B^c).$$

6.2 Derivatives and Functions

Here I list some standard mathematical derivatives. Below a, b are real-valued finite constants, unless otherwise stated.

$$\begin{aligned} \frac{d}{dx}(x^p) &= px^{p-1} \quad p \neq 0 \\ \frac{d}{dx}(ae^{bx}) &= abe^{bx} \\ \frac{d}{dx}(\log(x)) &= \frac{1}{x} \quad x > 0 \\ \frac{d}{dx}(a^x) &= \log(a)a^x \quad a > 0 \\ \frac{d}{dx}(\sin(ax)) &= a\cos(ax) \\ \frac{d}{dx}(\cos(ax)) &= -a\sin(ax) \\ \frac{d}{dx}(\tan(ax)) &= a\sec^2(ax) \end{aligned}$$

Also recall the product rule; for two functions $f(x)$ and $g(x)$, we have

$$\frac{d}{dx}(f(x)g(x)) = f(x)\frac{dg(x)}{dx} + g(x)\frac{df(x)}{dx}.$$

In addition the chain rule; for two functions $f(x)$ and $g(x)$, let $h(x) = f(g(x))$, then

$$\frac{dh(x)}{dx} = \frac{df(g(x))}{dg(x)}\frac{dg(x)}{dx}.$$

For example, suppose $f(x) = \log(x)$ and $g(x) = 2 + \sin(x)$ and set $h(x) = \log(2 + \sin(x))$, then

$$\frac{df(g(x))}{dg(x)} = \frac{1}{g(x)} \quad \frac{dg(x)}{dx} = \cos(x)$$

where df/dg is the derivative of $\log(g)$. Hence

$$\frac{dh(x)}{dx} = \frac{1}{g(x)}\cos(x) = \frac{\cos(x)}{2 + \sin(x)}.$$

Note that by $\log(x)$ I mean the inverse function of the exponential function, or natural logarithm:

$$\log(e^x) = x.$$

Recall that $\log(ab) = \log(a) + \log(b)$ and $\log(a^b) = b\log(a)$. Note that $\sin^2(x) + \cos^2(x) = 1$.

6.3 Integration by Parts

Recall the product rule for differentiation:

$$\frac{d}{dx}(u(x)v(x)) = u(x)\frac{dv(x)}{dx} + v(x)\frac{du(x)}{dx}.$$

Then, on rearranging:

$$u(x)\frac{dv(x)}{dx} = \frac{d}{dx}(u(x)v(x)) - v(x)\frac{du(x)}{dx}.$$

Now integrate both sides w.r.t. x and we have the *integration-by-parts formula*:

$$\int u(x)\frac{dv(x)}{dx}dx = u(x)v(x) - \int v(x)\frac{du(x)}{dx}dx$$

(note at this level we ignore the constant of integration). Thus, this method is designed for integrating a product of functions and really only of use if we *know* how to compute

$$v(x) \quad \int v(x)\frac{du(x)}{dx}dx$$

That is to say, that we need to choose $u(x)$ and $dv(x)/dx$ very carefully.

Example 6.3.1. *Consider*

$$\int xe^{-x}dx.$$

Now, here it seems that choosing $u(x) = x$ is a good idea as the derivative is 1. So, we choose

$$u(x) = x \quad \frac{dv(x)}{dx} = e^{-x}.$$

Then, we have

$$\frac{du(x)}{dx} = 1 \quad v(x) = -e^{-x}.$$

Applying the integration-by-parts formula:

$$\int xe^{-x}dx = -xe^{-x} - \int(-e^{-x})dx = -xe^{-x} - e^{-x}.$$

6.4 Integration by Substituion

This is a method of integration, which changes variables from one set to another. The basic idea is the variables which one changes variables to, produce an integral which you know how to do. Consider

$$I = \int_a^b f(x)dx$$

for $f : \mathbb{R} \to \mathbb{R}$, $-\infty \leq a < b \leq \infty$. Suppose we think $y = g(x)$ is a good choice to change variables to, where, g is a one-to-one differentiable and invertible function. Then:

$$x = g^{-1}(y) \quad \Rightarrow \frac{dx}{dy} = \frac{dg^{-1}(y)}{dy}.$$

So, we have

$$dx = \frac{dg^{-1}(y)}{dy}dy.$$

The term

$$\frac{dg^{-1}(y)}{dy}$$

is called the *Jacobian of transformation*. Now for mathematical reasons, we replace dx by

$$\Big|\frac{dg^{-1}(y)}{dy}\Big|dy.$$

So that the integral that is brought about by our substitution is

$$I = \int_{g(a)\wedge g(b)}^{g(a)\vee g(b)} f(g^{-1}(y))\Big|\frac{dg^{-1}(y)}{dy}\Big|dy$$

where $g(a) \wedge g(b) = \min\{g(a), g(b)\}$, $g(a) \vee g(b) = \max\{g(a), g(b)\}$. Note that the integration limits are induced by the fact that the function g may not be monotonic non-decreasing (so we do not know that $g(a) < g(b)$).

Example 6.4.1. *Consider*

$$\int_{-\infty}^{\infty} xe^{-(x-\mu)^2/2}dx$$

for μ some fixed finite and real constant. Let us make the change of variable

$$y = (x - \mu)$$

so, we have that

$$x = y + \mu$$

and so the jacobian of transformation is

$$\frac{dx}{dy} = 1.$$

Thus, our integral is equal to

$$\int_{-\infty}^{\infty} [y + \mu]e^{-y^2/2}dy = \int_{-\infty}^{\infty} ye^{-y^2/2}dy + \mu\int_{-\infty}^{\infty} e^{-y^2/2}dy.$$

Note, the integral of a finite sum, is equal to the finite sum of the integrals. One can show

$$\int_{-\infty}^{\infty} e^{-y^2/2}dy = \sqrt{2\pi}.$$

In addition, we have

$$\int_{-\infty}^{\infty} ye^{-y^2/2}dy = \Big[-e^{-y^2/2}\Big]_{-\infty}^{\infty}.$$

Now recall

$$\lim_{x\to\pm\infty} e^{-y^2/2} = 0.$$

So, collecting the above calculations, we have shown that

$$I = \mu\sqrt{2\pi}.$$

6.5 Double Integration

Suppose we want to calculate

$$\int\int f(x,y)dxdy$$

then this is just the same as integrating first w.r.t. x, holding y constant and then integrating w.r.t. y.

Example 6.5.1. *Consider*

$$\int_0^\infty \int_0^\infty e^{-x}e^{-y}dxdy.$$

First, we calculate

$$\int_0^\infty e^{-x}e^{-y}dx = e^{-y}\int_0^\infty e^{-x}dx = e^{-y}\Big[-e^{-x}\Big]_0^\infty = e^{-y}\Big[-\lim_{x\to\infty} e^{-x} + e^0\Big] = e^{-y}$$

where we use the fact that $\lim_{x\to\infty} e^{-x} = 0$. *Thus, to compute our integral, we have*

$$\int_0^\infty \int_0^\infty e^{-x}e^{-y}dxdy = \int_0^\infty e^{-y}dy = \Big[-e^{-y}\Big]_0^\infty = 1.$$

6.6 Summation

Recall that $\sum_{x=k_1}^{k_2} f(x)$, for some integer $-\infty \geq k_1 \leq k_2 \leq \infty$, with $f : \mathbb{Z} \to \mathbb{R}$, means

$$f(k_1) + f(k_1+1) + f(k_1+2) + \cdots + f(k_2-1) + f(k_2).$$

For example $k_1 = 1, k_2 = 3$, $f(x) = x$, then

$$\sum_{x=1}^{3} x = 1 + 2 + 3 = 6.$$

Recall that for any finite $c \in \mathbb{R}$:

$$\sum_{x=k_1}^{k_2} cf(x) = c\sum_{x=k_1}^{k_2} f(x).$$

Also for two real-valued functions $f(x)$ and $g(x)$:

$$\sum_{x=k_1}^{k_2} \Big(f(x) + g(x)\Big) = \sum_{x=k_1}^{k_2} f(x) + \sum_{x=k_1}^{k_2} g(x).$$

Some standard summations that you may have seen in the past are listed below:

$$\begin{aligned}
\frac{1}{1-z} &= \sum_{k=0}^{\infty} z^k \quad |z| < 1 \quad \text{GEOMETRIC} \\
e^z &= \sum_{k=0}^{\infty} \frac{z^k}{k!} \quad z \in \mathbb{R} \quad \text{EXPONENTIAL} \\
(1+z)^n &= \sum_{k=0}^{n} \binom{n}{k} z^k \quad n \in \mathbb{Z}^+ \quad \text{BINOMIAL} \\
(1+z)^{-n} &= \sum_{k=0}^{n} \binom{n+k}{k} z^k \quad n \in \mathbb{Z}^+, |z| < 1 \quad \text{NEGATIVE BINOMIAL} \\
-\log(1-z) &= \sum_{k=1}^{\infty} \frac{z^k}{k} \quad |z| < 1 \quad \text{LOGARITHMIC} \\
\log(1+z) &= \sum_{k=1}^{\infty} (-1)^{k+1}\frac{z^k}{k} \quad |z| < 1 \quad \text{LOGARITHMIC}
\end{aligned}$$

Note that, suppose $|z| < 1$, then we have

$$\sum_{k=j}^{n} z^k = z^j \frac{1 - z^{n-j+1}}{1 - z},$$

A standard trick that is used in this course is the reversal of summation and differentiation. For example, consider $f : \mathbb{Z} \times \Theta \to \mathbb{R}$, where $\Theta \subseteq \mathbb{R}$, then it is valid to assume in this course that:

$$\frac{d}{d\theta}\Big(\sum_{x=k_1}^{k_2} f(x,\theta)\Big) = \sum_{x=k_1}^{k_2} \frac{df(x,\theta)}{d\theta}.$$

In words 'the derivative of the sum is equal to the sum of the derivatives'. This trick is very useful for computing expectations w.r.t. e.g. geometric distributions. Note that this also applies to higher-order derivatives:

$$\frac{d^2}{d\theta^2}\Big(\sum_{x=k_1}^{k_2} f(x,\theta)\Big) = \sum_{x=k_1}^{k_2} \frac{d^2 f(x,\theta)}{d\theta^2}.$$

A *double summation* $\sum_{y=k_3}^{k_4}\sum_{x=k_1}^{k_2} f(x,y)$, with $f : \mathbb{Z}^2 \to \mathbb{R}$, $-\infty \geq k_1 \leq k_2 \leq \infty$, with $f : \mathbb{Z} \to \mathbb{R}$, $-\infty \geq k_3 \leq k_4 \leq \infty$, with $f : \mathbb{Z} \to \mathbb{R}$ means:

$$\begin{aligned}\sum_{y=k_3}^{k_4}\sum_{x=k_1}^{k_2} f(x,y) &= \sum_{y=k_3}^{k_4}\Big(f(k_1,y) + \cdots + f(k_2,y)\Big) \\ &= f(k_1,k_3) + f(k_1,k_3+1) + \cdots + f(k_1,k_4) + \cdots + f(k_2,k_3) + f(k_2,k_3+1) + \cdots + f(k_2,k_4).\end{aligned}$$

In words: 'keep y constant and sum over x, then for each x value sum over y'. Note again, that it will be mathematically valid to switch the order of summation in this course, i.e.:

$$\sum_{y=k_3}^{k_4}\sum_{x=k_1}^{k_2} f(x,y) = \sum_{x=k_1}^{k_2}\sum_{y=k_3}^{k_4} f(x,y).$$

You should be careful when switching the order of summation, when the first limit depends upon y, e.g.

$$\sum_{y=k_3}^{k_4}\sum_{x=k_1}^{y} f(x,y).$$

6.7 Exponential Function

For $x \in \mathbb{R}^+$:

$$\begin{aligned}\lim_{n\to\infty}\Big(1 + \frac{x}{n}\Big)^n &= e^x \\ \lim_{n\to\infty}\Big(1 - \frac{x}{n}\Big)^n &= e^{-x}.\end{aligned}$$

6.8 Taylor Series

Under assumptions that are in effect in this course, for infinitely differentiable $f : \mathbb{R} \to \mathbb{R}$:

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-c)^k}{k!} f^{(k)}(c) \quad c \in \mathbb{R}.$$

6.9 Integration Methods Continued

We have the following standard properties of integrals, for c real and finite and $f(x)$ integrable

$$\int cf(x)dx = c\int f(x)dx.$$

For two integrable functions $f(x)$ and $g(x)$ we have

$$\int \Big(f(x)+g(x)\Big)dx = \int f(x)dx + \int g(x)dx.$$

A table of integrals, essentially follows by reversing the derivatives that you saw previously (we omit the constant of integration which is not needed at this level):

$$\begin{aligned}
\int x^p dx &= \frac{x^{p+1}}{p+1} \quad p \neq -1 \\
\int ae^{bx} dx &= \frac{a}{b}e^{bx} \\
\int \log(x)dx &= x\log(x) - x \\
\int a^x dx &= \frac{1}{\log(a)}a^x \quad a > 0 \\
\int \sin(ax)dx &= -\frac{1}{a}\cos(ax) \\
\int \cos(ax)dx &= \frac{1}{a}\sin(ax) \\
\int \tan(ax)dx &= \log|\sec(ax)|.
\end{aligned}$$

Some ideas for integration:

$$\frac{d}{dx}[f(g(x))] = g'(x)f'(g(x)) \qquad \int g'(x)f'(g(x))dx = f(g(x)) + C.$$

Consider the following integration 'trick'. Suppose you want to compute

$$\int g(x)dx$$

where g is a positive function. Suppose that there is a PDF $f(x)$ such that

$$f(x) = cg(x)$$

where you know c. Then

$$\int g(x)dx = \int \frac{1}{c}f(x)dx = \frac{1}{c}.$$

Recall that

$$I = \int_{-\infty}^{\infty} e^{-u^2/2}du = \sqrt{2\pi}.$$

This can be established by the fact that

$$I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-u^2/2-v^2/2}dudv$$

and then changing to polar co-ordinates.

6.10 Distributions

A Table of discrete distributions can be found in Table 6.1 and continuous Distributions in Table 6.2. Recall that

$$\Gamma(a) = \int_0^{\infty} t^{a-1}e^{-t}dt \quad a > 0.$$

Note that one can show $\Gamma(a+1) = a\Gamma(a)$ and for $a \in \mathbb{Z}_+$, $\Gamma(a) = (a-1)!$. In addition that for $a, b > 0$

$$B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.$$

One can show also that

$$B(a,b) = \int_0^1 u^{a-1}(1-u)^{b-1}du.$$

	Support X	Par.	PMF	CDF	$\mathbb{E}[X]$	$\mathbb{V}\mathrm{ar}[X]$	MGF
$\mathcal{B}(1,p)$	$\{0,1\}$	$p \in (0,1)$	$p^x(1-p)^{1-x}$		p	$p(1-p)$	$(1-p)+pe^t$
$\mathcal{B}(n,p)$	$\{0,1,\dots,n\}$	$p \in (0,1)$, $n \in \mathbb{Z}^+$	$\binom{n}{x}p^x(1-p)^{n-x}$		np	$np(1-p)$	$((1-p)+pe^t)^n$
$\mathcal{P}(\lambda)$	$\{0,1,2,\dots\}$	$\lambda \in \mathbb{R}^+$	$\frac{\lambda^x e^{-\lambda}}{x!}$		λ	λ	$\exp\{\lambda(e^t-1)\}$
$\mathcal{G}e(p)$	$\{1,2,\dots\}$	$p \in (0,1)$	$(1-p)^{x-1}p$	$1-q^x$	$1/p$	$(1-p)/p^2$	$\frac{pe^t}{1-e^t(1-p)}$
$\mathcal{N}e(n,p)$	$\{n,n+1,\dots\}$	$p \in (0,1), n \in \mathbb{Z}^+$	$\binom{x-1}{n-1}(1-p)^{x-n}p^n$		n/p	$n(1-p)/p^2$	$\left(\frac{pe^t}{1-e^t(1-p)}\right)^n$

Table 6.1: Table of Discrete Distributions.Note that $q = 1-p$.

	Support X	Par.	PDF	CDF	$\mathbb{E}[X]$	$\mathbb{V}\mathrm{ar}[X]$	MGF
$\mathcal{U}_{[a,b]}$	$[a,b]$	$-\infty < a < b < \infty$	$\frac{1}{b-a}$	$\frac{x-a}{b-a}$	$(a+b)/2$	$(b-a)^2/12$	$\frac{e^{bt}-e^{at}}{t(b-a)}$
$\mathcal{E}(\lambda)$	$\mathbb{R}^+$	$\lambda \in \mathbb{R}^+$	$\lambda e^{-\lambda x}$	$1-e^{-\lambda x}$	$1/\lambda$	$1/\lambda^2$	$\frac{\lambda}{\lambda-t}$
$\mathcal{G}(a,b)$	$\mathbb{R}^+$	$a,b \in \mathbb{R}^+$	$\frac{b^a}{\Gamma(a)} x^{a-1} e^{-bx}$		a/b	a/b^2	$(\frac{b}{b-t})^a$
$\mathcal{N}(a,b)$	$\mathbb{R}$	$(a,b) \in \mathbb{R} \times \mathbb{R}^+$	$\frac{1}{\sqrt{2\pi b}} e^{-\frac{1}{2b}(x-a)^2}$		a	b	$e^{at+bt^2/2}$
$\mathcal{B}e(a,b)$	$[0,1]$	$(a,b) \in (\mathbb{R}^+)^2$	$B(a,b)^{-1} x^{a-1}(1-x)^{b-1}$		$a/(a+b)$	$\frac{ab}{(a+b)^2(a+b+1)}$	

Table 6.2: Table of Continuous Distributions. Note that $B(a,b)^{-1} = \Gamma(a+b)/[\Gamma(a)\Gamma(b)]$.

Bibliography

[1] Blitzstein, J, & Hwang, J. (2019). *Introduction to Probability*. New York: Taylor & Francis.

[2] Tijms, H. (2012). *Understanding Probability*. Third Edition. Cambridge: CUP.